

# Atlas — Complete Documentation

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## Multi-tenant SaaS for travel agencies

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- Live: <https://agencetool.vercel.app>
- Repo: [github.com/ouks1993/agence\\_tool](https://github.com/ouks1993/agence_tool)

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# Atlas — Documentation Index

Atlas is the operating system for modern travel agencies. Every decision should support that sentence.

Atlas is a **multi-tenant SaaS for travel agencies**. Each agency runs its clients, sales pipeline, proposals, bookings and finance in a fully isolated workspace; the vendor manages every agency from a platform console. Multilingual (EN/FR/AR with RTL), deployed on Vercel with GitHub auto-deploy.

- **Live:** <https://agencetool.vercel.app>
- **Repo:** [github.com/ouks1993/agence\\_tool](https://github.com/ouks1993/agence_tool)
- **Vendor console:** <https://agencetool.vercel.app/platform>

This is the reference index. The detailed reference is split into focused docs under `docs/`. See also `DESIGN.md` (design system) and `AGENTS.md/CLAUDE.md` (working conventions) — those remain the source of truth for UI and coding conventions respectively.

## Documentation map

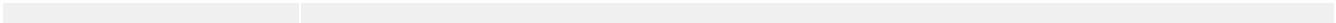
| Doc                 | Covers                                                                                           |
|---------------------|--------------------------------------------------------------------------------------------------|
| Vision              | What Atlas is, who it's for, operating model, features at a glance                               |
| Architecture        | Multi-tenancy · auth · platform admin · impersonation · roles · route map · module layout · i18n |
| Tech stack          | Framework, libraries, services                                                                   |
| Domain model        | Core entities, the Lead→...→Feedback chain, aggregates & ownership                               |
| Database            | Schema tables, tenancy columns, migrations, ID convention                                        |
| API integrations    | Duffel · Hotelbeds · Stripe (billing + Connect) · Resend · OpenRouter · Blob                     |
| Development guide   | Setup, env vars, commands, scripts, DB workflow                                                  |
| Coding standards    | → points to <code>AGENTS.md</code> ; conventions summary                                         |
| UI / UX             | → points to <code>DESIGN.md</code> ; product UX patterns                                         |
| Business rules      | RBAC, currency, booking lifecycle, commissions, vocabularies                                     |
| Deployment          | Vercel, env, prod migrations, demo accounts                                                      |
| Roadmap & changelog | Phase status, open items, commit changelog                                                       |
| Security            | Tenant isolation, auth, guards, webhooks, DB safety, known risks                                 |
| Analytics & BI      | Dashboards, currency-safe metrics, CSV/Excel export                                              |

|    |                                |
|----|--------------------------------|
| AI | Assistant + inline AI features |
|----|--------------------------------|

## At a glance

- **Stack:** Next.js 16 (App Router) · React 19 · TypeScript · Tailwind v4 + shadcn/ui · PostgreSQL (Neon) + Drizzle · Better Auth · next-intl · Stripe + Stripe Connect · Resend · Duffel (flights) · Hotelbeds (hotels) · Vercel AI SDK + OpenRouter · Vercel.
- **Tenancy:** every business row is scoped to an `agencyId`; enforced via `requireAgencyUser()`. Vendor platform admin sits above all tenants.
- **Roles:** admin · manager · finance · support · agent — each with a tailored landing and nav.
- **Deploy:** push to `main` → Vercel auto-deploys production.
- **DBs:** dev `ep-wandering-sunset-aitlty78` · prod `ep-misty-thunder-aixz34vy`. Run `POSTGRES_URL=<prod-url> npx drizzle-kit migrate` after each schema change.

Phases 1–3 + UX + Data-quality/BI + Sprint 1 booking architecture complete. Multi-tenant SaaS with live travel sourcing, billing, client portal, supplier/commission management, and inline AI features. Real supplier booking (Duffel + Hotelbeds) is fully wired — activate with production credentials. Migrations: 19 (latest 0019).



# Vision

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## Atlas is the operating system for modern travel agencies.

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Every decision should support that sentence.

It is the north star: a feature, a design choice, or a piece of scope earns its place only if it makes Atlas more of an agency's single operating system. If it doesn't support the sentence, it doesn't ship.

**No OTA knows agency behaviour. No agency knows market behaviour. Only Atlas sees both.**

That information asymmetry — across thousands of agencies, in real time — is the business. Not the software. Not the AI features. The information.

Today an agency runs on a patchwork of tools:

- Excel
- WhatsApp
- Email
- Booking portals
- Accounting software

Everything should happen inside Atlas instead — **one workspace**.

## Long-term goals

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- CRM
- Sales
- Quotations
- Flight booking
- Hotel booking
- Supplier management
- Accounting
- Customer portal
- AI assistant
- BI
- Automation

## Operating model

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- **Per-agency isolation.** Every business record is scoped to an `agencyId`; agencies never see each other's data. Reference numbers (`BKG-...`, `PRD-...`) are unique per agency. See [architecture.md](#) and [security.md](#).

- **Vendor + tenants.** The vendor (platform operator) sells Atlas to many agencies and bills them by subscription, managing all of them from a platform console. Each agency is a fully isolated tenant.
- **DZD-first, no FX.** The agency operates in Algerian Dinar. EUR/USD are supported for source data but Atlas never converts between currencies — analytics group by currency rather than summing across. See business-rules.md.
- **Graceful degradation.** Every external integration (flights, hotels, email, payments, AI) falls back to sample/logged behaviour when its keys are unset, so the app runs end-to-end with only a database and an auth secret. See api-integrations.md.

## Where it is today

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Atlas is a deployed, multi-tenant SaaS — multilingual (EN/FR/AR with RTL), live on Vercel with GitHub auto-deploy.

- **Live:** <https://agencetool.vercel.app>
- **Repo:** [github.com/ouks1993/agence\\_tool](https://github.com/ouks1993/agence_tool)
- **Vendor console:** <https://agencetool.vercel.app/platform>

Shipped against the long-term goals:

- **CRM** — clients with contacts and a unified funnel timeline.
- **Sales** — opportunities, pipeline stages, conversion analytics.
- **Quotations** — proposals with PDF + e-signature, one-click convert→booking, AI quote builder.
- **Flight booking** — live Duffel search (real order placement is pending).
- **Hotel booking** — live Hotelbeds search + content cache (real booking pending).
- **Supplier management** — directory, contracts & rates, supplier picker.
- **Customer portal** — magic-link login, trip view, online payments, proposal signing.
- **AI assistant** — agency-scoped chat + inline itinerary/quote/email/visa features. See ai.md.
- **BI** — decision dashboards + standardized CSV/Excel export. See analytics.md.
- **Accounting** — partial: payments, two-ledger commissions, AR aging. Full accounting is a long-term goal.
- **Automation** — early: auto-generated commissions, lifecycle guards. Broader automation (e.g. quote-on-stage-change, WhatsApp) is on the roadmap.

## Atlas principles

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The ten founding principles. Everything else — the design system, the never-rules, the golden workflow — is an expression of these.

1. **Client first.** The client record is the spine; every workflow starts there.
2. **Every entity belongs to an agency.** Multi-tenancy is non-negotiable — `agencyId` on every business row, enforced on every read/write.
3. **No duplicated data.** One source of truth; canonical values over free text.
4. **Automation over manual work.** Generate, don't ask — quotes, itineraries, commissions, documents.

5. **Every workflow has one happy path.** A single, obvious golden route through each flow (business-rules.md).
6. **Managers need insight.** Decision-oriented analytics where managers land.
7. **Agents need speed.** Fast, scoped surfaces — minimal clicks, minimal waiting.
8. **Data quality > flexibility.** Constrain inputs to keep reporting clean.
9. **Everything is reversible.** Prefer soft state and undo over destructive operations.
10. **Every page answers "What do I do next?"** Always surface the next action; never a dead end.

These are operationalized as UI guardrails in ui-ux.md, hard constraints in business-rules.md, and tracked against the code in the gap tracker.

See roadmap.md for phase status and open items, and the index for the full documentation map.

# Tech Stack

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| Layer       | Choice                                                                             |
|-------------|------------------------------------------------------------------------------------|
| Framework   | Next.js 16 (App Router), React 19, TypeScript                                      |
| Styling     | Tailwind CSS v4 ( <code>@theme inline</code> ), shadcn/ui (new-york), Lucide icons |
| Fonts       | Geist (sans/mono), IBM Plex Sans Arabic (RTL)                                      |
| Auth        | Better Auth (email/password, invitation-gated)                                     |
| Database    | PostgreSQL (Neon) + Drizzle ORM                                                    |
| i18n        | next-intl 4 (cookie-based, no URL routing)                                         |
| Charts      | recharts 3 (tokenized)                                                             |
| AI          | Vercel AI SDK + OpenRouter                                                         |
| Email       | Resend (transactional: invites, password reset, proposals)                         |
| Billing     | Stripe subscriptions (vendor → agency) + webhook                                   |
| Payments    | Stripe Connect (traveler → agency, destination charges)                            |
| Flights     | Duffel (Amadeus self-service kept only as legacy fallback)                         |
| Hotels      | Hotelbeds (APITUDE: availability + content)                                        |
| PDF         | <code>@react-pdf/renderer</code> (server-rendered proposals)                       |
| Data export | <code>exceljs</code> (XLSX) + built-in CSV (UTF-8 BOM) for BI/Power BI             |
| Storage     | Vercel Blob (optional)                                                             |
| Hosting     | Vercel + GitHub auto-deploy                                                        |

## Conventions tied to the stack

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- **Styling** follows the design system in `DESIGN.md`; see `ui-ux.md`.
- **Working conventions** (sub-agents, UUID ids, Drizzle workflow, testing) live in `AGENTS.md`; see `coding-standards.md`.
- **External integrations** and their env vars are documented in `api-integrations.md`. Every one degrades gracefully when unconfigured.
- **Tailwind v4** is configured CSS-first via `@theme inline` in `globals.css` — there is no `tailwind.config.ts`.



# Architecture

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Tech choices live in tech-stack.md. This doc covers structure: multi-tenancy, auth, roles, the route map, the module layout, and i18n plumbing.

## Multi-tenancy

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Every business table carries `agencyId` (tenant roots) or inherits it through a parent (children). **All** reads/writes are scoped by agency, enforced in actions and pages via `requireAgencyUser()` → `user.agencyId`. References (BKG-..., PRD-...) are unique **per agency**. Re-verified by `scripts/test-tenant-isolation.ts`. See `security.md` for the isolation guarantees and `database.md` for tenancy columns.

## Auth & onboarding

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- **Better Auth** (`src/lib/auth.ts`) — email/password. The `user.create.before` hook makes signup **invitation-only**: it requires a pending `agency_invite` matching the email, stamps `agencyId` + `role`, and rejects everyone else (blocks the raw signup endpoint too).  
`BETTER_AUTH_URL/baseURL` + `trustedOrigins` set so the deployed domain is trusted.
- **Guards** (`src/lib/permissions.ts`): `requireUser`, `requireAgencyUser` (tenant + agency-suspension lockout), `requireManager`, `requireCapability`, `requirePlatformAdmin`.

## Platform admin (vendor)

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A user with `isPlatformAdmin = true`, `agencyId = null` — above all tenants, routed to `/platform`. Cannot enter a tenant app except via impersonation.

## Impersonation (cookie-driven, platform-admin only)

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- `viewAsAgency(agencyId)` → cookie `platform_view_agency` → acts as agency **admin**.
- `viewAsUser(userId)` → cookie `platform_view_user` (takes precedence) → adopts that user's identity, agency and **role** (full fidelity — an agent sees only their work).
- Resolved in `requireUser`; `user.impersonating = "agency" | "user" | null` drives the exit banner. `exitAgencyView()` clears both cookies.

## Per-role landing

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`roleHome(role)` routes `finance` → `/finance`, `support` → `/support`, `else` `/dashboard` (which itself adapts: agency-wide for admin/manager, scoped "Your work" for agents). App-shell nav is role-aware via a `show(role)` predicate per item.

## Roles & permissions

Defined in `src/lib/domain.ts`. Full capability semantics are in `business-rules.md`.

| Role    | Sees all data | Team mgmt | Payments | Finance view | Support view | Delete | Home                |
|---------|---------------|-----------|----------|--------------|--------------|--------|---------------------|
| admin   | ✓             | ✓         | ✓        | ✓            | ✓            | ✓      | /dashboard          |
| manager | ✓             | ✓         | ✓        | ✓            | ✓            | ✓      | /dashboard          |
| finance | ✓             | —         | ✓        | ✓            | —            | —      | /finance            |
| support | ✓             | —         | —        | —            | ✓            | —      | /support            |
| agent   | own only      | —         | —        | —            | —            | —      | /dashboard (scoped) |

Capability helpers: `seesAllData`, `canManageTeam`, `canAssignAdmin`, `canManagePayments`, `canViewFinance`, `canViewSupport`, `canDeleteRecords`, `roleHome`. Only an **admin** can assign/change the admin role.

## Scale targets

The capacity Atlas is designed toward, with implications for the current architecture (single Neon Postgres + Vercel serverless).

| Target                         | Implication / current state                                                                                                              |
|--------------------------------|------------------------------------------------------------------------------------------------------------------------------------------|
| 1,000 agencies                 | Multi-tenant single DB; fine with <code>agencyId</code> indexes (present).                                                               |
| 50,000 users                   | Better Auth + Postgres; fine at this scale.                                                                                              |
| 5,000,000 bookings             | Needs solid indexing + pagination (currently hardcoded <code>limits</code> — see tracker).                                               |
| 500,000,000 activities         | <code>activity_log</code> at this size needs <b>partitioning + archival/retention</b> — not in place today (single unpartitioned table). |
| 99.9% uptime                   | Relies on Vercel + Neon SLAs; no DR runbook yet (see <code>security.md</code> ).                                                         |
| 100 concurrent hotel searches  | Bound by Hotelbeds quota; content cache serves photos quota-free. No concurrency/queue control or rate limiting yet.                     |
| 100 concurrent flight searches | Bound by Duffel quota; <code>safeSearch</code> degrades to sample data on error. No queue/backpressure yet.                              |

These are design targets. The gating work to reach them — table partitioning for `activity_log`, real pagination, rate limiting/backpressure on search, and a DR plan — is in the gap tracker.

## Route map

**Authenticated app** (`(app) /`, gated by `requireAgencyUser`): `dashboard`, `finance`, `commissions`

(canViewFinance), support, bookings (+ new, [id], [id]/edit, [id]/itinerary), clients (+ new, [id], [id]/edit), opportunities (+ new, [id], [id]/edit), products (+ new, [id], [id]/edit), suppliers (+ new, [id], [id]/edit) (canManageTeam), operations (Pipeline board), reports (canViewFinance — BI export hub), search, hotels (+ [code] details), assistant, team (canManageTeam), billing (admin-only), settings, profile.

**Client portal** (portal/, gated by requirePortalSession): portal (trip list), portal/bookings/[id] (detail + pay), portal/proposals (list), portal/proposals/[id] (view + sign), portal/login.

**Platform** (platform/, gated by requirePlatformAdmin): platform, platform/agencies/new, platform/agencies/[id].

**Auth** ((auth)/): login, register (invite-only notice), forgot-password, reset-password. **Accept invite:** invite/[token].

**Public / docs:** i/[token] (shareable itinerary, unauth), p/[token] (public signable proposal) + p/[token]/pdf, proposal/[id] (internal preview) + proposal/[id]/pdf, booking-docs/[id]/voucher, booking-docs/[id]/invoice.

**API:** api/auth/[...all] (Better Auth), api/chat (AI assistant), api/export/[entity] (BI export: CSV/XLSX per dataset + workbook = all sheets), api/stripe/webhook (subscription reconciliation), api/stripe/connect-webhook (Connect payment reconciliation), api/portal/auth/request · verify · signout (portal magic-link flow).

## Module layout

src/lib/

- domain.ts — roles, capabilities, enums (statuses, stages, item types, currencies + controlled vocabularies: lead source, travel purpose, trip type, gender, title, lost reason, industry), roleHome, status/role metadata.
- analytics.ts — pure dashboard helpers (see analytics.md).
- reference/countries.ts — ISO 3166-1 list (name, demonym, derived flag) + normalizeCountry() alias mapping.
- export/ — csv.ts (RFC-4180 + UTF-8 BOM), xlsx.ts (exceljs workbook), datasets.ts (BI dataset registry with dual code/label columns).
- permissions.ts — auth guards + impersonation resolution.
- auth.ts / auth-client.ts — Better Auth config + client.
- invites.ts — create/find/accept invite tokens (7-day TTL).
- queries.ts — shared agency-scoped pickers.
- activity.ts — logActivity (agency-scoped audit).
- db.ts (validates env via getServerEnv), schema.ts, env.ts, config.ts, format.ts, utils.ts, itinerary.ts, storage.ts.
- notifications/email.ts (Resend adapter) + notifications/templates.ts (HTML).
- billing/stripe.ts — SaaS subscriptions (distinct from payments/stripe.ts).
- payments/stripe.ts — Stripe Connect: account, onboarding link, destination charges.
- suppliers/ — index.ts (getFlightSupplier/getHotelSupplier + safeSearch), duffel.ts, hotelbeds.ts, content-cache.ts, amadeus.ts (legacy), mock.ts, types.ts.

- `documents/proposal-pdf.tsx + proposal-data.ts` — proposal PDF rendering.
- `portal-session.ts` — `getPortalSession / requirePortalSession` (`httpOnly` cookie).

**src/lib/actions/** (server actions, all agency-scoped): `clients`, `opportunities`, `products`, `proposals-public`, `bookings`, `payments`, `notifications`, `team`, `invites`, `platform`, `billing`, `settings`, `search`, `suppliers`, `commissions`, `portal-payments`, `portal-proposals`, `portal-invite`, `onboarding`, `ai`.

**src/components/** — `app/` (shell, page-header, stat-card, status-badge, getting-started-card), `charts/` (insight-charts incl. `HBarInsight`, `FunnelInsight`), `reference/` (country-combobox, city-input), `reports/`, `settings/`, `team/`, `platform/`, `auth/`, `bookings/` (lifecycle stepper, search sheet, board), `clients/` (portal invite, timeline), `products/` (convert-to-booking, AI quote builder), `opportunities/`, `billing/`, `commissions/`, `suppliers/`, `portal/`, `search/`, `documents/`, `ui/` (shadcn).

## Internationalization

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- **Locales:** `en`, `fr`, `ar` (Arabic = RTL). Cookie-based (`locale`), no URL-locale routing — all routes unchanged.
- `src/i18n/` — `config.ts` (locales, metadata, dir), `request.ts` (next-intl request config reading the `locale` cookie). Messages: `messages/{en,fr,ar}.json`.
- Root layout sets `<html lang dir>` and applies IBM Plex Sans Arabic for RTL (`html[dir="rtl"] body` in `globals.css`).
- Translated surfaces: `nav`, `login`, `dashboard`, `settings`. Others fall back to English. **To translate a string:** add the key to all three `messages/*.json` and swap to `t("...")` (`getTranslations` in server, `useTranslations` in client).
- Language is changed in **Settings**; choice is saved to `user.locale` and the cookie. **Known gap:** a fresh device without the cookie shows English until the user re-picks — `user.locale` isn't yet synced to the cookie on login (see roadmap.md open items).

# Domain Model

The core business entities and how they relate. This is the **data/relationship** view; the process view (states, who does what) is the golden workflow. Tables and columns live in database.md.

## The chain

```
Lead → Client → Opportunity → Proposal → Booking → Supplier →  
Invoice → Payment → Accounting → Travel → Feedback
```

The **client is the aggregate root** — everything hangs off it, and every entity is scoped to an `agencyId` (founding principle #2).

## Entities

| Domain concept | Table(s)                                                                                                                             | Status | Key relations                                                                                                                  |
|----------------|--------------------------------------------------------------------------------------------------------------------------------------|--------|--------------------------------------------------------------------------------------------------------------------------------|
| Lead           | client (status lead, source code)                                                                                                    | ✓      | becomes a Client; no separate table                                                                                            |
| Client         | client (+ client_contact)                                                                                                            | ✓      | root; owns opportunities, proposals, bookings                                                                                  |
| Opportunity    | opportunity                                                                                                                          | ✓      | belongs to a client; assignedToId; links to a proposal                                                                         |
| Proposal       | product (+ product_item)                                                                                                             | ✓      | belongs to client; e-sign; converts → Booking                                                                                  |
| Booking        | booking (+ booking_traveller, booking_item, booking_day, booking_supplier_ref, booking_event, booking_document, booking_idempotency) | ✓      | belongs to client; lifecycle states; items reference suppliers; Sprint 1 tables power real booking, audit log, and idempotency |
| Supplier       | supplier (+ supplier_contract, supplier_rate)                                                                                        | ✓      | referenced by booking_item.supplierId + product_item                                                                           |
| Invoice        | — (on-demand PDF: booking-docs/[id]/invoice)                                                                                         | ●      | no managed invoice entity yet (planned module)                                                                                 |
| Payment        | payment (child of booking)                                                                                                           | ✓      | deposits/installments; Stripe Connect                                                                                          |
| Accounting     | commission ledger only                                                                                                               | ●      | partial; no GL/accounting entity (planned module)                                                                              |

|                 |                                         |   |                                            |
|-----------------|-----------------------------------------|---|--------------------------------------------|
| <b>Travel</b>   | booking (state completed) + booking_day | ✓ | itinerary timeline; vouchers               |
| <b>Feedback</b> | — (activity_log / notes)                | ● | no review/feedback entity (planned module) |

## Aggregates & ownership

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- **Client aggregate** — `client` → `client_contact`; the spine that opportunities, proposals, and bookings all reference by `clientId`.
- **Booking aggregate** — `booking` → `booking_traveller`, `booking_item`, `booking_day`, `payment`, `booking_supplier_ref`, `booking_event`, `booking_document`, `booking_idempotency`. Children carry **no** `agencyId`; they inherit tenancy through the parent booking (scoped via the parent on every query). `booking_event` is append-only (audit + analytics); `booking_idempotency` prevents duplicate supplier orders on retry.
- **Supplier aggregate** — `supplier` → `supplier_contract` → `supplier_rate`; `commission` rows link a booking item back to the earning supplier/agent.
- **Reference data** — `hotel_content` is **global** (not tenant-scoped); ISO countries/vocabularies are canonical lookups, not per-agency rows.

## Where the chain is aspirational

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Three links are not yet first-class domain entities — they're the bridge to the planned modules:

- **Invoice** — today an on-demand PDF; the Accounting module turns it into a numbered, ledgered record.
- **Accounting** — only the two-ledger `commission` exists; no general ledger.
- **Feedback** — not modelled; "Travel completed → request review" is an unbuilt automation trigger.

See the entity standard for the baseline fields every one of these should carry.

# Database

Schema lives in `src/lib/schema.ts`. PostgreSQL (Neon) + Drizzle ORM. Tenancy column shown where present; see `architecture.md` and `security.md` for how scoping is enforced.

## Entity standard

Every business entity should carry this baseline. It operationalizes the never rules (one source of truth, no hard delete, no cross-tenant exposure) at the schema level.

| Field        | Purpose                                               | Current state                                                        |
|--------------|-------------------------------------------------------|----------------------------------------------------------------------|
| UUID         | Primary key, randomly generated (non-Better-Auth IDs) | ✓ convention in place                                                |
| Reference    | Human-facing id unique per agency (BKG-..., PRD-...)  | ⚠ only <code>booking</code> + <code>product</code> today             |
| createdAt    | Creation timestamp                                    | ✓ widespread                                                         |
| updatedAt    | Last-modified timestamp                               | ⚠ on roots, missing on some children                                 |
| createdBy    | <code>createdById</code> → user                       | ✓ on main entities                                                   |
| agencyId     | Tenant scope (root) or inherited (child)              | ✓ enforced everywhere                                                |
| Status       | Lifecycle/state enum                                  | ⚠ per-entity, not universal                                          |
| Activity log | Audit trail of changes                                | ⚠ shared <code>activity_log</code> table, not wired for every entity |
| Soft delete  | <code>deletedAt</code> — never hard delete            | ✗ <b>no soft-delete column exists yet</b>                            |
| Notes        | Free-form internal notes                              | ⚠ only some entities                                                 |

**Gap to close:** soft delete is **not implemented** — there is no `deletedAt` anywhere, so the "never hard delete" rule is currently aspirational. Adding a nullable `deletedAt` + filtering it out of reads (and a `reference` on the remaining entities) would need a migration. Until then, deletes are real.

## Tables

| Table  | Tenancy | Notes                                                                                                                                                                                                                                                                                                                                                                                                            |
|--------|---------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| agency | (root)  | name, slug, status (active/suspended); <b>Stripe billing</b> : <code>stripeCustomerId</code> , <code>stripeSubscriptionId</code> , <code>subscriptionStatus</code> , <code>priceId</code> , <code>currentPeriodEnd</code> , <code>trialEndsAt</code> ; <b>Connect</b> : <code>stripeConnectAccountId</code> , <code>stripeConnectOnboarded</code> ;<br><code>onboardingDismissedAt</code> (getting-started card) |

|                                                                                                 |                                       |                                                                                                                                                                                                                                              |
|-------------------------------------------------------------------------------------------------|---------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| agency_invite                                                                                   | agencyId                              | email, role, token, status, expiresAt                                                                                                                                                                                                        |
| user                                                                                            | agencyId (nullable)                   | + isPlatformAdmin, role, active, locale, commissionRatePercent (Better Auth)                                                                                                                                                                 |
| session, account, verification                                                                  | via user                              | Better Auth                                                                                                                                                                                                                                  |
| portal_session                                                                                  | via client                            | passwordless client portal sessions; token + expiresAt                                                                                                                                                                                       |
| client                                                                                          | agencyId                              | + client_contact (child); source (lead-source code), industry (code), country stored as canonical ISO name                                                                                                                                   |
| opportunity                                                                                     | agencyId                              | pipeline stage, value, currency; travel_purpose code; lost_reason code                                                                                                                                                                       |
| product                                                                                         | agencyId                              | proposal; ref unique per agency; + product_item (child; supplierId FK optional); <b>e-sign</b> : shareToken (unique), acceptedAt/declinedAt, signerName/Email, signatureData, signerIp/ UserAgent                                            |
| booking                                                                                         | agencyId                              | ref unique per agency; shareToken; travel_purpose + trip_type codes                                                                                                                                                                          |
| booking_traveller (+ title, gender codes), booking_item (+ supplierId FK), payment, booking_day | via booking                           | children; booking_traveller carries email + phone (required by supplier APIs for the lead passenger)                                                                                                                                         |
| booking_supplier_ref                                                                            | via booking + booking_item            | structured supplier confirmation: providerId, confirmationNumber, pnr, supplierOrderId, rawPayload; replaces untyped JSONB in booking_item.details                                                                                           |
| booking_event                                                                                   | bookingId + agencyId                  | append-only event log (audit + analytics); stable event codes: search_initiated, offer_selected, price_validated, price_changed, booking_submitted, booking_confirmed, booking_failed, booking_cancelled, payment_started, payment_completed |
| booking_document                                                                                | via booking (+ optional booking_item) | documents: type (voucher/ticket/invoice/itinerary/receipt), url (Vercel Blob), rawData (supplier payload for re-generation)                                                                                                                  |
| booking_idempotency                                                                             | via booking (+ optional booking_item) | idempotency key registry; key = sha256(bookingId+itemId+offerId); prevents duplicate supplier orders on retry; expiresAt (24 h TTL)                                                                                                          |
| notification                                                                                    | agencyId                              | comms log                                                                                                                                                                                                                                    |



|                   |              |                                                                                                                                        |
|-------------------|--------------|----------------------------------------------------------------------------------------------------------------------------------------|
| activity_log      | agencyId     | audit trail                                                                                                                            |
| supplier          | agencyId     | managed supplier directory (hotels, airlines, DMC, etc.)                                                                               |
| supplier_contract | agencyId     | commission basis/rate, validity dates, file URL                                                                                        |
| supplier_rate     | via contract | structured per-product rates within a contract                                                                                         |
| commission        | agencyId     | earnings ledger: supplier→agency and agency→agent; bookingId, supplierId, agentUserId, basis, rate, amount, status                     |
| hotel_content     | (global)     | Hotelbeds content cache (photos, facilities, coords) — shared reference data, <b>not</b> tenant-scoped; PK is the Hotelbeds hotel code |
|                   |              |                                                                                                                                        |

## ID convention

All ID columns **not** related to Better Auth use UUIDs, randomly generated. See coding-standards.md.

## Migrations

| Migration | What it adds                                                                                                                                                     |
|-----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0006      | Tenancy + backfill (agencyId on all tables)                                                                                                                      |
| 0007      | agency_invite table                                                                                                                                              |
| 0008      | user.locale                                                                                                                                                      |
| 0009      | Agency Stripe billing columns                                                                                                                                    |
| 0010      | Product e-signature columns                                                                                                                                      |
| 0011      | hotel_content cache                                                                                                                                              |
| 0012      | Stripe Connect columns on agency                                                                                                                                 |
| 0013      | portal_session table                                                                                                                                             |
| 0014      | supplier, supplier_contract, supplier_rate tables; supplierId FK on booking_item + product_item                                                                  |
| 0015      | commission table; user.commissionRatePercent                                                                                                                     |
| 0016      | agency.onboardingDismissedAt                                                                                                                                     |
| 0017      | Controlled-vocab columns: client.industry, opportunity.travel_purpose, booking.travel_purpose/trip_type, booking_traveller.title/gender (all nullable, additive) |
| 0018      | (see existing)                                                                                                                                                   |

|      |                                                                                                                                                            |
|------|------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0019 | Sprint 1 booking architecture: booking_supplier_ref, booking_event, booking_document, booking_idempotency tables; booking_traveller.email + .phone columns |
|------|------------------------------------------------------------------------------------------------------------------------------------------------------------|

Migrations 0000–0005 predate the multi-tenant rework (the pre-tenancy base schema) and are not itemized here.

## Workflow

```
db:generate → db:migrate # NEVER db:push
```

Run on prod after deploy: `POSTGRES_URL=<prod-url> npx drizzle-kit migrate`. Branches: dev ep-wandering-sunset-aitlty78 · **prod** ep-misty-thunder-aixz34vy. Full setup in development-guide.md.

# Schema Reference

Full per-table, per-column reference. Schema is Drizzle ORM + PostgreSQL (Neon). Source of truth: `src/lib/schema.ts`. High-level overview: `database.md`.

Convention — `id` is always `uuid PRIMARY KEY DEFAULT gen_random_uuid()` on business tables unless noted. Better Auth tables use `text` PKs. All timestamps are `timestampz`.

## Auth — Better Auth managed

**user**

| Column                               | Type                      | Constraints                              | Notes                                                                                                        |
|--------------------------------------|---------------------------|------------------------------------------|--------------------------------------------------------------------------------------------------------------|
| <code>id</code>                      | <code>text</code>         | PK                                       | Better Auth managed                                                                                          |
| <code>name</code>                    | <code>text</code>         | NOT NULL                                 | Display name                                                                                                 |
| <code>email</code>                   | <code>text</code>         | NOT NULL UNIQUE                          | Login credential                                                                                             |
| <code>email_verified</code>          | <code>boolean</code>      | NOT NULL<br>DEFAULT <code>false</code>   |                                                                                                              |
| <code>image</code>                   | <code>text</code>         |                                          | Avatar URL                                                                                                   |
| <code>agency_id</code>               | <code>uuid</code>         | FK → <code>agency</code><br>CASCADE      | <code>null</code> = platform admin or not yet onboarded                                                      |
| <code>is_platform_admin</code>       | <code>boolean</code>      | NOT NULL<br>DEFAULT <code>false</code>   | Vendor console access                                                                                        |
| <code>role</code>                    | <code>text</code>         | NOT NULL<br>DEFAULT <code>"agent"</code> | <code>admin</code>   <code>manager</code>   <code>finance</code>   <code>support</code>   <code>agent</code> |
| <code>locale</code>                  | <code>text</code>         |                                          | <code>"en"</code>   <code>"fr"</code>   <code>"ar"</code> — UI language preference                           |
| <code>active</code>                  | <code>boolean</code>      | NOT NULL<br>DEFAULT <code>true</code>    | Soft-disable without deleting history                                                                        |
| <code>commission_rate_percent</code> | <code>numeric(5,2)</code> |                                          | Default commission % this agent earns                                                                        |
| <code>created_at</code>              | <code>timestamp</code>    | NOT NULL<br>DEFAULT <code>now()</code>   |                                                                                                              |
| <code>updated_at</code>              | <code>timestamp</code>    | NOT NULL                                 | Auto-set on update                                                                                           |

**Indexes:** `user_email_idx`, `user_agency_idx`

## session

| Column     | Type      | Notes             |
|------------|-----------|-------------------|
| id         | text      | PK                |
| expires_at | timestamp | NOT NULL          |
| token      | text      | NOT NULL UNIQUE   |
| ip_address | text      |                   |
| user_agent | text      |                   |
| user_id    | text      | FK → user CASCADE |
| created_at | timestamp | NOT NULL          |
| updated_at | timestamp | NOT NULL          |

## account

OAuth/credential accounts linked to a user. Better Auth managed.

| Column                                            | Type                   |
|---------------------------------------------------|------------------------|
| id                                                | text PK                |
| account_id                                        | text NOT NULL          |
| provider_id                                       | text NOT NULL          |
| user_id                                           | text FK → user CASCADE |
| access_token, refresh_token, id_token             | text                   |
| access_token_expires_at, refresh_token_expires_at | timestamp              |
| scope, password                                   | text                   |

## verification

Magic-link / email verification tokens. Better Auth managed.

| Column     | Type               |
|------------|--------------------|
| id         | text PK            |
| identifier | text NOT NULL      |
| value      | text NOT NULL      |
| expires_at | timestamp NOT NULL |

## Agency & onboarding

## agency

| Column                    | Type      | Notes                                              |
|---------------------------|-----------|----------------------------------------------------|
| id                        | uuid      | PK                                                 |
| name                      | text      | NOT NULL                                           |
| slug                      | text      | NOT NULL UNIQUE                                    |
| status                    | text      | "active"   "suspended" DEFAULT "active"            |
| <b>Stripe billing</b>     |           |                                                    |
| stripe_customer_id        | text      |                                                    |
| stripe_subscription_id    | text      |                                                    |
| subscription_status       | text      | Stripe status string                               |
| price_id                  | text      | Stripe price for the current plan                  |
| current_period_end        | timestamp | Subscription period end                            |
| trial_ends_at             | timestamp | 14-day trial expiry on first provision             |
| <b>Stripe Connect</b>     |           |                                                    |
| stripe_connect_account_id | text      | Agency's Express account                           |
| stripe_connect_onboarded  | boolean   | DEFAULT false — true after Connect flow            |
| <b>Onboarding</b>         |           |                                                    |
| onboarding_dismissed_at   | timestamp | Set when agency dismisses the getting-started card |
| created_at                | timestamp | NOT NULL                                           |
| updated_at                | timestamp | NOT NULL                                           |

## agency\_invite

| Column        | Type      | Notes                                                |
|---------------|-----------|------------------------------------------------------|
| id            | uuid      | PK                                                   |
| agency_id     | uuid      | FK → agency CASCADE                                  |
| email         | text      | NOT NULL                                             |
| role          | text      | NOT NULL — target role after sign-up                 |
| token         | text      | NOT NULL UNIQUE — unguessable magic link             |
| status        | text      | "pending"   "accepted"   "expired" DEFAULT "pending" |
| invited_by_id | text      | FK → user SET NULL                                   |
| expires_at    | timestamp | NOT NULL                                             |
| created_at    | timestamp | NOT NULL                                             |

# CRM

## client

| Column        | Type      | Notes                                           |
|---------------|-----------|-------------------------------------------------|
| id            | uuid      | PK                                              |
| agency_id     | uuid      | FK → agency CASCADE                             |
| name          | text      | NOT NULL                                        |
| type          | text      | "individual"   "corporate" DEFAULT "individual" |
| status        | text      | "lead"   "active"   "inactive" DEFAULT "active" |
| email         | text      |                                                 |
| phone         | text      |                                                 |
| company       | text      | Corporate name                                  |
| address       | text      |                                                 |
| city          | text      |                                                 |
| country       | text      | Canonical ISO 3166-1 name (full name, not code) |
| source        | text      | Lead-source vocab code                          |
| industry      | text      | Industry vocab code (corporate clients)         |
| notes         | text      |                                                 |
| owner_id      | text      | FK → user SET NULL — relationship owner         |
| created_by_id | text      | FK → user SET NULL                              |
| created_at    | timestamp | NOT NULL                                        |
| updated_at    | timestamp | NOT NULL                                        |

**Indexes:** client\_agency\_idx, client\_owner\_idx, client\_status\_idx, client\_name\_idx, client\_agency\_created\_idx **Unique:** none (names may repeat)

## client\_contact

Additional contacts for corporate clients.

| Column    | Type                              |
|-----------|-----------------------------------|
| id        | uuid PK                           |
| client_id | uuid FK → client CASCADE NOT NULL |
| name      | text NOT NULL                     |

|            |                                |
|------------|--------------------------------|
| job_title  | text                           |
| email      | text                           |
| phone      | text                           |
| is_primary | boolean DEFAULT false NOT NULL |
| created_at | timestamp NOT NULL             |

## Sales pipeline

### opportunity

| Column              | Type          | Notes                                                                        |
|---------------------|---------------|------------------------------------------------------------------------------|
| id                  | uuid          | PK                                                                           |
| agency_id           | uuid          | FK → agency CASCADE                                                          |
| title               | text          | NOT NULL                                                                     |
| client_id           | uuid          | FK → client CASCADE NOT NULL                                                 |
| stage               | text          | "lead"   "qualified"   "proposal"   "booked"   "won"   "lost" DEFAULT "lead" |
| value               | numeric(12,2) | DEFAULT 0                                                                    |
| currency            | text          | DEFAULT "EUR"                                                                |
| probability         | integer       | 0–100 DEFAULT 10                                                             |
| destination         | text          |                                                                              |
| travel_start_date   | timestamp     |                                                                              |
| travel_end_date     | timestamp     |                                                                              |
| pax_count           | integer       | DEFAULT 1                                                                    |
| travel_purpose      | text          | Vocab code                                                                   |
| expected_close_date | timestamp     |                                                                              |
| lost_reason         | text          | Vocab code — only when stage = "lost"                                        |
| notes               | text          |                                                                              |
| assigned_to_id      | text          | FK → user SET NULL                                                           |
| created_by_id       | text          | FK → user SET NULL                                                           |
| created_at          | timestamp     | NOT NULL                                                                     |
| updated_at          | timestamp     | NOT NULL                                                                     |

**Indexes:** opportunity\_agency\_idx, opportunity\_client\_idx, opportunity\_stage\_idx, opportunity\_assigned\_idx, opportunity\_agency\_stage\_idx, opportunity\_agency\_created\_idx

## Proposals (Products)

product

| Column                       | Type          | Notes                                                                  |
|------------------------------|---------------|------------------------------------------------------------------------|
| id                           | uuid          | PK                                                                     |
| agency_id                    | uuid          | FK → agency CASCADE                                                    |
| reference                    | text          | NOT NULL — e.g. "PRD-1042" unique per agency                           |
| title                        | text          | NOT NULL                                                               |
| client_id                    | uuid          | FK → client SET NULL                                                   |
| opportunity_id               | uuid          | FK → opportunity SET NULL                                              |
| status                       | text          | "draft"   "sent"   "accepted"   "rejected"   "expired" DEFAULT "draft" |
| destination                  | text          |                                                                        |
| start_date, end_date         | timestamp     |                                                                        |
| pax_count                    | integer       | DEFAULT 1                                                              |
| currency                     | text          | DEFAULT "EUR"                                                          |
| markup_percent               | numeric(5,2)  | DEFAULT 0                                                              |
| total_cost                   | numeric(12,2) | DEFAULT 0                                                              |
| total_price                  | numeric(12,2) | DEFAULT 0                                                              |
| summary                      | text          | Client-facing narrative (AI-generated)                                 |
| valid_until                  | timestamp     |                                                                        |
| E-signature                  |               |                                                                        |
| share_token                  | text          | UNIQUE — public /p/[token] link                                        |
| accepted_at, declined_at     | timestamp     |                                                                        |
| signer_name, signer_email    | text          | Non-repudiation                                                        |
| signature_data               | text          | Typed name or drawn-signature data URL                                 |
| signer_ip, signer_user_agent | text          |                                                                        |



|                        |           |                    |
|------------------------|-----------|--------------------|
| created_by_id          | text      | FK → user SET NULL |
| created_at, updated_at | timestamp | NOT NULL           |

**Unique:** product\_agency\_reference\_unique (agency\_id, reference)

**product\_item**

Line items within a proposal.

| Column                | Type          | Notes                                                                |
|-----------------------|---------------|----------------------------------------------------------------------|
| id                    | uuid          | PK                                                                   |
| product_id            | uuid          | FK → product CASCADE NOT NULL                                        |
| supplier_id           | uuid          | FK → supplier SET NULL (optional)                                    |
| type                  | text          | "flight"   "hotel"   "activity"   "transfer"   "insurance"   "other" |
| title                 | text          | NOT NULL                                                             |
| description, supplier | text          |                                                                      |
| quantity              | integer       | DEFAULT 1 NOT NULL                                                   |
| unit_cost, unit_price | numeric(12,2) | DEFAULT 0 NOT NULL                                                   |
| currency              | text          | DEFAULT "EUR"                                                        |
| start_date, end_date  | timestamp     |                                                                      |
| details               | jsonb         | Raw supplier offer payload                                           |
| sort_order            | integer       | DEFAULT 0                                                            |
| created_at            | timestamp     | NOT NULL                                                             |

**Bookings**

**booking**

| Column    | Type | Notes                                        |
|-----------|------|----------------------------------------------|
| id        | uuid | PK                                           |
| agency_id | uuid | FK → agency CASCADE                          |
| reference | text | NOT NULL — e.g. "BKG-1042" unique per agency |
| client_id | uuid | FK → client SET NULL                         |

|                             |               |                                                                                                        |
|-----------------------------|---------------|--------------------------------------------------------------------------------------------------------|
| status                      | text          | "draft"   "awaiting_payment"   "confirmed"   "ticketed"   "completed"   "cancelled"<br>DEFAULT "draft" |
| destination                 | text          |                                                                                                        |
| depart_date,<br>return_date | timestamp     |                                                                                                        |
| travel_purpose              | text          | Vocab code                                                                                             |
| trip_type                   | text          | "one_way"   "round_trip"   "multi_city"                                                                |
| currency                    | text          | DEFAULT "EUR"                                                                                          |
| notes                       | text          |                                                                                                        |
| total_amount                | numeric(12,2) | DEFAULT 0                                                                                              |
| share_token                 | text          | UNIQUE — shareable itinerary /i/[token]                                                                |
| created_by_id               | text          | FK → user SET NULL                                                                                     |
| created_at,<br>updated_at   | timestamp     | NOT NULL                                                                                               |

#### Lifecycle prerequisite rules (server-side enforced):

- confirmed requires: trip items present AND zero outstanding balance.
- ticketed requires: trip items present.

**Unique:** booking\_agency\_reference\_unique (agency\_id, reference)

#### booking\_traveller

| Column          | Type      | Notes                                    |
|-----------------|-----------|------------------------------------------|
| id              | uuid      | PK                                       |
| booking_id      | uuid      | FK → booking CASCADE NOT NULL            |
| full_name       | text      | NOT NULL                                 |
| title           | text      | Vocab code (mr   mrs   ms   dr   prof)   |
| gender          | text      | Vocab code (male   female   unspecified) |
| passport_number | text      |                                          |
| passport_expiry | timestamp |                                          |
| nationality     | text      | ISO country name                         |
| date_of_birth   | timestamp |                                          |

|                                              |                     |                                                 |
|----------------------------------------------|---------------------|-------------------------------------------------|
| passport_issue_date,<br>passport_issue_place | timestamp /<br>text |                                                 |
| email                                        | text                | Required by Duffel for lead passenger           |
| phone                                        | text                | Required by Duffel for lead passenger           |
| is_lead                                      | boolean             | NOT NULL DEFAULT false — first passenger = lead |
| sort_order                                   | integer             | DEFAULT 0                                       |
| created_at                                   | timestamp           | NOT NULL                                        |

### booking\_item

A purchased line item (flight, hotel, transfer, fee, etc.).

| Column                   | Type          | Notes                                                                            |
|--------------------------|---------------|----------------------------------------------------------------------------------|
| id                       | uuid          | PK                                                                               |
| booking_id               | uuid          | FK → booking CASCADE NOT NULL                                                    |
| supplier_id              | uuid          | FK → supplier SET NULL (optional)                                                |
| type                     | text          | "flight"   "hotel"   "excursion"  <br>"transfer"   "insurance"   "fee"   "other" |
| title                    | text          | NOT NULL                                                                         |
| description,<br>supplier | text          |                                                                                  |
| booking_ref              | text          | Supplier confirmation / PNR reference                                            |
| start_date, end_date     | timestamp     |                                                                                  |
| quantity                 | integer       | DEFAULT 1                                                                        |
| amount                   | numeric(12,2) | DEFAULT 0 — price charged to client (per unit)                                   |
| currency                 | text          | DEFAULT "EUR"                                                                    |
| item_status              | text          | "pending"   "confirmed"   "ticketed"  <br>"cancelled" DEFAULT "pending"          |
| confirmation_number      | text          | Supplier confirmation number once booked                                         |
| day_index                | integer       | Itinerary day (0-based from departure; null = auto by date)                      |
| details                  | jsonb         | Raw supplier offer payload                                                       |
| sort_order               | integer       | DEFAULT 0                                                                        |
| created_at               | timestamp     | NOT NULL                                                                         |

### payment

| Column            | Type          | Notes                                                               |
|-------------------|---------------|---------------------------------------------------------------------|
| id                | uuid          | PK                                                                  |
| booking_id        | uuid          | FK → booking CASCADE NOT NULL                                       |
| amount            | numeric(12,2) | NOT NULL                                                            |
| currency          | text          | DEFAULT "EUR"                                                       |
| kind              | text          | "deposit"   "installment"   "payment"   "refund" DEFAULT "payment"  |
| method            | text          | "manual"   "card"   "transfer"   "cash"   "stripe" DEFAULT "manual" |
| status            | text          | "pending"   "completed"   "failed"   "refunded" DEFAULT "completed" |
| reference         | text          | Stripe payment id or manual reference                               |
| stripe_session_id | text          | Stripe Checkout Session id (Connect flow)                           |
| checkout_url      | text          | Hosted Checkout URL sent to client                                  |
| note              | text          |                                                                     |
| created_by_id     | text          | FK → user SET NULL                                                  |
| created_at        | timestamp     | NOT NULL                                                            |

### booking\_day

Per-day itinerary titles and notes.

| Column     | Type                               |
|------------|------------------------------------|
| id         | uuid PK                            |
| booking_id | uuid FK → booking CASCADE NOT NULL |
| day_index  | integer NOT NULL (0-based)         |
| title      | text                               |
| notes      | text                               |

### notification

Communications log.

| Column     | Type | Notes                           |
|------------|------|---------------------------------|
| id         | uuid | PK                              |
| agency_id  | uuid | FK → agency CASCADE             |
| booking_id | uuid | FK → booking CASCADE (nullable) |

|               |           |                                                                                            |
|---------------|-----------|--------------------------------------------------------------------------------------------|
| channel       | text      | "email"   "sms"   "whatsapp"   "push" DEFAULT "email"                                      |
| recipient     | text      | NOT NULL — email address or phone                                                          |
| subject       | text      |                                                                                            |
| body          | text      |                                                                                            |
| kind          | text      | "confirmation"   "voucher"   "receipt"   "custom"   "invite"   "proposal" DEFAULT "custom" |
| status        | text      | "sent"   "failed"   "logged" DEFAULT "sent"                                                |
| error         | text      | Error message when status = "failed"                                                       |
| created_by_id | text      | FK → user SET NULL                                                                         |
| created_at    | timestamp | NOT NULL                                                                                   |

## Sprint 1 — Booking lifecycle tables

### booking\_supplier\_ref

Structured supplier confirmation. Replaces ad-hoc JSONB in `booking_item.details`.

| Column              | Type      | Notes                                                              |
|---------------------|-----------|--------------------------------------------------------------------|
| id                  | uuid      | PK                                                                 |
| booking_id          | uuid      | FK → booking CASCADE NOT NULL                                      |
| booking_item_id     | uuid      | FK → booking_item CASCADE NOT NULL                                 |
| provider_id         | text      | NOT NULL — "duffel"   "hotelbeds"   "mock"                         |
| confirmation_number | text      | NOT NULL — shown to client                                         |
| pnr                 | text      | Airline PNR / record locator                                       |
| supplier_order_id   | text      | Provider's internal order id (e.g. Duffel <code>order_xxx</code> ) |
| raw_payload         | jsonb     | Full raw response for debugging                                    |
| created_at          | timestamp | NOT NULL                                                           |

**Indexes:** `booking_supplier_ref_booking_idx`, `booking_supplier_ref_item_idx`, `booking_supplier_ref_provider_idx`

### booking\_event

Append-only audit + analytics log. Never update or delete rows.

| Column | Type | Notes |
|--------|------|-------|
| id     | uuid | PK    |

|                |           |                                                               |
|----------------|-----------|---------------------------------------------------------------|
| booking_id     | uuid      | FK → booking CASCADE NOT NULL                                 |
| agency_id      | uuid      | FK → agency CASCADE NOT NULL (for agency-scoped analytics)    |
| event          | text      | NOT NULL — stable event code (see booking-architecture.md §6) |
| provider_id    | text      | "duffel", "hotelbeds", "mock", or null for UI events          |
| correlation_id | text      | Request-level trace id                                        |
| metadata       | jsonb     | Structured payload (offer id, price, error code, etc.)        |
| created_at     | timestamp | NOT NULL                                                      |

**Event codes:** search\_initiated, offer\_selected, price\_validated, price\_changed, booking\_submitted, booking\_confirmed, booking\_failed, booking\_cancelled, payment\_started, payment\_completed

**Indexes:** booking\_event\_booking\_idx, booking\_event\_agency\_idx, booking\_event\_event\_idx, booking\_event\_created\_idx, booking\_event\_booking\_created\_idx

### booking\_document

Documents generated for a booking (vouchers, tickets, invoices, itineraries).

| Column          | Type      | Notes                                                                 |
|-----------------|-----------|-----------------------------------------------------------------------|
| id              | uuid      | PK                                                                    |
| booking_id      | uuid      | FK → booking CASCADE NOT NULL                                         |
| booking_item_id | uuid      | FK → booking_item SET NULL (null = booking-level doc)                 |
| type            | text      | NOT NULL — "voucher"   "ticket"   "invoice"   "itinerary"   "receipt" |
| provider_id     | text      | Supplier that issued the document                                     |
| url             | text      | Vercel Blob URL or supplier CDN link                                  |
| raw_data        | jsonb     | Supplier payload for re-generation                                    |
| generated_at    | timestamp | When the document was issued                                          |
| created_at      | timestamp | NOT NULL                                                              |

### booking\_idempotency

Key registry preventing duplicate supplier orders on retry.

| Column          | Type | Notes                                 |
|-----------------|------|---------------------------------------|
| key             | text | PK — sha256(bookingId:itemId:offerId) |
| booking_id      | uuid | FK → booking CASCADE NOT NULL         |
| booking_item_id | uuid | FK → booking_item CASCADE             |

|              |           |                                                    |
|--------------|-----------|----------------------------------------------------|
| provider_id  | text      | NOT NULL — provider the key was sent to            |
| status       | text      | "pending"   "success"   "failed" DEFAULT "pending" |
| supplier_ref | text      | Confirmation number when status = "success"        |
| created_at   | timestamp | NOT NULL                                           |
| expires_at   | timestamp | NOT NULL — 24h TTL; safe to clean up after expiry  |

**Indexes:** booking\_idempotency\_booking\_idx, booking\_idempotency\_expires\_idx

## Supplier management

### supplier

Agency's managed supplier directory (hotels, airlines, DMCs, etc.).

| Column                 | Type      | Notes                                                                           |
|------------------------|-----------|---------------------------------------------------------------------------------|
| id                     | uuid      | PK                                                                              |
| agency_id              | uuid      | FK → agency CASCADE                                                             |
| name                   | text      | NOT NULL — unique per agency                                                    |
| type                   | text      | "hotel"   "airline"   "car_rental"   "transfer"   "dmc"   "insurance"   "other" |
| status                 | text      | "active"   "inactive" DEFAULT "active"                                          |
| email, phone, website  | text      |                                                                                 |
| address, city, country | text      |                                                                                 |
| contact_name           | text      |                                                                                 |
| notes                  | text      |                                                                                 |
| created_by_id          | text      | FK → user SET NULL                                                              |
| created_at, updated_at | timestamp | NOT NULL                                                                        |

**Unique:** supplier\_agency\_name\_unique (agency\_id, name)

### supplier\_contract

Commercial contracts with commission terms.

| Column | Type | Notes |
|--------|------|-------|
|--------|------|-------|

|                        |              |                                                                      |
|------------------------|--------------|----------------------------------------------------------------------|
| id                     | uuid         | PK                                                                   |
| supplier_id            | uuid         | FK → <code>supplier</code> CASCADE NOT NULL                          |
| agency_id              | uuid         | FK → <code>agency</code> CASCADE (denormalized for query efficiency) |
| name                   | text         | NOT NULL                                                             |
| reference              | text         | Contract number                                                      |
| commission_basis       | text         | "percent"   "fixed"   "net"                                          |
| commission_rate        | numeric(5,2) | % or fixed amount depending on basis                                 |
| valid_from, valid_to   | timestamp    |                                                                      |
| file_url               | text         | Vercel Blob URL for the PDF contract                                 |
| notes                  | text         |                                                                      |
| created_at, updated_at | timestamp    | NOT NULL                                                             |

#### supplier\_rate

Structured per-product rates within a contract.

| Column               | Type          | Notes                                                |
|----------------------|---------------|------------------------------------------------------|
| id                   | uuid          | PK                                                   |
| contract_id          | uuid          | FK → <code>supplier_contract</code> CASCADE NOT NULL |
| name, description    | text          |                                                      |
| rate_type            | text          | Type of rate (product/season/tier)                   |
| amount               | numeric(12,2) |                                                      |
| currency             | text          |                                                      |
| valid_from, valid_to | timestamp     |                                                      |
| created_at           | timestamp     | NOT NULL                                             |

#### commission

Two-ledger earnings record. Auto-generated on booking confirm/ticket.

| Column     | Type | Notes                             |
|------------|------|-----------------------------------|
| id         | uuid | PK                                |
| agency_id  | uuid | FK → <code>agency</code> CASCADE  |
| booking_id | uuid | FK → <code>booking</code> CASCADE |



|                           |               |                                                |
|---------------------------|---------------|------------------------------------------------|
| booking_item_id           | uuid          | FK → booking_item (nullable)                   |
| supplier_id               | uuid          | FK → supplier SET NULL                         |
| agent_user_id             | text          | FK → user SET NULL                             |
| ledger                    | text          | "supplier_to_agency"   "agency_to_agent"       |
| basis                     | text          | "percent"   "fixed"   "net"                    |
| rate                      | numeric(5,2)  |                                                |
| amount                    | numeric(12,2) | NOT NULL — computed commission                 |
| currency                  | text          |                                                |
| status                    | text          | "pending"   "confirmed"   "paid"   "cancelled" |
| notes                     | text          |                                                |
| created_at,<br>updated_at | timestamp     | NOT NULL                                       |

## Client portal

### portal\_session

Passwordless sessions for the client-facing traveler portal. Entirely separate from the Better Auth staff session system.

| Column     | Type      | Notes                                                                |
|------------|-----------|----------------------------------------------------------------------|
| id         | uuid      | PK                                                                   |
| client_id  | uuid      | FK → client CASCADE NOT NULL                                         |
| token      | text      | NOT NULL UNIQUE — magic-link token (15 min) → session token (7 days) |
| expires_at | timestamp | NOT NULL                                                             |
| created_at | timestamp | NOT NULL                                                             |

Magic-link flow: token is short-lived (15 min); on verification the row is updated with a rotated long-lived (7-day) token stored in an httpOnly cookie.

## Reference data (global, not tenant-scoped)

### hotel\_content

Hotelbeds Content API cache. PK is the Hotelbeds hotel code (text), **not** UUID. Shared across all agencies — intentionally not scoped to agency\_id.

| Column                              | Type          | Notes                                    |
|-------------------------------------|---------------|------------------------------------------|
| code                                | text          | PK — Hotelbeds hotel code (e.g. "12345") |
| name                                | text          | NOT NULL                                 |
| stars                               | integer       | DEFAULT 0                                |
| hotel_type                          | text          |                                          |
| description                         | text          |                                          |
| address, city, country, postal_code | text          |                                          |
| latitude, longitude                 | numeric(10,6) |                                          |
| destination_code                    | text          | Hotelbeds destination code (e.g. "BCN")  |
| segments                            | jsonb         | Marketing segment tags: string[]         |
| facilities                          | jsonb         | Amenity names: string[]                  |
| images                              | jsonb         | { url: string; roomCode?: string }[]     |
| updated_at                          | timestamp     | NOT NULL — sync timestamp                |

**Index:** hotel\_content\_destination\_idx (destination\_code)

## Audit

|              |
|--------------|
| activity_log |
|--------------|

Records every meaningful action for manager oversight.

| Column       | Type      | Notes                                                                           |
|--------------|-----------|---------------------------------------------------------------------------------|
| id           | uuid      | PK                                                                              |
| agency_id    | uuid      | FK → agency CASCADE                                                             |
| user_id      | text      | FK → user SET NULL                                                              |
| action       | text      | "created"   "updated"   "deleted"   "stage_changed"   "sent"   "status_changed" |
| entity_type  | text      | "client"   "opportunity"   "product"   "user"                                   |
| entity_id    | text      |                                                                                 |
| entity_label | text      | Human-readable name kept even if entity is deleted                              |
| metadata     | jsonb     | Diff / context payload                                                          |
| created_at   | timestamp | NOT NULL                                                                        |

**Indexes:** activity\_agency\_idx, activity\_user\_idx, activity\_entity\_idx (entity\_type, entity\_id), activity\_created\_idx

## Migrations summary

| Migration | Description                                                                                                                                                  |
|-----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0000-0005 | Pre-tenancy base schema (not itemized)                                                                                                                       |
| 0006      | Multi-tenant agencyId backfill                                                                                                                               |
| 0007      | agency_invite                                                                                                                                                |
| 0008      | user.locale                                                                                                                                                  |
| 0009      | Agency Stripe billing columns                                                                                                                                |
| 0010      | Product e-signature columns                                                                                                                                  |
| 0011      | hotel_content cache                                                                                                                                          |
| 0012      | Stripe Connect columns on agency                                                                                                                             |
| 0013      | portal_session                                                                                                                                               |
| 0014      | supplier, supplier_contract, supplier_rate, supplier_id FK on booking_item + product_item                                                                    |
| 0015      | commission, user.commission_rate_percent                                                                                                                     |
| 0016      | agency.onboarding_dismissed_at                                                                                                                               |
| 0017      | Controlled-vocab columns (additive, nullable): client.industry, opportunity.travel_purpose, booking.travel_purpose/trip_type, booking_traveller.title/gender |
| 0018      | (see git log)                                                                                                                                                |
| 0019      | Sprint 1: booking_supplier_ref, booking_event, booking_document, booking_idempotency, booking_traveller.email + .phone                                       |

# API Integrations

---

Every external integration **degrades gracefully** to sample or logged behaviour when its keys are unset, so the app runs end-to-end with only `POSTGRES_URL` + `BETTER_AUTH_SECRET`. Env vars are summarized in `development-guide.md`.

## Provider architecture

---

Booking providers plug into **one abstraction** so adding Hotelbeds, Amadeus, TravelgateX, Booking.com, or Expedia never touches calling code. Interfaces + registry live in `src/lib/suppliers/providers/`. Environment resolution lives in `src/lib/suppliers/config.ts` — the single entry point for credential/hostname config; no other file reads `process.env` for supplier credentials directly.

- **Capability-segmented interfaces** (`types.ts`) — a provider implements only what it offers: `HotelSearchCapable`, `HotelBookingCapable`, `FlightSearchCapable`, `FlightBookingCapable`, `CancelCapable`. A hotels-only aggregator implements the hotel interfaces and nothing else (no empty stubs).
- **Full booking lifecycle** — `search` → `quote` (re-validate price/rate) → `book` (idempotent) → `cancel`. Real bedbank/GDS rates expire, so `quoteHotel/quoteFlight` re-price immediately before `book*`; book requests carry an `idempotencyKey` so replays never double-book.
- **Normalized errors** — every provider throws `ProviderError` with a stable code (`rate_expired`, `sold_out`, `rate_limited`, `provider_unavailable`, ...) and a `retryable` flag, so callers branch on the code, not provider strings.
- **Tenant-aware context** — every call takes a `ProviderContext` (`agencyId`, `currency`, `locale`, `correlationId`, abort signal); providers never read globals.
- **Registry** (`registry.ts`) — adapters `register()` themselves at startup; callers resolve by vertical + capability + priority (`providerRegistry.pick("hotels", "search")`) with capability type-guards (`canSearchHotels`, `canBookFlights`, ...). Replaces the hardcoded `getFlightSupplier()/getHotelSupplier()` if/else.
- **Registration** (`register.ts`, Wave 2) — `registerBuiltInProviders()` wires the three shipped adapters into the registry: `MockBookingProvider` (both verticals, priority 0, always configured — the fallback), `DuffelBookingProvider` (flights, priority 50) and `HotelbedsBookingProvider` (hotels, priority 50). It is idempotent (module-level guard) and runs as a side effect when the providers barrel is imported, so the registry is self-wiring. The mock outranks nothing, so `pick()` returns a real provider whenever one is configured and falls back to the mock otherwise.
- **Provider catalog** (`PROVIDER_CATALOG`) — one logic-free source of truth for which providers exist, their verticals/capabilities, env vars, and status (`live` / `legacy` / `planned`):

| Provider | Verticals        | Status |
|----------|------------------|--------|
| Mock     | flights + hotels | live   |
| Duffel   | flights          | live   |

|                 |                    |         |
|-----------------|--------------------|---------|
| Amadeus         | flights (+ hotels) | legacy  |
| Hotelbeds       | hotels             | live    |
| TravelgateX     | hotels + flights   | planned |
| Booking.com     | hotels             | planned |
| Expedia (Rapid) | hotels             | planned |

**Migration complete (Wave 3):** the booking actions now go through the registry and the full `quote → book → cancel` lifecycle. `getFlightSupplier/ getHotelSupplier` remain for search; booking uses `serviceBookFlight/ serviceBookHotel` (`src/lib/suppliers/booking-service.ts`) which picks the best configured provider via `providerRegistry.pick(...)`, runs `quote*` (price revalidation), then `book*` (idempotent), and writes to `booking_supplier_ref`, `booking_event`, and `booking_idempotency`. Open item #1 (real supplier booking) is now closed — swap in production credentials and booking flows go live.

## Flights — Duffel

---

- `src/lib/suppliers/duffel.ts`, behind `getFlightSupplier() + safeSearch()`.
- Airport autocomplete, one-way/round-trip, flight codes, connecting airports.
- Falls back to sample data when `DUFFEL_API_TOKEN` is unset.
- Amadeus self-service (`amadeus.ts`) is kept **only as a legacy fallback**.
- **Sprint 1:** `DuffelBookingProvider` implements `FlightSearchCapable + FlightBookingCapable` through the registry, adding `quoteFlight` (price revalidation) + `bookFlight` (idempotent order creation) to the existing search capability.

## Hotels — Hotelbeds (APITUDE)

---

- `src/lib/suppliers/hotelbeds.ts + content-cache.ts`.
- Availability + content APIs. Booking.com-style search/results/details, dynamic occupancy pricing, filters, room photos, compare.
- **Content cache** (`hotel_content` table, global/non-tenant) serves real photos, facilities and coords quota-free; synced via `scripts/sync-hotel-content.ts`.
- **Search by hotel name** resolves matching hotels via the Content API (`searchHotelbedsHotelsByName`) then prices them live; falls back to estimated rates.
- **Keys:** `HOTELBEDS_API_KEY / HOTELBEDS_SECRET`. **Hostname:** `HOTELBEDS_HOSTNAME` (defaults to test endpoint; set to `https://api.hotelbeds.com` for production). Falls back to sample data when keys are unset.
- **Sprint 1:** `HotelbedsBookingProvider` implements `HotelSearchCapable + HotelBookingCapable` through the registry, adding `quoteHotel + bookHotel`.

## Supplier abstraction

---

`src/lib/suppliers/index.ts` exposes per-vertical `getFlightSupplier / getHotelSupplier` and a `safeSearch` wrapper that catches provider errors and returns mock results, so the UI never hard-fails.

`mock.ts` provides the sample data; `types.ts` the shared shapes. This is separate from the managed **supplier directory** (the `supplier` table — agency's own hotel/airline/DMC contacts; see `business-rules.md`).

## Billing — Stripe subscriptions (vendor → agency)

---

- `src/lib/billing/stripe.ts`. Vendor bills agencies; 14-day trial on provision.
- Checkout, billing portal, and **manual webhook signature verification**.
- `api/stripe/webhook` reconciles subscription status; `requireAgencyUser` gates on a lapsed subscription.
- Keys: `STRIPE_SECRET_KEY`, `STRIPE_PRICE_ID`, `STRIPE_WEBHOOK_SECRET`.

## Payments — Stripe Connect (traveler → agency)

---

- `src/lib/payments/stripe.ts` — **distinct** from `billing/stripe.ts`.
- Agencies onboard a connected Express account to receive traveler payments directly (**destination charges**); the platform takes a configurable fee.
- `api/stripe/connect-webhook` reconciles Connect payments. Powers the client portal "Pay now".
- Keys: `STRIPE_CONNECT_WEBHOOK_SECRET`, `STRIPE_PLATFORM_FEE_PERCENT`.

## Email — Resend

---

- `src/lib/notifications/email.ts` (adapter) + `templates.ts` (HTML).
- Sends invite emails, password-reset, proposal acceptance, portal invites.
- When unconfigured, logs to console + the `notification` table; invite/portal links are made copyable so the flow still works.
- Keys: `RESEND_API_KEY`, `EMAIL_FROM`.

## AI — Vercel AI SDK + OpenRouter

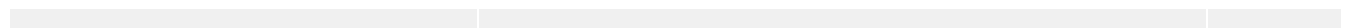
---

- Powers the assistant and inline AI features; see `ai.md`.
- Key: `OPENROUTER_API_KEY`. AI features no-op/degrade when unset.

## Storage — Vercel Blob (optional)

---

- Supplier contract PDF uploads. Key: `BLOB_READ_WRITE_TOKEN`.



# Booking Architecture (Sprint 1)

---

Detailed technical reference for the provider abstraction system, booking service, idempotency model, and event log introduced in Sprint 1. For a high-level overview see [api-integrations.md](#).

---

## 1. Overview

---

The booking architecture introduces a **three-layer** stack on top of the existing search-only supplier adapters:



The server action collects intent (which booking item to fulfil), builds a `ProviderContext`, and calls `serviceBookFlight` or `serviceBookHotel`. The booking service runs the full lifecycle — idempotency check, quote, book, event logging, supplier-ref persistence — without any booking-specific code leaking into the action layer.

---

## 2. Provider abstraction

---

### 2.1 Capability-segmented interfaces

`src/lib/suppliers/providers/types.ts`

A provider implements **only the capabilities it actually offers**. There are no empty stubs. The capability contract:

| Interface                       | Methods                                                                                                                                         | Who implements        |
|---------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------|
| <code>ProviderDescriptor</code> | <code>id</code> , <code>label</code> , <code>verticals</code> , <code>capabilities</code> , <code>priority</code> , <code>isConfigured()</code> | <b>every</b> provider |

|                      |                                               |                 |
|----------------------|-----------------------------------------------|-----------------|
| HotelSearchCapable   | searchHotels(params, ctx)                     | Hotelbeds, Mock |
| HotelBookingCapable  | quoteHotel(offer, ctx), bookHotel(req, ctx)   | Hotelbeds, Mock |
| FlightSearchCapable  | searchFlights(params, ctx)                    | Duffel, Mock    |
| FlightBookingCapable | quoteFlight(offer, ctx), bookFlight(req, ctx) | Duffel, Mock    |
| CancelCapable        | cancel(ref, ctx)                              | (future)        |

A concrete provider is typed as:

```
type BookingProvider = ProviderDescriptor &
  Partial<HotelSearchCapable & HotelBookingCapable &
    FlightSearchCapable & FlightBookingCapable & CancelCapable>
```

Callers narrow capabilities using type-guard functions:

```
canSearchHotels(p)    // p is HotelSearchCapable
canBookHotels(p)      // p is HotelBookingCapable
canSearchFlights(p)   // p is FlightSearchCapable
canBookFlights(p)     // p is FlightBookingCapable
```

## 2.2 ProviderContext

Passed to every provider call so providers never read globals:

```
type ProviderContext = {
  agencyId:      string;           // tenant scope + per-agency cred lookup
  currency?:    string;           // ISO code, e.g. "DZD"
  locale?:      string;           // e.g. "en"
  correlationId?: string;          // traces a request across providers and logs
  signal?:      AbortSignal;      // timeout / user cancellation
}
```

## 2.3 Booking lifecycle DTOs

**RateQuote** — returned by quoteHotel / quoteFlight:

| Field      | Type                 | Purpose                         |
|------------|----------------------|---------------------------------|
| quoteId    | string               | Opaque token passed to book*    |
| providerId | ProviderId           | Which provider issued the quote |
| vertical   | "flights"   "hotels" | —                               |
| priceTotal | number               | Price after re-validation       |



|                       |              |                                            |
|-----------------------|--------------|--------------------------------------------|
| currency              | string       | —                                          |
| refundable            | boolean      | —                                          |
| expiresAt             | string (ISO) | Quote expiry deadline                      |
| cancellationDeadline? | string (ISO) | Free-cancel deadline (when refundable)     |
| priceChanged?         | boolean      | True when price differs from search result |

**HotelBookingRequest** — sent to `bookHotel`:

| Field            | Type           | Purpose                                     |
|------------------|----------------|---------------------------------------------|
| offer            | HotelOffer     | Original search offer                       |
| guests           | GuestDetails[] | Guest roster (lead first)                   |
| idempotencyKey   | string         | Caller-generated; same key = no double-book |
| quoteId?         | string         | Fresh rate key from <code>quoteHotel</code> |
| agencyReference? | string         | Cross-linking reference (e.g. BKG-1001)     |

**FlightBookingRequest** — sent to `bookFlight`:

| Field            | Type              | Purpose                                     |
|------------------|-------------------|---------------------------------------------|
| offer            | FlightOffer       | Original search offer                       |
| passengers       | FlightPassenger[] | Passenger roster                            |
| idempotencyKey   | string            | Caller-generated; same key = no double-book |
| quoteId?         | string            | Fresh quote token                           |
| agencyReference? | string            | Cross-linking reference                     |

**BookingResult** — returned by `book*` on success:

| Field                  | Type                            |
|------------------------|---------------------------------|
| ref.providerId         | ProviderId                      |
| ref.confirmationNumber | string                          |
| ref.raw?               | unknown (raw supplier response) |
| status                 | "confirmed"   "pending"         |
| priceTotal             | number                          |
| currency               | string                          |

## 2.4 Normalized error model

All providers throw `ProviderError` with a stable `code`:

| Code                 | Meaning                                 | Retryable |
|----------------------|-----------------------------------------|-----------|
| auth                 | Bad/expired credentials                 | false     |
| validation           | Malformed request                       | false     |
| rate_expired         | Quote/rate no longer valid → re-quote   | true      |
| sold_out             | Availability gone                       | false     |
| rate_limited         | Provider throttled us → backoff         | true      |
| provider_unavailable | Upstream down/timeout                   | true      |
| not_supported        | Capability not offered by this provider | false     |
| unknown              | Unclassified                            | false     |

Callers branch on `code`, never on provider-specific error strings:

```

} catch (err) {
  if (err instanceof ProviderError && err.code === "rate_expired") {
    // re-quote then retry
  }
}

```

## 3. Provider registry

`src/lib/suppliers/providers/registry.ts`

The registry is a singleton. Providers self-register at startup and callers resolve by vertical + capability.

### 3.1 API

```

// Register (called by each provider at startup):
providerRegistry.register(provider: BookingProvider): void

// Resolve (returns highest-priority configured provider for vertical+capability):
providerRegistry.pick(vertical: ProviderVertical, capability: ProviderCapability): B

// Introspect:
providerRegistry.list(): BookingProvider[]
providerRegistry.all(vertical, capability): BookingProvider[]

```

`pick()` returns the highest-priority `isConfigured()` provider for the requested vertical + capability. If no configured provider exists, it returns `undefined` — the caller must handle that gracefully (the mock always fills this gap since it is always configured at priority 0).

### 3.2 Registration (`register.ts`)

`src/lib/suppliers/providers/register.ts` wires all three built-in adapters into the registry as a

**module side effect** — importing the providers barrel triggers registration automatically:

```
export function registerBuiltInProviders(): void {
  if (registered) return;           // module-level idempotency guard
  registered = true;
  providerRegistry.register(new MockBookingProvider());           // priority 0 — always
  providerRegistry.register(new DuffelBookingProvider());         // priority 50
  providerRegistry.register(new HotelbedsBookingProvider());       // priority 50
}
registerBuiltInProviders();           // runs on import
```

### 3.3 Provider catalog (PROVIDER\_CATALOG)

Logic-free metadata record used to drive UI ("Available integrations") and future dynamic registration. The catalog is not the registry — it is a static description of what *could* be wired:

| Provider     | Verticals          | Capabilities        | Status  | Env vars                                 |
|--------------|--------------------|---------------------|---------|------------------------------------------|
| mock         | flights + hotels   | search, quote, book | live    | (none)                                   |
| duffel       | flights            | search, quote, book | live    | DUFFEL_API_TOKEN                         |
| amadeus      | flights (+ hotels) | search              | legacy  | AMADEUS_CLIENT_ID, AMADEUS_CLIENT_SECRET |
| hotelbeds    | hotels             | search, quote, book | live    | HOTELBEDS_API_KEY, HOTELBEDS_SECRET      |
| travelgatemx | hotels + flights   | search              | planned | —                                        |
| booking_com  | hotels             | search              | planned | —                                        |
| expedia      | hotels             | search              | planned | —                                        |

### 3.4 Priority resolution

When multiple providers are configured for the same vertical:

- **Highest priority wins** (higher number = higher priority).
- `MockBookingProvider` is always priority **0** — the unconditional fallback.
- `DuffelBookingProvider` and `HotelbedsBookingProvider` are priority **50**.
- Add a future aggregator at **100** to override the defaults.

## 4. Booking service

`src/lib/suppliers/booking-service.ts`

The orchestration layer. Server actions call `serviceBookFlight` or `serviceBookHotel` and receive a

ServiceBookingResult:

```
type ServiceBookingResult =  
  | { confirmed: true;   confirmationNumber: string; providerId: string }  
  | { confirmed: false; confirmationNumber: string; reason: string }
```

**The booking service never throws.** Failures are returned as { confirmed: false } so the server action can write a provisional reference and surface a user-facing error without crashing the booking flow.

## 4.1 Flight booking flow (serviceBookFlight)

1. Derive idempotency key  
sha256("bookingId:bookingItemId:offerId")
2. Replay check  
IF booking\_idempotency.status = "success" THEN return cached result (no supplier)
3. Pick provider  
providerRegistry.pick("flights", "book") → DuffelBookingProvider (or mock)  
IF no provider → return { confirmed: false, reason: "No configured flight booking"
4. Register idempotency key as "pending" (ON CONFLICT DO NOTHING)  
→ prevents duplicate calls even if two requests race to the same booking
5. Quote (price revalidation) – non-fatal  
provider.quoteFlight(offer, ctx)  
→ emit "price\_validated" or "price\_changed" event
6. Emit "booking\_submitted" event
7. Book  
provider.bookFlight({ offer, passengers, idempotencyKey, agencyReference }, ctx)
8. On success:
  - INSERT booking\_supplier\_ref (confirmationNumber, pnr, rawPayload)
  - UPDATE booking\_idempotency → status = "success", supplierRef = confirmationNumber
  - emit "booking\_confirmed" event
  - return { confirmed: true, confirmationNumber, providerId }
9. On failure:
  - UPDATE booking\_idempotency → status = "failed"
  - emit "booking\_failed" event
  - return { confirmed: false, confirmationNumber: "REF-<timestamp>", reason }

## 4.2 Hotel booking flow (serviceBookHotel)

Same as 4.1 with two differences:

1. **offerId** is offer.rateKey ?? offer.id (Hotelbeds uses rate keys).
2. **Quote carries the fresh rateKey** — quoteHotel returns a new quoteId (the refreshed Hotelbeds rateKey), which is passed as quoteId in the HotelBookingRequest. This ensures

the book call always uses the freshest rate, never the one from the initial search.

### 4.3 Quote step — non-fatal by design

If `quoteFlight/quoteHotel` throws, the booking service logs a warning and proceeds to book. This prevents a quote-API timeout from blocking an entire booking attempt — the supplier's `book*` endpoint will reject a stale rate explicitly if it expired, surfacing as a `ProviderError("rate_expired")` from the book step itself.

For hotels, if the quote fails the `quoteId` is `undefined` and the book request uses the original `rateKey` from the search offer.

## 5. Idempotency model

### 5.1 Key derivation

```
sha256(`${bookingId}:${bookingItemId}:${offerId}`)
```

- `offerId` is the supplier's offer/rate identifier (e.g. Duffel offer id, Hotelbeds `rateKey`).
- The same triple always produces the same 64-hex-char key.
- Keys are written with `ON CONFLICT DO NOTHING`, so two concurrent requests for the same booking item cannot both insert a "pending" row.

### 5.2 Lifecycle states

| State   | Meaning                                                                 |
|---------|-------------------------------------------------------------------------|
| pending | Row inserted before the supplier call                                   |
| success | Supplier confirmed; <code>supplierRef</code> is the confirmation number |
| failed  | Supplier rejected; safe to retry with a fresh key (new offer/rate)      |

### 5.3 Replay semantics

- On success (`status = "success"`): return `supplierRef` immediately — no supplier call. This handles browser re-submits, network retries, and serverless double-invocations.
- On pending (`status = "pending"`): the prior call is still in flight (or crashed mid-flight). `ON CONFLICT DO NOTHING` means the new attempt skips the insert and proceeds to quote/book. The supplier's own idempotency key prevents double-booking at the supplier level.
- On failed (`status = "failed"`): the key represents a bad offer; the agent should select a fresh offer (which will have a different `offerId`) and book again with a new key.

### 5.4 TTL

Keys expire after **24 hours** (`expiresAt` column). A future cron job can clean up expired rows. Expired

rows do not block new bookings — expiry is advisory.

## 6. Event log (`booking_event`)

Append-only. Never update or delete rows.

### 6.1 Event types

| Event             | When                              | Metadata                                             |
|-------------------|-----------------------------------|------------------------------------------------------|
| search_initiated  | Agent starts a search             | { destination, checkIn, nights, guests }             |
| offer_selected    | Agent picks an offer from results | { offerId, priceTotal, currency }                    |
| price_validated   | Quote returned same price         | { priceTotal, currency }                             |
| price_changed     | Quote returned different price    | { priceTotal, currency }                             |
| booking_submitted | Just before calling<br>book*      | { offerId, priceChanged }                            |
| booking_confirmed | book* returned success            | { confirmationNumber, status, priceTotal, currency } |
| booking_failed    | book* threw                       | { reason }                                           |
| booking_cancelled | Cancellation succeeded            | { penaltyAmount, currency }                          |
| payment_started   | Stripe Checkout session created   | { sessionId, amount }                                |
| payment_completed | Stripe webhook confirmed          | { sessionId, amount }                                |

### 6.2 Dual purpose

- **Compliance audit trail** — every state transition is permanently recorded with timestamp, providerId, and correlationId. Never deleted (even if the booking itself is cancelled).
- **Analytics source** — booking funnel metrics (search→select, select→book, book→confirm, confirm→cancel) can be derived directly from this table without a third-party SDK.

### 6.3 Schema

| Column    | Type | Notes        |
|-----------|------|--------------|
| id        | uuid | PK           |
| bookingId | uuid | FK → booking |

|               |           |                                                      |
|---------------|-----------|------------------------------------------------------|
| agencyId      | uuid      | FK → agency (for efficient agency-scoped analytics)  |
| event         | text      | Stable event code (see 6.1)                          |
| providerId    | text?     | "duffel", "hotelbeds", "mock", or null for UI events |
| correlationId | text?     | Traces a request across logs                         |
| metadata      | jsonb?    | Structured payload (offer id, price, error, etc.)    |
| createdAt     | timestamp | Append timestamp                                     |

**Indexes:** bookingId, agencyId, event, createdAt, composite (bookingId, createdAt).

## 7. Supplier reference (booking\_supplier\_ref)

Structured record of the supplier's confirmation for a booking item. Replaces the ad-hoc JSONB in booking\_item.details so references are **queryable** for status polling, cancellation, and voucher retrieval.

| Column             | Type      | Notes                                               |
|--------------------|-----------|-----------------------------------------------------|
| id                 | uuid      | PK                                                  |
| bookingId          | uuid      | FK → booking                                        |
| bookingItemId      | uuid      | FK → booking_item                                   |
| providerId         | text      | "duffel", "hotelbeds", etc.                         |
| confirmationNumber | text      | Shown to client                                     |
| pnr                | text?     | Airline PNR / record locator                        |
| supplierOrderId    | text?     | Provider's internal order id                        |
| rawPayload         | jsonb?    | Full raw response for debugging + future re-parsing |
| createdAt          | timestamp | —                                                   |

Multiple refs per bookingItemId are possible (e.g. re-booking after a failure leaves both the failed attempt and the successful one).

## 8. Document store (booking\_document)

Documents generated for a booking: vouchers, e-tickets, invoices, itineraries.

| Column    | Type | Notes        |
|-----------|------|--------------|
| id        | uuid | PK           |
| bookingId | uuid | FK → booking |

|               |            |                                                            |
|---------------|------------|------------------------------------------------------------|
| bookingItemId | uuid?      | FK → booking_item (null = booking-level doc)               |
| type          | text       | "voucher"   "ticket"   "invoice"   "itinerary"   "receipt" |
| providerId    | text?      | Supplier that issued the document                          |
| url           | text?      | Vercel Blob URL or supplier CDN link                       |
| rawData       | jsonb?     | Supplier's payload for re-generation                       |
| generatedAt   | timestamp? | When the document was issued                               |
| createdAt     | timestamp  | —                                                          |

## 9. Adding a new provider

1. **Create the provider class** in `src/lib/suppliers/providers/<name>-provider.ts`.

- Implement `ProviderDescriptor` on every provider.
- Implement only the capability interfaces the supplier supports.
- Throw `ProviderError` (never native `Error`) on failure.
- Return `true` from `isConfigured()` only when all required env vars are set.

2. **Export from the barrel** — add to `src/lib/suppliers/providers/index.ts`:

```
export * from "./<name>-provider";
```

3. **Register in `register.ts`**:

```
providerRegistry.register(new MyNewProvider());
```

4. **Add to `PROVIDER_CATALOG`** — one entry in `registry.ts` with `id`, `label`, `verticals`, `capabilities`, `envVars`, and `status: "live"`.

5. **Set priority** — choose a number relative to existing providers. If it should win over Duffel/Hotelbeds (priority 50), use `> 50`. Use `< 50` if it should only fill the gap when the primary provider is unconfigured.

No calling-code changes are needed — the registry and booking service automatically route to the new provider when it is configured.

## 10. Configuration and activation

All provider credentials are resolved in `src/lib/suppliers/config.ts`:



|                       |                                  |
|-----------------------|----------------------------------|
| ATLAS_ENV=development | → mock only (default when unset) |
| ATLAS_ENV=sandbox     | → provider sandbox endpoints     |
| ATLAS_ENV=production  | → live endpoints                 |

| Provider           | Required env vars                   | Optional           |
|--------------------|-------------------------------------|--------------------|
| Duffel (flights)   | DUFFEL_API_TOKEN                    | —                  |
| Hotelbeds (hotels) | HOTELBEDS_API_KEY, HOTELBEDS_SECRET | HOTELBEDS_HOSTNAME |

When `ATLAS_ENV=production` and the credentials are set, `pick()` returns the real provider at priority 50. The mock continues to serve as a fallback only when the real provider's `isConfigured()` returns `false`.

# Server Actions Reference

---

All server-side mutations are Next.js Server Actions under `src/lib/actions/`. Every action:

1. Calls `requireAgencyUser()` to authenticate + `get (userId, agencyId)`.
2. Validates input with Zod (most actions; see gap tracker in roadmap.md).
3. Returns `ActionResult<T>` — `{ success: true; data: T }` or `{ success: false; error: string }`.

```
type ActionResult<T = void> =  
  | { success: true; data: T }  
  | { success: false; error: string }
```

---

## bookings.ts

---

The largest action file (≈1000 lines). Orchestrates the full booking lifecycle.

### `createBooking(input)`

Creates a new booking file.

**Input:** `{ clientId, destination?, departDate?, returnDate?, currency?, notes? }`

**Guards:** client must belong to the agency (tenant isolation). **Side effects:** auto-generates a human reference (BKG-NNN).

### `updateBooking(bookingId, input)`

Updates booking header fields (destination, dates, currency, notes, etc.).

**Guards:** booking must belong to the agency.

### `deleteBooking(bookingId)`

Hard-deletes a booking and all its children.

**Guards:** `canDeleteRecords(role)` (admin/manager only). **Note:** soft delete is not yet implemented — see database.md gap tracker.

### `advanceStatus(bookingId)`

Advances the booking to the next lifecycle state.

**Lifecycle:** `draft → awaiting_payment → confirmed → ticketed → completed → cancelled`

### Prerequisites enforced:

- `confirmed`: trip items present AND zero outstanding balance.

- `ticketed`: trip items present.

### Side effects:

- On `confirmed` or `ticketed`: `autoGenerateCommissions(bookingId, agencyId)` (idempotent).
- On `ticketed`: fires `confirmItemBooking` for all unconfirmed trip items (flights, hotels, transfers)  
→ calls `serviceBookFlight` / `serviceBookHotel` via the provider registry.

### `bookItem(bookingItemId)`

Attempts to confirm a single booking item with the supplier.

**Side effects:** calls `serviceBookFlight` or `serviceBookHotel` in `booking-service.ts`. Returns {  
`confirmed`: boolean; `confirmationNumber`: string; `reason?`: string }.

### `addTraveller(bookingId, input) / updateTraveller(travellerId, input) / deleteTraveller(travellerId)`

CRUD for `booking_traveller`. Input includes personal details (name, title, gender, passport, DOB, nationality) plus `email/phone` (required for the lead passenger by supplier APIs).

### `addBookingItem(bookingId, input) / updateBookingItem(itemId, input) / deleteBookingItem(itemId)`

CRUD for `booking_item`. `type` must be a `BOOKING_ITEM_TYPES` code.

### `updateItemOrder(bookingId, orderedIds)`

Reorders booking items by updating `sort_order`.

### `assignDay(itemId, dayIndex)`

Assigns a `booking_item` to a specific itinerary day.

### `updateBookingDay(bookingId, dayIndex, input)`

Upserts a `booking_day` title/notes row for a given day index.

### `shareBooking(bookingId) / unshareBooking(bookingId)`

Generates or clears the `booking.shareToken` (itinerary public link `/i/[token]`).

### `generateVoucher(bookingId) / generateInvoice(bookingId)`

Server-rendered PDF generation. Guards: booking must have trip services. Sends the PDF link via `notification log`.

### `sendConfirmationEmail(bookingId)`

Sends a booking confirmation email to the client via Resend. Falls back to logging when Resend is unconfigured.

---

## clients.ts

---

**createClient (input)**

Creates a client record. `type` defaults to "individual". Validates country against the ISO reference list.

**updateClient (clientId, input)**

Updates client fields. Validates country, source, and industry codes.

**deleteClient (clientId)**

Hard-deletes the client and all descendant records. **Guards:** `canDeleteRecords (role)`.

**addContact (clientId, input) / updateContact (contactId, input) / deleteContact (contactId)**

CRUD for `client_contact`.

**convertToClient (clientId)**

Flips a lead (`status = "lead"`) to an active client.

---

## opportunities.ts

---

**createOpportunity (input)**

Creates an opportunity. `stage` defaults to "lead".

**updateOpportunity (opportunityId, input)**

Updates opportunity fields including stage, value, and travel details.

**deleteOpportunity (opportunityId)**

**Guards:** `canDeleteRecords (role)`.

**advanceOpportunityStage (opportunityId)**

Advances through: lead → qualified → proposal → booked → won. won is terminal; lost can be set via `updateOpportunity`.

---

## products.ts (proposals)

---

**createProduct (input)**

Creates a proposal draft. Auto-generates PRD-NNN reference.

**updateProduct (productId, input)**

Updates proposal fields including markup and dates.

**deleteProduct (productId)**

**Guards:** `canDeleteRecords(role)`.

**addProductItem(productId, input) / updateProductItem(itemId, input) / deleteProductItem(itemId)**

CRUD for `product_item` line items.

**sendProposal (productId)**

Sets `status = "sent"` and generates a `shareToken` for the public link. Sends a notification email to the client.

**shareProposal (productId) / unshareProposal (productId)**

Generates or clears the `product.shareToken`.

**convertProposalToBooking (productId)**

One-click conversion: creates a `booking` from an accepted `product`, copying client, dates, and line items. Flips the linked opportunity to "won".

---

## proposals-public.ts

---

Actions callable from the **public proposal page** (`/p/[token]`) — no auth.

**acceptProposal (token, input)**

- Validates token; finds product.
- Sets `acceptedAt`, stamps signer name/email/IP/UA and `signatureData`.
- If `opportunityId` is linked, advances opportunity to "won".

**declineProposal (token)**

Sets `declinedAt`.

---

## payments.ts

---

**createPayment**(bookingId, input)

Records a payment against a booking. `kind` must be `PAYMENT_KINDS` code. Re-calculates outstanding balance.

**deletePayment**(paymentId)

**Guards:** `canManagePayments(role)`.

**createStripeCheckout**(bookingId, amount)

Creates a Stripe Checkout Session (Connect: destination charge to agency). Returns { `checkoutUrl` }.

---

## commissions.ts

---

**autoGenerateCommissions**(bookingId, agencyId)

Idempotent — safe to call multiple times. Generates two commission rows per bookable item: `supplier_to_agency` (from the supplier contract) and `agency_to_agent` (from `user.commissionRatePercent`). Called automatically by `advanceStatus` on `confirmed` / `ticketed`.

**updateCommission**(commissionId, input)

Adjusts `status` (`confirmed` → `paid`, etc.) or `amount` / `notes`.

**deleteCommission**(commissionId)

**Guards:** `canViewFinance(role)`.

---

## suppliers.ts

---

**createSupplier**(input) / **updateSupplier**(supplierId, input) /  
**deleteSupplier**(supplierId)

CRUD for the managed supplier directory.

**createSupplierContract**(supplierId, input) / **updateSupplierContract**(contractId,  
input) / **deleteSupplierContract**(contractId)

CRUD for supplier contracts.

```
createSupplierRate(contractId, input) / updateSupplierRate(rateId, input) /  
deleteSupplierRate(rateId)
```

CRUD for per-product rates within a contract.

```
uploadContractFile(contractId, formData)
```

Uploads a PDF contract to Vercel Blob and saves the URL.

---

## notifications.ts

---

```
sendNotification(bookingId, input)
```

Sends an email/SMS/WhatsApp/push notification to a recipient. Logs to `notification` table regardless of delivery status.

---

## invites.ts

---

```
inviteTeamMember(input)
```

Creates an `agency_invite` row with a signed, expiring token. Sends an invitation email via Resend (or logs + returns the link when unconfigured).

```
revokeInvite(inviteId)
```

Marks the `invite` `status` = "expired".

---

## team.ts

---

```
updateTeamMember(userId, input)
```

Updates `role`, `active`, `commissionRatePercent`. Role changes are gated: only an admin can assign/change the `admin` role (`canAssignAdmin(role)`).

```
deactivateTeamMember(userId)
```

Sets `active` = `false`.

---

## settings.ts

---

```
updateAgencySettings(input)
```

Updates agency `name`, `slug`.

**`updateUserProfile(input)`**

Updates the current user's `name`, `locale`.

---

## onboarding.ts

---

**`dismissOnboarding()`**

Sets `agency.onboardingDismissedAt` to close the getting-started card.

---

## search.ts

---

**`searchFlights(params)` / `searchHotels(params)`**

Calls `safeSearch()` from `src/lib/suppliers/index.ts`. Returns offers or mock data on provider failure. No mutation; safe to call from Server Components or actions.

---

## ai.ts

---

**`generateItinerary(bookingId, prompt)`**

Calls `OpenRouter` (streamed) to generate a day-by-day itinerary. Returns a `ReadableStream` — consumed by the assistant UI.

**`generateQuote(opportunityId, prompt)`**

AI-assisted quote: generates product items + summary text.

**`draftEmail(context, prompt)`**

Drafts a client communication email. Output is shown in a preview step before sending.

**`visaAssistant(destination, nationality)`**

Returns visa requirement guidance. Advisory only — never authoritative.

---

## billing.ts

---

**`createCheckoutSession()`**

Starts a Stripe subscription Checkout for an agency. Redirect to Stripe.



# Development Guide

---

## Setup

---

```
npm install --legacy-peer-deps    # better-auth peer range
npm run dev                       # http://localhost:3000
npm run check                     # lint + typecheck
npm run build:ci                  # next build
```

After completing any change, run `npm run check` and `npm run build:ci` to verify code quality (lint, typecheck, build). See `coding-standards.md`.

## Environment (`.env`)

---

**Required:** `POSTGRES_URL`, `BETTER_AUTH_SECRET`.

**Optional** (all integrations degrade gracefully to sample/logged behaviour when unset — see `api-integrations.md`):

- `OPENROUTER_API_KEY` — AI features
- `BLOB_READ_WRITE_TOKEN` — uploads
- `GOOGLE_CLIENT_ID` / `GOOGLE_CLIENT_SECRET`
- `NEXT_PUBLIC_APP_URL`
- `RESEND_API_KEY` / `EMAIL_FROM` — email
- `STRIPE_SECRET_KEY` / `STRIPE_PRICE_ID` / `STRIPE_WEBHOOK_SECRET` — billing
- `STRIPE_CONNECT_WEBHOOK_SECRET` / `STRIPE_PLATFORM_FEE_PERCENT` — Connect payments
- `DUFFEL_API_TOKEN` — flights
- `HOTELBEDS_API_KEY` / `HOTELBEDS_SECRET` — hotels
- `PROTECTED_DB_HOSTS` — makes destructive scripts refuse a prod DB (override with `ALLOW_PROD=1`). See `security.md`.

## Database

---

- Branches: `dev` `ep-wandering-sunset-aitlty78` · `prod` `ep-misty-thunder-aixz34vy`.
- After schema changes: `db:generate` → `db:migrate` (**never** `db:push`). `db:studio` to browse. See `database.md`.
- **Always run migrations on prod after deploy:** `POSTGRES_URL=<prod-url> npx drizzle-kit migrate`.

## Maintenance scripts

---

```
# Promote an existing account to the platform super-admin
npx tsx --env-file=.env scripts/make-platform-admin.ts <email>
```

```
# Seed / reset the Demo Agency (idempotent — wipes its data, keeps users, reseeds)
npx tsx --env-file=.env scripts/seed-demo-data.ts

# Cross-tenant isolation test (seeds 2 agencies, asserts no leak, cleans up)
npx tsx --env-file=.env scripts/test-tenant-isolation.ts

# Normalize legacy free-text country values to canonical ISO names
npx tsx --env-file=.env scripts/backfill-countries.ts

# Sync Hotelbeds hotel content (photos/facilities/coords) into the cache table
npx tsx --env-file=.env scripts/sync-hotel-content.ts # curated destinations
npx tsx --env-file=.env scripts/sync-hotel-content.ts BCN MAD # specific codes
```

## Testing

---

Use the project's testing tools/scripts (e.g. `test-tenant-isolation.ts`) to verify changes — never assume changes work. If no tooling covers a change, ask whether testing should be skipped. See [coding-standards.md](#).

# Coding Standards

**Source of truth:** `AGENTS.md` (aliased by `CLAUDE.md`) holds the enforced working conventions. This doc summarizes and points there — it does not restate or override it.

## Working conventions (from AGENTS.md)

- **Responses** — keep concise and to the point unless asked otherwise.
- **Planning** — always ask clarifying questions; never assume design, stack or features; use deep-dive sub-agents for research and to review plans.
- **Change/edit mode** — prefer sub-agents to implement features; act as a coordinator; parallelize independent work; use premium models for complex tasks (coding) and mid-tier for simpler ones (documentation).
- **Quality gates** — after completing features, run lint, type check and `next build` (`npm run check`, `npm run build:ci`). See `development-guide.md`.
- **Testing** — use available testing tools; never assume changes work; if no tooling exists for a change, ask whether to skip.

## Engineering rules

Hard rules for how code is written. Status reflects the current codebase (✓ holds · ● partial).

| Rule                                | Status | Notes                                                                                                                              |
|-------------------------------------|--------|------------------------------------------------------------------------------------------------------------------------------------|
| Never duplicate logic               | ●      | Shared helpers in <code>src/lib</code> ( <code>queries</code> , <code>analytics</code> , <code>domain</code> ); enforce by review. |
| Always use server actions           | ✓      | 20/21 files in <code>src/lib/actions</code> are <code>"use server"</code> ; mutations go through them.                             |
| Always validate input               | ●      | Validated where it matters, not universally.                                                                                       |
| Always use Zod                      | ●      | Zod v4 present, used in ~13 files; not every action yet.                                                                           |
| No business logic inside components | ●      | Logic lives in <code>actions/lib</code> ; enforce by review.                                                                       |
| No direct SQL in UI                 | ✓      | Pages call Drizzle queries server-side / via actions, not raw SQL in client components.                                            |
| Every action logs activity          | ●      | <code>logActivity</code> in ~11/21 action files — not yet every mutation (see <code>security.md</code> ).                          |
| Every mutation is tenant scoped     | ✓      | All actions go through <code>requireAgencyUser()</code> → <code>agencyId</code> scope (see <code>business-rules.md</code> ).       |

The ● rows (universal Zod validation, universal activity logging) are in the spec-vs-reality gap tracker.

## Database changes

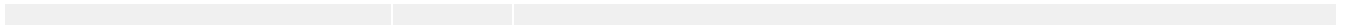
---

- After any schema change, run **drizzle generate + migrate**. **Never** `db:push`. Workflow detail in `database.md`.
- All ID columns **not** related to Better Auth use **UUIDs**, randomly generated.

## UI

---

- Follow the design system in `DESIGN.md` when creating or reviewing components/pages. See `ui-ux.md`.



# UI / UX

---

**Source of truth:** `DESIGN.md` defines the full visual design system — colors (oklch tokens), typography, spacing, radius, shadows, animations, layout patterns, shadcn/ui component conventions, and dark mode. All new components and pages **must** follow it. This doc points there and captures product-level UX patterns only; it does not duplicate the token tables.

## Atlas design principles

---

The UI-level expression of the ten founding Atlas principles. The product guardrails every page and feature is measured against:

1. **Never ask the user twice for the same information.** Reuse what's already captured (controlled vocabularies, reference data, client records) instead of re-prompting.
2. **Everything starts from the client.** The client record is the spine — leads, opportunities, proposals, bookings and payments all hang off it.
3. **Every action must be reversible.** Prefer soft states, confirmations and undo over destructive, irreversible operations.
4. **One source of truth.** No duplicated or conflicting data; canonical values (ISO countries, enums, references unique per agency) over free text.
5. **Automation before manual work.** Default to generating (commissions, itineraries, quotes, documents) rather than asking the user to do it by hand.
6. **Every page should answer "What should I do next?"** Surface the next action — stepper advance, CTA, empty-state prompt — never leave a dead end.
7. **Managers want insights; agents want speed.** Tailor each role's surface — analytics-dense for managers, fast and scoped for agents.
8. **No empty pages.** Every list/chart/detail has an empty state with a clear next step, never blank space.
9. **Data quality is more important than flexibility.** Constrain inputs (dropdowns, pickers, validation) to keep reporting clean.
10. **Everything should be mobile friendly.** Layouts work on small screens and in both LTR and RTL.

These principles inform the patterns below and the business rules. See also the product vision.

## Design system summary

---

- **Stack:** Next.js + Tailwind v4 (CSS-first `@theme inline`, no config file), shadcn/ui (new-york, neutral), Lucide icons, Geist fonts, next-themes dark mode.
- **Tokens:** semantic oklch color tokens, `--radius 10px` base, Tailwind-default shadows, custom `fade-in/fade-up/scale-in` animations. Use `cn()` for class merging.
- See `DESIGN.md` for every value.

## Performance budgets

---

Target response times — what "fast enough" means per surface. These back design principle #6 (agents want speed) and dictate where to lean on caching, streaming, and loading states.

| Surface                       | Budget   |
|-------------------------------|----------|
| Dashboard                     | < 1 s    |
| Search (in-app lookup)        | < 500 ms |
| Booking creation              | < 2 s    |
| Hotel search (live Hotelbeds) | < 5 s    |
| Reports / export              | < 10 s   |

Budgets are **targets, not yet measured** — there's no perf instrumentation in place. The slow surfaces (hotel search, reports) are I/O-bound on external APIs and the export workbook; the content cache already keeps hotel photos quota- and latency-free (api-integrations.md). Every surface over its budget must show a loading state.

## Data table standard

---

The target capabilities for every data table (clients, bookings, opportunities, proposals, suppliers, commissions). These belong in **one reusable DataTable component**, not re-implemented per page:

- **Sorting** — click a column header to sort; multi-column where useful.
- **Filtering** — per-column filters plus the page-level search/filter bar.
- **Column chooser** — show/hide columns; remember the user's choice.
- **Export** — export the current view (respecting sort/filter) to CSV/Excel via the existing `/api/export` endpoint.
- **Infinite scrolling** — load more on scroll instead of hard `limit` truncation (fixes the silent-cap issue in the page audit).
- **Sticky header** — header stays visible while the body scrolls.
- **Keyboard shortcuts** — arrow-key row navigation, Enter to open, / to focus search.
- **Context menu** — right-click a row for quick actions (open, edit, duplicate, delete — gated by permissions).

**Current state:** tables today are plain `shadcn Table` markup with none of the above — no sorting, column chooser, infinite scroll, sticky header, shortcuts or context menu, and a hardcoded row `limit`. This standard is the target for a shared `DataTable`; building it also clears most of the systemic gaps in the page audit (export, pagination, bulk actions).

## Page requirements checklist

---

Every list/index page must ship with all of these — a page missing any of them is incomplete, not "v1":

- [ ] **Search** — free-text lookup over the page's records
- [ ] **Filters** — by the page's key dimensions (status, type, date range, owner)
- [ ] **Export** — CSV/Excel of the current view (see analytics.md)
- [ ] **Bulk actions** — multi-select + an action that applies to the selection
- [ ] **Empty state** — a prompt with a clear next step (never blank — principle #8)
- [ ] **Loading state** — skeleton/spinner while data resolves
- [ ] **Error state** — a recoverable message, not a crash
- [ ] **Pagination** — bounded result sets, never an unbounded dump
- [ ] **Mobile layout** — works on small screens, LTR and RTL (principle #10)
- [ ] **Permissions** — role-scoped data + role-gated actions (see business-rules.md)

This operationalizes the design principles at the page level. Detail pages additionally answer "what should I do next?" via a primary CTA or lifecycle action.

## Product UX patterns

---

These are app-specific patterns layered on the design system:

- **Getting-started checklist** — dismissible 4-step card on the dashboard for new agencies; dismissed state persists in DB (`agency.onboardingDismissedAt`) across devices and team members.
- **Lifecycle stepper** — horizontal progress bar on booking detail with an "Advance to [next]" button and soft/hard prerequisite guards (see business-rules.md).
- **Role-gated nav** — locked nav items shown dimmed with a tooltip for non-admin roles; nav is filtered by a `show(role)` predicate.
- **List/Board toggle** — Bookings and Pipeline share data with a table↔kanban switch.
- **Inline search sheet** — "Search flights/hotels" on the trip-services panel opens a sheet pre-scoped to the booking's destination.
- **Consolidated sharing** — proposal sharing lives under one "Share with client ▼" dropdown.
- **Empty states** — charts and lists show empty-state badges/CTAs rather than blank space.

## RTL & i18n

---

Arabic is RTL (IBM Plex Sans Arabic, `html[dir="rtl"]`). Layout must work in both directions. i18n plumbing is in architecture.md.

# Business Rules

Domain logic and constraints. Enums and capability helpers are defined in `src/lib/domain.ts`; server-side enforcement lives in the actions under `src/lib/actions/`.

## Automation triggers

The event → action automation layer Atlas is targeting (design principle #5, *automation before manual work*). Each is a domain event that should fire a follow-up action:

| Event                                | Action             | Current state                                                                                              |
|--------------------------------------|--------------------|------------------------------------------------------------------------------------------------------------|
| Client created                       | Send welcome email | ✗ not implemented (Resend only sends invites/reset/proposal/portal)                                        |
| Opportunity won                      | Generate proposal  | ✗ manual (new proposal + AI quote builder)                                                                 |
| Proposal accepted                    | Generate booking   | ⚠ one-click <i>convert proposal</i> → <i>booking</i> , not automatic                                       |
| Booking confirmed                    | Generate invoice   | ✗ invoices are on-demand PDFs, not auto-generated                                                          |
| Payment received / booking confirmed | Update accounting  | ⚠ partial — <code>autoGenerateCommissions</code> fires on confirm/ticket (bookings.ts); no full accounting |
| Travel completed                     | Request review     | ✗ not implemented                                                                                          |

**Current state:** the only true event-driven automation today is commission generation on booking confirm/ticket (idempotent). The rest are manual or on-demand. Building this trigger layer is the heart of the "Automation" pillar in the vision and the automated-quotation item in the roadmap.

## Never rules (hard constraints)

Invariants Atlas must never violate. Each maps to an enforcement point — a violation is a bug, not a preference.

| Never                                      | Why / where enforced                                                                                             |
|--------------------------------------------|------------------------------------------------------------------------------------------------------------------|
| Never use free text when a dropdown exists | Controlled vocabularies + reference pickers keep reporting clean (data quality > flexibility). See vocabularies. |
| Never duplicate client information         | One source of truth — the client record is the spine; reuse it, don't re-enter it.                               |
| Never create a booking without a client    | Every booking hangs off a client; the spine is mandatory.                                                        |



|                                             |                                                                                                                                                                               |
|---------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Never hard delete data</b>               | Reversibility — prefer soft state / archival + the activity log; deletes are gated to admin/manager only.                                                                     |
| <b>Never expose another tenant</b>          | Every read/write is scoped by <code>agencyId</code> via <code>requireAgencyUser()</code> ; verified by <code>test-tenant-isolation.ts</code> . See <code>security.md</code> . |
| <b>Never sum different currencies</b>       | No FX — group by currency with a DZD headline. See <code>Currency</code> and <code>analytics.md</code> .                                                                      |
| <b>Never make users navigate five pages</b> | Surface work where it happens — inline search sheets, detail-page panels, "what next?" CTAs.                                                                                  |
| <b>Never require more than three clicks</b> | Key actions (advance lifecycle, convert proposal→booking, share, pay) are one-to-three clicks.                                                                                |

These are the enforcement edge of the design principles.

## Golden workflow

The canonical end-to-end happy path Atlas is built around. Everything starts from the client (design principle #2) and ends by feeding the next trip — a loop, not a line.

```
Lead → Client → Opportunity → Proposal → Customer accepts → Booking →
Supplier reservation → Payment → Ticketing → Travel → Feedback → Repeat client
```

| Step                        | In Atlas                                                                                     | Notes                                                                                                                                                 |
|-----------------------------|----------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Lead</b>                 | <code>client</code> with a <code>source</code> (lead-source code)                            | captured as an early-stage client                                                                                                                     |
| <b>Client</b>               | <code>client</code> (+ contacts)                                                             | the spine; all records hang off it                                                                                                                    |
| <b>Opportunity</b>          | <code>opportunity</code> (pipeline stage, <code>value</code> , <code>travel_purpose</code> ) | shown on the Pipeline board                                                                                                                           |
| <b>Proposal</b>             | <code>product</code> (PDF + e-sign)                                                          | shareable <code>/p/[token]</code> or in-portal                                                                                                        |
| <b>Customer accepts</b>     | e-sign stamps signer + flips opportunity to <b>won</b>                                       | also acceptable in the client portal                                                                                                                  |
| <b>Booking</b>              | <code>booking</code> , one-click <b>convert proposal → booking</b>                           | lifecycle starts at <code>draft</code>                                                                                                                |
| <b>Supplier reservation</b> | <code>booking_item</code> + <code>supplier picker</code> + <code>booking_service.ts</code>   | live Duffel/Hotelbeds search; real booking wired via provider registry (quote → book → idempotency → event log); activate with production credentials |
| <b>Payment</b>              | <code>payment</code> (deposit/installments, Stripe Connect)                                  | <code>confirmed</code> requires zero balance                                                                                                          |
| <b>Ticketing</b>            | <code>lifecycle</code> <code>ticketed</code>                                                 | requires trip items; auto-generates commissions                                                                                                       |
| <b>Travel</b>               | <code>lifecycle</code> <code>completed</code> ; <code>itinerary /i/[token]</code>            | day-by-day timeline, vouchers/invoices                                                                                                                |

|                      |                                               |                               |
|----------------------|-----------------------------------------------|-------------------------------|
| <b>Feedback</b>      | activity log / notes on the client timeline   | closes the loop               |
| <b>Repeat client</b> | back to <b>Opportunity</b> on the same client | retention, not re-acquisition |

This maps onto the booking lifecycle below and the role landings in architecture.md.

## Roles & capabilities

Five roles: **admin**, **manager**, **finance**, **support**, **agent**. The capability matrix is in architecture.md. Key rules:

- `seesAllData` — all roles except **agent** (agents see only their own records).
- `canManageTeam`, `canManagePayments` — admin/manager.
- `canViewFinance` — admin/manager/finance (gates `/finance`, `/commissions`, `/reports`).
- `canViewSupport` — admin/manager/support.
- `canDeleteRecords` — admin/manager.
- `canAssignAdmin` — **only an admin** can assign or change the admin role.

## Onboarding & access

- **Invitation-only signup** — enforced at the auth layer (`user.create.before` hook). No invite ⇒ no account. See security.md.
- **Agency suspension** — a suspended agency locks out its users via `requireAgencyUser`.
- **Lapsed subscription** — gates access via `requireAgencyUser` (14-day trial on provision). See api-integrations.md.

## Currency

- Supported: **DZD (default)**, EUR, USD.
- **No FX conversion**. The agency operates in DZD; Atlas never sums across currencies. Analytics group **by currency** with a DZD headline. See analytics.md.

## References

- BKG-... (bookings) and PRD-... (proposals) reference numbers are unique **per agency**, not globally.

## Booking lifecycle

States: `draft` → `awaiting_payment` → `confirmed` → `ticketed` → `completed`. A visual stepper drives it with an "Advance to [next]" button. **Hard prerequisites are enforced server-side:**

- `confirmed` requires trip items **and** zero outstanding balance.

- `ticketed` requires trip items.
- Vouchers/invoices are **blocked** when a booking has no trip services.

## Proposals & e-signature

---

- Server-rendered PDF; shareable via public tokenized `/p/[token]` (no login) and in-portal signing.
- E-sign stamps signer name/email/IP/UA and flips the linked opportunity to **won**.
- **Convert accepted proposal → booking** in one click. (A `convertedBookingId` guard column is a pending open item — see roadmap.md.)

## Commissions

---

- **Two ledgers:** `supplier_to_agency` (agency earns from a supplier per booking item) and `agency_to_agent` (agent earns from the agency per booking).
- **Auto-generated** when a booking is confirmed or ticketed; **idempotent**.
- Agent rate from `user.commissionRatePercent`; supplier basis from the supplier contract (percent / fixed / net).

## Controlled vocabularies

---

Seven fields are enum dropdowns (codes stored, labels shown) for clean reporting: lead source, travel purpose, trip type, gender, title, lost reason, industry. Country/nationality use canonical ISO 3166-1 values (full name stored) so spellings never drift; city uses curated autocomplete suggestions.

## Client portal

---

- Magic-link passwordless sessions scoped to **one client**, separate from staff auth (`portal_session`).
  - Agents send the invite from a client or booking detail page; the link is copyable when email is unconfigured.
  - Online "Pay now" appears only when the agency has onboarded Stripe Connect.
  - Clients can accept/decline proposals in-portal with the same e-sign audit as the public flow.
-

# Deployment

---

## Vercel

---

- Project `atlasproject/agence_tool`, connected to GitHub.
- **Auto-deploy:** `git push origin main` → builds & ships production.
- **Manual:** `npx vercel deploy --prod --yes`.
- `vercel.json` → `buildCommand: npm run build:ci; .npmrc` → `legacy-peer-deps=true`.

## Vercel environment

---

`POSTGRES_URL`, `BETTER_AUTH_SECRET`, `NEXT_PUBLIC_APP_URL`, `BETTER_AUTH_URL`,  
`PROTECTED_DB_HOSTS=ep-misty-thunder-aixz34vy` (plus integration keys as needed — see `api-integrations.md`).

## Migrations on prod

---

**Always run migrations on prod after a deploy that changed the schema:**

```
POSTGRES_URL=<prod-url> npx drizzle-kit migrate
```

Branches: `dev` `ep-dawn-voice-ai8d6q3o` · `prod` `ep-misty-thunder-aixz34vy`.

`PROTECTED_DB_HOSTS` makes destructive scripts refuse the prod DB unless `ALLOW_PROD=1`. See `security.md` and `database.md`.

## Demo accounts

---

Throwaway accounts on the **Demo Agency**, all on the live site.

| Role                    | Email                                     | Password                  |
|-------------------------|-------------------------------------------|---------------------------|
| Platform admin (vendor) | <code>ouksili.abdelmalek@gmail.com</code> | <code>Atlas!2026</code>   |
| Manager                 | <code>yasmine@agence.test</code>          | <code>Agency!2026</code>  |
| Finance                 | <code>finance@demo.test</code>            | <code>Finance!2026</code> |
| Support                 | <code>support@demo.test</code>            | <code>Support!2026</code> |
| Agent                   | <code>karim@demo.test</code>              | <code>Agent!2026</code>   |
| Agent                   | <code>lina@demo.test</code>               | <code>Agent!2026</code>   |
| Agent                   | <code>omar@demo.test</code>               | <code>Agent!2026</code>   |

**🔑 DEMO CREDENTIALS ONLY** — rotate or delete before any real production use. These live

passwords are checked into the repo; treat as a known risk (see security.md).

**Suggested demo flow:** sign in as vendor → /platform → View as Demo Agency → manager dashboard → switch to finance/support/agent views → Settings → switch to Français or العربية.

# Roadmap & Changelog

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## Phase status

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**Phase 1 — Sellable ✓ COMPLETE** Email (Resend) · Stripe SaaS billing · Separate Neon DB branches (dev/prod) · PDF proposals · E-signature · Live flights (Duffel) · Live hotels (Hotelbeds + content cache).

**Phase 2 — Competitive ✓ COMPLETE** Supplier management (/suppliers, contracts, rates, picker) · Commissions tracking (auto-generated, two-ledger, /commissions) · Client portal (magic-link login, trip view, online payments via Stripe Connect, in-portal proposal signing + portal invite from agent).

**Phase 3 — AI differentiation ✓ COMPLETE** AI itinerary generation · AI quote builder · AI email drafting · AI visa assistant — all inline, embedded where the work happens. See ai.md.

**UX overhaul ✓ COMPLETE** (20 changes across 4 size categories) Getting-started checklist · Lifecycle stepper · Board view toggle · Inline search · Portal invite · Convert proposal→booking · Client funnel timeline · Booking hard guards · Vocabulary pass · Role nav tooltips · Empty-state badges.

**Data quality & BI ✓ COMPLETE** (4 phases) DZD-only currency (no FX) · **P1** richer analytics (revenue/conversion/pipeline funnel/AR aging/margin) · **P2** controlled-vocabulary enums (7 fields, codes+labels) · **P3** ISO country/nationality reference data + city suggestions · **P4** standardized CSV/Excel export (/reports, dual code+label columns). Hotel search-by-name added. See analytics.md.

## Module maturity

---

Maturity levels: **Production** (stable, in daily use) · **Beta** (works, hardening) · **Alpha** (functional, evolving) · **Planned** (not built).

| Module          | Status     |
|-----------------|------------|
| CRM             | Production |
| Sales           | Production |
| Hotels          | Beta       |
| Flights         | Beta       |
| Accounting      | Planned    |
| AI              | Alpha      |
| Supplier Portal | Planned    |

**Flights/Hotels nuance:** live *search* is shipped for both (Duffel + Hotelbeds); the maturity above reflects the end-to-end **booking** flow — Hotels is further along (Beta), real Flights booking is Planned (Duffel orders). See api-integrations.md and Open item #1.

## Open items

---

1. **Real supplier booking** — ✓ architecture complete (Sprint 1 Waves 1–3). Provider registry, quote→book→cancel lifecycle, idempotency, event log, and supplier-ref tables are all wired. Activate by setting `DUFFEL_API_TOKEN` (flights) and `HOTELBEDS_API_KEY/HOTELBEDS_SECRET + ATLAS_ENV=production` (hotels).
2. **Translate deeper pages** — bookings, clients, finance, support, platform (i18n plumbing ready).
3. **Cross-device locale** — sync `user.locale` → cookie on login.
4. **WhatsApp integration** — Meta Cloud API adapter (skeleton ready; Meta Business account needed).
5. **Automated quotations** — trigger quote generation from opportunity stage change.
6. **Agent performance scoring** — leaderboard from commission + booking counts.
7. **Convert proposal to booking guard** — add `convertedBookingId` column (needs migration).

## Planned modules

---

The longer-horizon module backlog, grouped. Some already have partial foundations (noted) — the rest are net-new.

### Accounting

- Accounting · General Ledger · Payroll — net-new finance suite.
- Invoices — △ on-demand invoice PDFs exist today; this is the move to managed, numbered, ledgered invoices.

### Marketing

- Marketing · Email campaigns — net-new (transactional email via Resend exists; campaigns do not).
- WhatsApp — △ adapter skeleton ready (open item #4); needs a Meta Business account.
- SMS — net-new channel adapter.

### Documents

- Document management — net-new (today: PDF generation + Vercel Blob uploads for supplier contracts).

### Travel products

- eVisa · Visa tracking — △ an AI visa assistant exists; these add structured application + tracking.
- Insurance — △ exists as a trip-service item type; this is a managed insurance product/flow.
- Vehicle rental · Cruises — net-new verticals (alongside flights/hotels).
- Package builder — net-new (bundle multi-service trips).

### Ecosystem

- Supplier portal — net-new (suppliers self-serve, parallel to the client portal).
- Customer loyalty · Affiliate program — net-new growth modules.

- API marketplace — net-new (expose/consume third-party integrations).

These are directional, not scheduled. Phase status and the active gap list are above; this is the "what's next after the core is hardened" backlog.

## Spec vs. reality gap tracker

The design system (principles, never-rules, page/table/entity standards, AI guardrails, automation triggers, perf budgets) describes a **target state**. This table consolidates every item where the spec is ahead of the code, so the gaps are one punch list instead of scattered "current state" notes.

Status: ✓ done · ● partial · ● not started

| Item                                                                                                            | Spec              | Status | Notes                                                                                                           |
|-----------------------------------------------------------------------------------------------------------------|-------------------|--------|-----------------------------------------------------------------------------------------------------------------|
| Agent visibility scoped to own records on list pages                                                            | security          | ✓      | Fixed — clients/bookings/opportunities/products now scope by owner column (commit 134ceb3).                     |
| Quality gate <code>npm run check</code> works (pnpm v11)                                                        | development-guide | ✓      | Fixed — config moved to <code>pnpm-workspace.yaml</code> ; 0 lint/type errors (commit 7d5f4e1).                 |
| Soft delete ( <code>deletedAt</code> , filtered reads)                                                          | database          | ●      | No soft-delete column anywhere; "never hard delete" is currently aspirational. Needs migration.                 |
| reference on all entities                                                                                       | database          | ●      | Only <code>booking + product</code> ; missing on clients/opportunities/suppliers. Needs migration.              |
| Consistent <code>updatedAt/status/notes</code> on children                                                      | database          | ●      | Present on roots, inconsistent on child tables.                                                                 |
| Page systemic-five: per-page <b>Export</b>                                                                      | ui-ux             | ●      | Endpoint exists ( <code>/api/export</code> ); list pages lack buttons.                                          |
| Page systemic-five: <b>Bulk actions</b>                                                                         | ui-ux             | ●      | No <code>Checkbox</code> primitive exists; nothing built.                                                       |
| Page systemic-five: <b>Loading</b> states                                                                       | ui-ux             | ●      | Only <code>assistant/dashboard</code> have <code>loading.tsx</code> . <code>skeleton.tsx</code> exists, unused. |
| Page systemic-five: <b>Error</b> states                                                                         | ui-ux             | ●      | Only <code>assistant</code> has <code>error.tsx</code> .                                                        |
| Page systemic-five: <b>Pagination</b>                                                                           | ui-ux             | ●      | Hardcoded <code>limit</code> (200/500) silently truncates.                                                      |
| Shared <b>DataTable</b> (sort, filter, column chooser, infinite scroll, sticky header, shortcuts, context menu) | ui-ux             | ●      | Tables are plain shadcn markup. Building this clears most of the systemic-five.                                 |



|                                                                                                          |                             |   |                                                                                          |
|----------------------------------------------------------------------------------------------------------|-----------------------------|---|------------------------------------------------------------------------------------------|
| Mobile-friendly tables ( <code>overflow-x-auto</code> )                                                  | ui-ux                       | ● | Tables overflow on phones; only <code>assistant</code> wraps for scroll.                 |
| Automation triggers (welcome email, won→proposal, accepted→booking, confirmed→invoice, completed→review) | business-rules              | ● | Only commission auto-gen on confirm/ticket fires; rest manual/on-demand.                 |
| AI mutation behind a hard confirm step                                                                   | ai                          | ● | <code>createBooking</code> is intent-gated by prompt only, not an enforced confirm gate. |
| Performance instrumentation vs budgets                                                                   | ui-ux                       | ● | Budgets defined; nothing measured.                                                       |
| Real supplier booking (Duffel orders, Hotelbeds book)                                                    | api-integrations            | ● | Search-only today (also open item #1 above).                                             |
| Cross-device locale sync                                                                                 | architecture                | ● | English until re-pick on a fresh device (also open item #3 above).                       |
| Rate limiting (auth + API)                                                                               | security                    | ● | No throttling anywhere.                                                                  |
| GDPR subject export/erasure + consent                                                                    | security                    | ● | No flow; erasure also blocked by missing soft delete.                                    |
| Disaster recovery runbook + RTO/RPO                                                                      | security                    | ● | Neon has provider backups; no documented DR plan.                                        |
| Universal audit-log coverage                                                                             | security · coding-standards | ● | <code>logActivity</code> in ~11/21 action files; not all mutations logged.               |
| Universal Zod input validation                                                                           | coding-standards            | ● | Zod in ~13/21 action files; not every action validates.                                  |
| <code>activity_log</code> partitioning + retention (500M-row target)                                     | architecture                | ● | Single unpartitioned table; no archival.                                                 |
| Search concurrency/backpressure (100 concurrent)                                                         | architecture                | ● | No queue or rate control on hotel/flight search.                                         |

## Changelog

| Commit  | Summary                                                                                                  |
|---------|----------------------------------------------------------------------------------------------------------|
| b7eab1b | Sprint 1 Wave 3: booking-service — quote→book lifecycle, idempotency, event log, supplier ref            |
| cc65ce1 | Sprint 1 Wave 2: DuffelBookingProvider, HotelbedsBookingProvider, MockBookingProvider, registry wiring   |
| bb571d2 | Sprint 1 Wave 1: booking_supplier_ref/event/document/idempotency schema + config module (migration 0019) |
| a6def26 | Phase 4: standardized BI data export (CSV + Excel) at <code>/reports</code>                              |



# Security

## Security controls

The control areas Atlas is held to, with honest current status (✓ in place · ● partial / provider-level · ○ not implemented). Details for the implemented ones follow in the sections below.

| Control            | Status | Notes                                                                                                                    |
|--------------------|--------|--------------------------------------------------------------------------------------------------------------------------|
| Tenant isolation   | ✓      | <code>agencyId</code> scoping via <code>requireAgencyUser()</code> ; verified by <code>test-tenant-isolation.ts</code> . |
| RBAC               | ✓      | 5 roles + capability helpers; agent visibility now scoped on list pages too.                                             |
| Audit logs         | ●      | <code>activity_log</code> + <code>logActivity</code> used in ~11 action files; not yet universal.                        |
| Encryption         | ●      | TLS in transit + Neon at-rest (provider); Better Auth hashes passwords. No app-level field encryption.                   |
| Password policy    | ●      | Better Auth defaults only; no enforced strength/rotation policy.                                                         |
| Secrets management | ●      | Vercel env vars; <b>demo credentials committed in the repo</b> (rotate before prod). No vault.                           |
| Backups            | ●      | Neon automatic backups + branching (provider-level); not app-managed or restore-tested.                                  |
| Rate limiting      | ●      | None on auth or API routes.                                                                                              |
| GDPR               | ●      | No subject data-export/erasure flow or consent tracking; right-to-erasure also blocked by missing soft delete.           |
| Disaster recovery  | ●      | No documented DR runbook or RTO/RPO targets.                                                                             |

The ●/● rows are tracked in the spec-vs-reality gap tracker.

## Tenant isolation

Every business table carries `agencyId` (root) or inherits it via a parent (children). **All** reads/writes are scoped by agency through `requireAgencyUser()` → `user.agencyId`. Reference numbers (BKG-..., PRD-...) are unique per agency. Cross-tenant leakage is re-verified by `scripts/test-tenant-isolation.ts` (seeds two agencies, asserts no leak, cleans up). See `architecture.md`.

## Authentication

- Better Auth (email/password). Signup is **invitation-only**: the `user.create.before` hook requires a matching pending `agency_invite`, stamps `agencyId` + `role`, and rejects everyone else — this also blocks the raw signup endpoint.

- `BETTER_AUTH_URL / baseURL + trustedOrigins` restrict trusted origins to the deployed domain.
- Invite tokens have a 7-day TTL (`src/lib/invites.ts`).

## Authorization guards

---

`src/lib/permissions.ts`: `requireUser`, `requireAgencyUser` (tenant + agency-suspension + lapsed-subscription logout), `requireManager`, `requireCapability`, `requirePlatformAdmin`. Capability rules in `business-rules.md`. Only an **admin** can assign/change the admin role.

## Platform admin & impersonation

---

- Platform admin: `isPlatformAdmin = true`, `agencyId = null`; sits above all tenants and **cannot enter a tenant except via impersonation**.
- Impersonation is **cookie-driven and platform-admin only** (`platform_view_agency / platform_view_user`), resolved in `requireUser`, with a persistent exit banner. See `architecture.md`.

## Client portal sessions

---

Passwordless magic-link sessions (`portal_session`) are scoped to a single client and **separate** from staff auth, stored in an `httpOnly` cookie (`portal-session.ts`).

## Webhooks

---

Stripe webhooks use **manual signature verification** before acting: `api/stripe/webhook` (subscriptions) and `api/stripe/connect-webhook` (Connect payments). Secrets: `STRIPE_WEBHOOK_SECRET`, `STRIPE_CONNECT_WEBHOOK_SECRET`.

## Database safety

---

`PROTECTED_DB_HOSTS` makes destructive scripts **refuse to run against a protected (prod) host** unless explicitly overridden with `ALLOW_PROD=1`. In production it is set to the prod branch host `ep-misty-thunder-aixz34vy`. Schema changes go through `db:generate` → `db:migrate`; **never** `db:push`.

## Known risks

---

- **Demo credentials in the repo.** `deployment.md` lists live demo passwords — including the platform super-admin account. They must be rotated or deleted before any real production use. See `deployment.md`.
- **Locale cookie gap** — cosmetic only; a fresh device shows English until the user re-picks (see `roadmap.md` open items).

# Analytics & BI

---

Decision-oriented dashboards plus standardized data export, all currency-safe.

## Dashboards

---

Surfaced on the agency dashboard and `/finance`:

- Revenue evolution (monthly)
- Top destinations / clients / markets by revenue
- Revenue per agent
- Conversion rate, average booking value, MoM growth
- Pipeline funnel **by value** + weighted forecast / win rate
- AR aging buckets
- Margin %
- Commission-by-supplier

## Currency-safe by design

---

No FX conversion (see `business-rules.md`). Every metric **groups by currency** rather than summing across, with a **DZD headline**.

`src/lib/analytics.ts`

---

Pure helpers (no I/O), unit-testable:

- `sumByCurrency` — totals grouped by currency code
- `monthlyBuckets` — time-series bucketing
- `topN / countBy` — rankings and tallies
- `conversionRate, growthPct` — rate/growth metrics
- `agingBuckets` — AR aging

Charts render via `src/components/charts/` (recharts 3, tokenized), including `HBarInsight` and `FunnelInsight`.

## BI export (`/reports`)

---

Gated by `canViewFinance` (admin/manager/finance). One-click CSV + Excel export of every core dataset, plus a full multi-sheet workbook.

- `src/lib/export/` — `csv.ts` (RFC-4180 + UTF-8 BOM, Power BI friendly), `xlsx.ts` (exceljs workbook), `datasets.ts` (BI dataset registry).
- Each enum field emits **both** `*_code` and `*_label` columns; amounts in DZD, dates in ISO; a date-range filter applies.

- Served by `api/export/[entity]` — CSV/XLSX per dataset, or `workbook` = all sheets.

# AI Features

---

Built on the **Vercel AI SDK + OpenRouter** (`OPENROUTER_API_KEY`). All features degrade/no-op gracefully when the key is unset. See `api-integrations.md`.

## AI must never

---

Hard guardrails for every AI feature. These extend the never rules to the AI surface:

- **Never invent prices** — quote only live supplier rates or stored values, never a guessed number.
- **Never invent bookings** — only create a booking when explicitly asked.
- **Never invent clients** — never fabricate a client record.
- **Never invent payments** — never record or imply a payment that didn't happen.
- **Never modify data without confirmation** — mutations are explicit and reviewable, not silent side effects of a chat reply.
- **Never hide errors** — surface failures to the user; don't paper over a failed tool call with a plausible answer.
- **Never perform destructive actions silently** — no deletes/overwrites without a clear, confirmed user action.

**Current state:** all assistant tools are agency-scoped. Of them only `createBooking` mutates — it creates a **draft** booking (zero balance, unconfirmed), and the system prompt forbids inventing supplier confirmations or PNRs and restricts creation to explicit requests. Everything else (`searchFlights`, `searchHotels`, `findClients`, `bookingsSummary`) is read-only. Gap: creation is intent-gated by the prompt, not yet behind a hard confirm step — the "modify without confirmation" guard is a convention, not an enforced gate.

## AI assistant (`/assistant`)

---

Chat backed by `api/chat` with **agency-scoped tools**: find clients, bookings summary, create booking, search flights/hotels. Tools respect tenant scoping like every other action.

## Inline features

---

Embedded where the work happens, not in a separate AI section:

- **AI itinerary generation** — one-click generation from booking items; saves to `booking_day` rows. Shareable via `/i/[token]`.
- **AI quote builder** — natural-language brief → structured proposal line items, pre-filled in the new-proposal form.
- **AI email drafting** — generate subject + body for confirmation, voucher, follow-up, or custom emails from the booking messages panel.
- **AI visa assistant** — per-nationality visa requirement summary from traveller passport

nationalities + booking destination.

## Implementation

---

- Server actions in `src/lib/actions/ai.ts`.
- Itinerary helpers in `src/lib/itinerary.ts`.
- The assistant's tool definitions are agency-scoped wrappers over existing server actions, so AI cannot bypass security.md tenant isolation.



# Agentic Coding Starter Kit

---

A production-oriented starter kit for building AI-powered web apps with an agentic development workflow. It gives you a working Next.js app, authentication, PostgreSQL, Drizzle ORM, AI SDK integration, shadcn/ui components, and project instructions that help coding agents plan, split, implement, review, and verify changes.

The goal is simple: install the starter, describe the product you want to build, and let your coding agent help turn the boilerplate into your actual POC, MVP, or internal tool.

## What You Get

---

- **Next.js 16 and React 19** with the App Router
- **TypeScript** and a strict project setup
- **Better Auth** with email/password enabled by default
- **PostgreSQL and Drizzle ORM** for schema and migrations
- **AI SDK and OpenRouter** for chat and AI features
- **shadcn/ui, Tailwind CSS, and Lucide icons** for the UI foundation
- **Local or Vercel Blob file storage** through one storage abstraction
- **Agent instructions** through `AGENTS.md` and `CLAUDE.md`
- **Agent skills** for specs, implementation, reviews, security scans, UI work, and shipping

## Quick Start

---

Create a new app with the CLI:

```
npx create-agentic-app@latest my-app
cd my-app
```

Or create the app in the current directory:

```
npx create-agentic-app@latest .
```

Then configure and run the app:

```
cp env.example .env
docker compose up -d
pnpm db:migrate
pnpm dev
```

Open <http://localhost:3000>.

The CLI copies the starter files, installs dependencies with your selected package manager, and prepares the environment file. If you use `npm`, replace the `pnpm` commands above with `npm run`.

## Guided Setup with Claude Code (Optional)

---

If you use Claude Code, you can install the `create-agentic-app` skill and have Claude walk you through the entire setup — folder strategy, package manager, PostgreSQL (Docker / Neon / Vercel / BYO), `.env` config, migrations, optional integrations (OpenRouter, Vercel Blob, Polar, email), build verification, and dev-server check — ending at a verified `http://localhost:3000`.

Install the skill:

```
npx skills add leonvanzyl/agentic-coding-starter-kit@create-agentic-app --agent clau
```

The `--agent claude-code` flag is required. Without it, the installer drops the skill into 37+ IDE adapter folders at the project root.

Once installed, ask Claude something like:

```
Scaffold a new Agentic Coding Starter Kit project here.
```

Claude will run the skill end-to-end and ask you the few decisions it actually needs to make.

## Prerequisites

---

- Node.js 18 or newer
- Git
- PostgreSQL, either through the included Docker Compose file or a hosted provider
- A package manager: `pnpm`, `npm`, or `yarn`
- Optional: an OpenRouter API key for AI chat features
- Optional: a Vercel account for deployment, hosted Postgres, and Blob storage

## Environment Variables

---

Start from `env.example` and update values for your environment:

```
# Database
POSTGRES_URL=postgresql://dev_user:dev_password@localhost:5432/postgres_dev

# Authentication - Better Auth
BETTER_AUTH_SECRET=your-random-secret

# AI Integration via OpenRouter
OPENROUTER_API_KEY=
OPENROUTER_MODEL="openai/gpt-5-mini"

# Optional - for vector search only
OPENAI_EMBEDDING_MODEL="text-embedding-3-large"

# App URL
NEXT_PUBLIC_APP_URL="http://localhost:3000"
```

```
# File storage
BLOB_READ_WRITE_TOKEN=

# Polar payment processing
POLAR_WEBHOOK_SECRET=polar_
POLAR_ACCESS_TOKEN=polar_
```

For local development, the default database URL works with the included `docker-compose.yml`. For production, use the database URL from your hosting provider.

Generate a strong `BETTER_AUTH_SECRET` before deploying. The starter ships with a development value only so you can get moving quickly.

## Default Auth

---

The starter now defaults to **email and password authentication** through Better Auth. This keeps the first setup small and helps you start building POCs and MVPs without creating OAuth credentials up front.

The current auth setup includes:

- user registration
- email/password login
- protected routes
- password reset flow
- email verification flow

In development, verification and password reset links are logged to the terminal instead of being sent through an email provider. When you are ready for production, ask your coding agent to connect an email service and update the Better Auth email callbacks.

## Adding Google OAuth

Google OAuth is no longer the default, but adding it back is straightforward. Ask your coding agent:

```
Add Google OAuth to this Better Auth setup. Keep email/password login enabled, add t
```

Your agent should update the Better Auth config, add the required `GOOGLE_CLIENT_ID` and `GOOGLE_CLIENT_SECRET` variables, and adjust the login UI.

## Build With an Agent

---

This starter is designed to be used with coding agents. The generated project includes instructions that tell agents how to plan, ask questions, split work, use sub-agents when useful, follow the design system, and verify changes.

- `AGENTS.md` is the main instruction file for Codex, Cursor, and other agent-compatible tools.

- `CLAUDE.md` points Claude users to the same project guidance.
- `.agents/skills/` and `.claude/skills/` include optional workflows for more specialized tasks.
- `DESIGN.md` defines the UI design system agents should follow.

The default workflow does not require slash commands or a separate spec file.

## Recommended Default Workflow

1. Install the starter and open the project in your coding-agent environment.
2. Switch your agent tool to planning mode.
3. Describe the app you want to build in plain language.
4. Let the agent ask clarifying questions and shape a clear plan.
5. Confirm the plan once the goal, scope, constraints, and success criteria are clear.
6. Switch your agent tool to edit mode.
7. Ask the agent to implement the approved plan.
8. The main agent should split the work into parallel streams, silos, or feature chunks that fit within context.
9. The agent should use sub-agents to implement those chunks in parallel where useful, then coordinate the results.
10. The agent should run quality checks such as lint, typecheck, and build.
11. Review the result in the browser and iterate.

You do not need a special command for this default workflow. The project instructions already tell the agent how to plan, split implementation work, use sub-agents, and verify the result.

## Starter Prompt

---

Use this as a first message to your coding agent after installing the starter:

```
I am using the Agentic Coding Starter Kit. Treat the existing app as boilerplate tha

Use the project instructions in AGENTS.md or CLAUDE.md. During planning, ask clarify

What I want to build:
[Describe your app here]
```

For example:

```
What I want to build:
A lightweight CRM for solo consultants. It should let users manage clients, track de
```

## When to Use Specs

---

For most POCs and MVPs, the normal agent workflow is enough. Use a spec when the feature is large, long-running, risky, or needs to be split across multiple implementation sessions.

The starter includes two skills for that workflow:

- `create-spec`: turns a planning conversation into `specs/{feature}/` with requirements, task files, dependency waves, and manual action notes.
- `implement-feature`: reads a spec folder and coordinates implementation wave by wave with review gates.

Use this workflow when:

- the feature spans many files or modules
- multiple agents should work in parallel
- the implementation may take more than one session
- you need resumable progress tracking
- you want a written implementation record before coding starts

Example agent request:

```
Create a spec for the billing and subscriptions feature we just planned. Break it in
```

Then:

```
Implement the billing and subscriptions spec from specs/billing-subscriptions.
```

## Project Structure

---

```
src/
├── app/
│   ├── (auth)/
│   │   ├── forgot-password/
│   │   ├── login/
│   │   ├── register/
│   │   └── reset-password/
│   ├── api/
│   │   ├── auth/
│   │   ├── chat/
│   │   └── diagnostics/
│   ├── chat/
│   ├── dashboard/
│   ├── profile/
│   ├── layout.tsx
│   └── page.tsx
├── components/
│   ├── auth/
│   ├── ui/
│   ├── site-footer.tsx
│   └── site-header.tsx
├── hooks/
└── lib/
    ├── auth.ts
    ├── auth-client.ts
    ├── db.ts
    ├── env.ts
    └── schema.ts
```

```
|— session.ts
|— storage.ts
└— utils.ts
```

Important root files:

- `AGENTS.md`: coding-agent behavior rules
- `CLAUDE.md`: Claude entrypoint for the same guidance
- `DESIGN.md`: UI design system and component guidance
- `drizzle.config.ts`: Drizzle migration configuration
- `docker-compose.yml`: local PostgreSQL service
- `env.example`: environment variable template
- `components.json`: shadcn/ui configuration

## Available Scripts

---

```
pnpm dev          # Start the development server with Turbopack
pnpm build         # Run migrations, then build for production
pnpm build:ci      # Build without running migrations
pnpm start        # Start the production server
pnpm lint          # Run ESLint
pnpm typecheck     # Run TypeScript without emitting files
pnpm check         # Run lint and typecheck
pnpm format        # Format the repository
pnpm format:check  # Check formatting
pnpm setup         # Run the setup script
pnpm db:generate   # Generate Drizzle migrations
pnpm db:migrate    # Run Drizzle migrations
pnpm db:studio     # Open Drizzle Studio
```

The repository also contains Drizzle push/reset helper scripts for local experimentation. For schema changes you intend to keep, prefer:

```
pnpm db:generate
pnpm db:migrate
```

Do not use schema push as a replacement for migrations in real project work.

## Database Workflow

---

For local development:

```
docker compose up -d
pnpm db:migrate
```

When your app needs schema changes, ask your agent to update `src/lib/schema.ts`, generate a migration, and run it:

```
pnpm db:generate
pnpm db:migrate
```

If you deploy to Vercel or another hosted environment, set `POSTGRES_URL` in that environment before running migrations or building the app.

## AI Features

---

The starter uses the Vercel AI SDK with OpenRouter. Set these variables to enable AI chat:

```
OPENROUTER_API_KEY=sk-or-v1-your-key
OPENROUTER_MODEL="openai/gpt-5-mini"
```

OpenRouter lets you switch models without changing the application code. Update `OPENROUTER_MODEL` when you want to try a different model.

## File Storage

---

The starter includes a storage abstraction that can use local storage in development or Vercel Blob in production.

For local development, leave `BLOB_READ_WRITE_TOKEN` empty. Files are stored under `public/uploads/`.

For Vercel Blob:

1. Create a Blob store in Vercel.
2. Copy the `BLOB_READ_WRITE_TOKEN`.
3. Add it to your production environment variables.

The app chooses the storage backend based on whether `BLOB_READ_WRITE_TOKEN` is configured.

## Deployment

---

Vercel is the recommended deployment target.

```
npm install -g vercel
vercel --prod
```

Set the required production environment variables:

- `POSTGRES_URL`
- `BETTER_AUTH_SECRET`
- `NEXT_PUBLIC_APP_URL`
- `OPENROUTER_API_KEY`, if using AI features
- `OPENROUTER_MODEL`, if using AI features

- `BLOB_READ_WRITE_TOKEN`, if using Vercel Blob
- `POLAR_WEBHOOK_SECRET` and `POLAR_ACCESS_TOKEN`, if using Polar payments

The default `pnpm build` script runs database migrations before `next build`. If your CI or host should not run migrations during build, use `pnpm build:ci` and run migrations as a separate deployment step.

## Troubleshooting

---

### The app cannot connect to Postgres

Confirm Docker is running and start the database:

```
docker compose up -d
```

Then check that `POSTGRES_URL` in `.env` matches the database connection string.

### Auth reset or verification emails are not arriving

In development, links are logged to the terminal. This is intentional. Connect an email provider before using password reset or verification in production.

### AI chat is not working

Set `OPENROUTER_API_KEY` and restart the dev server. Also confirm `OPENROUTER_MODEL` is a model available to your OpenRouter account.

### My agent is preserving too much boilerplate

Tell the agent directly that the starter UI is scaffolding and should be replaced:

```
Replace the starter UI with the actual product UI. Do not keep setup checklists, pla
```

### I need Google login

Ask your agent to add Google OAuth through Better Auth while keeping email/password enabled. You will need Google OAuth credentials and production callback URLs.

## Video Tutorial

---

Watch the original walkthrough:

[Agentic Coding Boilerplate Tutorial](#)

Some details in older videos may differ from the current starter. This README is the source of truth for the current default workflow.



## Support This Project

---

If this starter kit helped you build something useful, you can support the project here:

[Buy me a coffee](#)

## Contributing

---

1. Fork this repository.
2. Create a feature branch.
3. Make your changes.
4. Run the relevant checks.
5. Open a pull request.

## License

---

This project is licensed under the MIT License.

## Need Help?

---

- Check the repository issues: [github.com/leonvanzyl/agent-coding-starter-kit/issues](https://github.com/leonvanzyl/agent-coding-starter-kit/issues)
- Review `AGENTS.md`, `CLAUDE.md`, and `DESIGN.md`
- Open a new issue with the exact setup steps, error output, and environment details

# Atlas — Travel Agency SaaS

A multi-tenant SaaS for travel agencies: each agency runs its bookings, clients, proposals and finance in an isolated workspace, while the vendor manages all agencies from a platform console. Built on the Agentic Coding Starter Kit.

**Live:** <https://agencetool.vercel.app>

## Tech stack

- **Framework:** Next.js 16 (App Router) + React 19 + TypeScript
- **Styling:** Tailwind CSS v4 (CSS-first, `@theme inline`) + shadcn/ui (new-york)
- **Auth:** Better Auth (email/password, invitation-gated signup)
- **DB:** PostgreSQL (Neon) + Drizzle ORM
- **i18n:** next-intl (EN / FR / AR, cookie-based, RTL)
- **Charts:** recharts (tokenized to the design system)
- **AI:** Vercel AI SDK + OpenRouter (assistant — needs `OPENROUTER_API_KEY`)
- **Hosting:** Vercel (GitHub auto-deploy on `main`)

See `DESIGN.md` for the design system and `AGENTS.md/CLAUDE.md` for working conventions.

## Architecture

### Multi-tenancy

Every business table carries an `agencyId` and **all** queries are scoped by it, so agencies never see each other's data. References (`BKG-...`, `PRD-...`) are unique **per agency**. Tenancy is enforced at the data layer and re-verified by a test harness (`scripts/test-tenant-isolation.ts`).

Tenant-root tables: `agency`, `client`, `opportunity`, `product`, `booking`, `notification`, `activity_log`, `agency_invite`, plus `user.agencyId`. Child tables (`travellers`, `items`, `payments`, `days`, `contacts`) inherit tenancy through their parent.

### Roles (RBAC)

Five roles, defined in `src/lib/domain.ts` with capability helpers:

| Role    | Sees all agency data | Manages team | Manages payments | Deletes records | Home       |
|---------|----------------------|--------------|------------------|-----------------|------------|
| admin   | ✓                    | ✓            | ✓                | ✓               | /dashboard |
| manager | ✓                    | ✓            | ✓                | ✓               | /dashboard |

|         |               |   |   |   |                        |
|---------|---------------|---|---|---|------------------------|
| finance | ✓             | — | ✓ | — | /finance               |
| support | ✓             | — | — | — | /support               |
| agent   | own work only | — | — | — | /dashboard<br>(scoped) |

Helpers: `seesAllData`, `canManageTeam`, `canAssignAdmin`, `canManagePayments`, `canViewFinance`, `canViewSupport`, `canDeleteRecords`, `roleHome`.

## Auth & onboarding

- **Invitation-only signup:** the Better Auth create hook (`src/lib/auth.ts`) rejects any signup without a matching pending invite and stamps `agencyId` + role from it. Accept flow lives at `/invite/[token]`.
- **Guards** (`src/lib/permissions.ts`): `requireUser`, `requireAgencyUser` (tenant + suspension lockout), `requireManager`, `requireCapability`, `requirePlatformAdmin`.

## Platform (vendor) console — `/platform`

The vendor is a user with `isPlatformAdmin = true` and `agencyId = null`, sitting above all tenants. From `/platform` they can:

- Create agencies (generates the first-admin invite link)
- Suspend / reactivate agencies (suspended → users locked out)
- **View as agency** (act as agency admin) or **View as user** (act as a specific user with *their* role + scoped data) — with an exit banner.

Impersonation is cookie-driven (`platform_view_agency` / `platform_view_user`) and only takes effect for a platform admin (resolved in `requireUser`).

## Per-role workspaces

- `/dashboard` — agency overview (admin/manager) or "Your work" (agent); manager view includes analytics charts (bookings by country, team performance, status, monthly trend) + finance KPIs.
- `/finance` — payments / accounts-receivable + revenue, with charts.
- `/support` — action queue (bookings needing attention) + clients + ops.
- `/team`, `/clients`, `/bookings`, `/opportunities`, `/products`, `/operations`, `/search`, `/assistant`, `/settings`, `/profile`.

## i18n

- English / French / Arabic, **cookie-based** (no URL-locale routing — routes unchanged). Arabic is full **RTL** with **IBM Plex Sans Arabic**.
- Config: `src/i18n/config.ts` + `src/i18n/request.ts`; messages in `messages/{en,fr,ar}.json`. Change language in **Settings**.
- Core surfaces (nav, login, dashboard, settings) are translated; deeper pages fall back to English. To translate more, add keys to **all three** message files and swap strings to `t("...")`.

---

## Local development

---

```
npm install --legacy-peer-deps      # better-auth peer range needs this
npm run dev                          # http://localhost:3000
npm run check                        # lint + typecheck
npm run build:ci                     # next build (no migrate)
```

Required env (.env): POSTGRES\_URL, BETTER\_AUTH\_SECRET. Optional (features off until set): OPENROUTER\_API\_KEY (AI chat), BLOB\_READ\_WRITE\_TOKEN (uploads), Google OAuth.

## Database

```
npm run db:generate    # after editing src/lib/schema.ts
npm run db:migrate     # apply migrations (NEVER db:push)
npm run db:studio
```

Migrations in drizzle/. Notable: 0006 (tenancy + backfill), 0007 (invites), 0008 (user.locale).

## Scripts

```
# Promote an existing account to platform super-admin (vendor)
npx tsx --env-file=.env scripts/make-platform-admin.ts <email>

# Seed / reset the Demo Agency with rich demo data (idempotent)
npx tsx --env-file=.env scripts/seed-demo-data.ts

# Verify cross-tenant isolation (seeds 2 agencies, asserts no leak)
npx tsx --env-file=.env scripts/test-tenant-isolation.ts
```

---

## Deployment

---

Hosted on **Vercel** (atlasproject/agence\_tool), connected to GitHub (ouks1993/agence\_tool).

- **Auto-deploy:** git push origin main builds & ships production.
- **Manual:** npx vercel deploy --prod --yes
- **Config in vercel.json** (buildCommand: npm run build:ci) + .npmrc (legacy-peer-deps=true).
- **Env vars on Vercel:** POSTGRES\_URL, BETTER\_AUTH\_SECRET, NEXT\_PUBLIC\_APP\_URL, BETTER\_AUTH\_URL (the last fixes INVALID\_ORIGIN on the deployed domain).

⚠ Production and local dev currently **share one Neon database**. Split prod onto its own Neon branch before onboarding real customers.

---

## Demo

The **Demo Agency** is seeded for customer demos (re-run the seed script to reset). Accounts (Demo Agency unless noted):

| Role                    | Email                                           |
|-------------------------|-------------------------------------------------|
| Platform admin (vendor) | ouksili.abdelmalek@gmail.com                    |
| Manager                 | yasmine@agence.test                             |
| Finance                 | finance@demo.test                               |
| Support                 | support@demo.test                               |
| Agents                  | karim@demo.test, lina@demo.test, omar@demo.test |

Demo credentials are throwaway and set by the seed/admin scripts — **rotate or remove them before any real production use**. (Agent accounts use the seed default; reset any password via the scripts above.)

**Suggested demo flow:** sign in as the vendor → `/platform` → create an agency (or *View as Demo Agency*) → tour the manager dashboard charts → switch logins to show the finance, support and agent views → open Settings and switch to Français or العربية (RTL).

## Roadmap / open items

- **Split prod database** onto a dedicated Neon branch (highest priority).
- **Translate deeper pages** (bookings, clients, finance, support, platform).
- **Cross-device locale:** apply a user's saved `locale` on a fresh device (`sync user.locale` → cookie on login).
- **Email delivery** for invite links (currently copy-paste; module logs to console).
- **Billing / subscriptions** (per-agency plans).
- **Traveler portal** (end-customer login to view their trips).

# CRITICAL RULES - MUST FOLLOW

---

## RESPONSES

---

- Keep responses concise and to the point - unless the user asks otherwise

## PLANNING MODE

---

- Always ask clarifying questions
- Never assume design, tech stack or features
- Use deep-dive sub-agents to assist with research
- Use deep-dive sub-agents to review the different aspects of your plan before presenting to the user

## CHANGE / EDIT MODE

---

- Never implement features yourself when possible - use sub-agents!
- Identify changes from the plan that can be implemented in parallel, and use sub-agents to implement the features efficiently
- When using sub-agents to implement features, act as a coordinator only
- Use the best model for the task - premium models for complex tasks (like coding) and mid-tier models for simpler tasks, like documentation
- After completing features (large or small), always run commands like lint, type check and next build to check code quality

## DATABASE SCHEMA CHANGES

---

- Whenever you make changes to the database schema, ALWAYS run the drizzle generate and migrate commands
- NEVER run drizzle push!
- For all ID columns NOT related to BetterAuth, use UUID for the ID columns and be randomly generated

## TESTING

---

- Use any testing tools, libraries available to the project for testing your changes
- Never assume your changes simply work, always test!
- If the project does not have any testing tools, scripts, MCP tools, skills, etc. available for testing, ask the user whether testing should be skipped.

## UI DESIGN

---

- Always follow the UI design system when creating or reviewing components or pages.
- Design System: @DESIGN.md

## FEATURE FILTER RULE

---

Before any feature is committed to the roadmap, it must satisfy at least one of:

1. **Reduces time to booking** — the agent completes the booking workflow faster
2. **Increases booking conversion** — more proposals become bookings, more searches become proposals
3. **Increases agency retention** — the agency becomes more operationally dependent on Atlas

If a proposed feature cannot clearly satisfy at least one criterion, it does not move forward. Apply this rule before design, before estimation, before any engineering discussion.

Examples:

- Supplier booking ✓ (reduces time to booking + increases conversion)
- Client portal depth ✓ (increases conversion + increases retention)
- Marketing campaigns module ✗ (does not directly satisfy any criterion — do not build)
- WhatsApp messaging platform ✗ (build as output channel/integration, not a messaging system)
- Accounting workflows (invoice generation + export) ✓ (reduces time to booking)
- Full GL accounting module ✗ (does not satisfy any criterion — partner instead)

## PRE-COMMIT DOCUMENTATION RULE

---

Before every commit, you MUST:

1. **Review all modified files** — understand the full scope of what changed.
2. **Update affected documentation** — if the change touches behaviour, architecture, business rules, APIs, DB schema, UI patterns, or config, update the relevant file(s) in `docs/`, `atlas.md`, `DESIGN.md`, or `AGENTS.md` to stay in sync.
3. **Ensure no contradictions** — the docs must not describe something that the code no longer does. If they conflict, fix the docs before committing.
4. **Record architectural/business decisions** — if the change introduces a new architectural pattern, a significant tech choice, or a business-rule change, create a record under `docs/decisions/` using the template at `docs/decisions/_template.md`.
5. **Summarize doc updates in the commit message** — include a "Docs:" line listing which documentation files were updated and why.

Example commit message footer:

```
Docs: updated docs/business-rules.md (new booking prerequisite),  
      docs/decisions/0001-soft-delete-strategy.md (new decision record)
```

# Design System

This document defines the visual design system for the project. All new components and pages **must** follow these tokens, patterns, and conventions.

## Stack

- **Framework:** Next.js (App Router) + React + TypeScript
- **Styling:** Tailwind CSS v4 (CSS-first config via `@theme inline` in `globals.css` — no `tailwind.config.ts`)
- **Components:** shadcn/ui (new-york style, neutral base)
- **Icons:** Lucide React
- **Fonts:** Geist (sans) + Geist Mono (mono) via `next/font/google`
- **Dark mode:** next-themes (class-based, system default)
- **Utilities:** `cn()` from `@/lib/utils` (clsx + tailwind-merge)

## Colors

All values use the **oklch** color space. Colors are defined as CSS custom properties in `globals.css` and bridged to Tailwind via `@theme inline`.

## Semantic Tokens

| Token                | Light                                   | Dark                                    | Usage                        |
|----------------------|-----------------------------------------|-----------------------------------------|------------------------------|
| background           | <code>oklch(1 0 0)</code>               | <code>oklch(0.141 0.005 285.823)</code> | Page background              |
| foreground           | <code>oklch(0.141 0.005 285.823)</code> | <code>oklch(0.985 0 0)</code>           | Primary text                 |
| primary              | <code>oklch(0.21 0.034 270)</code>      | <code>oklch(0.92 0.02 270)</code>       | Buttons, links, accents      |
| primary-foreground   | <code>oklch(0.985 0 0)</code>           | <code>oklch(0.21 0.006 285.885)</code>  | Text on primary              |
| secondary            | <code>oklch(0.967 0.001 286.375)</code> | <code>oklch(0.274 0.006 286.033)</code> | Secondary buttons, subtle bg |
| secondary-foreground | <code>oklch(0.21 0.006 285.885)</code>  | <code>oklch(0.985 0 0)</code>           | Text on secondary            |
| muted                | <code>oklch(0.967 0.001 286.375)</code> | <code>oklch(0.274 0.006 286.033)</code> | Subdued backgrounds          |



|                    |                            |                            |                               |
|--------------------|----------------------------|----------------------------|-------------------------------|
| muted-foreground   | oklch(0.552 0.016 285.938) | oklch(0.705 0.015 286.067) | Subdued text, placeholders    |
| accent             | oklch(0.96 0.012 270)      | oklch(0.28 0.018 270)      | Hover backgrounds, highlights |
| accent-foreground  | oklch(0.21 0.006 285.885)  | oklch(0.985 0 0)           | Text on accent                |
| destructive        | oklch(0.577 0.245 27.325)  | oklch(0.704 0.191 22.216)  | Error states, delete actions  |
| card               | oklch(1 0 0)               | oklch(0.21 0.006 285.885)  | Card backgrounds              |
| card-foreground    | oklch(0.141 0.005 285.823) | oklch(0.985 0 0)           | Card text                     |
| popover            | oklch(1 0 0)               | oklch(0.21 0.006 285.885)  | Popover/dropdown bg           |
| popover-foreground | oklch(0.141 0.005 285.823) | oklch(0.985 0 0)           | Popover/dropdown text         |
| border             | oklch(0.92 0.004 286.32)   | oklch(1 0 0 / 10%)         | Borders, dividers             |
| input              | oklch(0.92 0.004 286.32)   | oklch(1 0 0 / 15%)         | Input borders                 |
| ring               | oklch(0.705 0.06 270)      | oklch(0.552 0.05 270)      | Focus rings                   |

## Chart Colors

| Token   | Light                     | Dark                       |  |
|---------|---------------------------|----------------------------|--|
| chart-1 | oklch(0.646 0.222 41.116) | oklch(0.488 0.243 264.376) |  |
| chart-2 | oklch(0.6 0.118 184.704)  | oklch(0.696 0.17 162.48)   |  |
| chart-3 | oklch(0.398 0.07 227.392) | oklch(0.769 0.188 70.08)   |  |
| chart-4 | oklch(0.828 0.189 84.429) | oklch(0.627 0.265 303.9)   |  |
| chart-5 | oklch(0.769 0.188 70.08)  | oklch(0.645 0.246 16.439)  |  |

## Sidebar Colors

| Token              | Light                      | Dark                      |
|--------------------|----------------------------|---------------------------|
| sidebar            | oklch(0.985 0 0)           | oklch(0.21 0.006 285.885) |
| sidebar-foreground | oklch(0.141 0.005 285.823) | oklch(0.985 0 0)          |

|                            |                            |                            |
|----------------------------|----------------------------|----------------------------|
| sidebar-primary            | oklch(0.21 0.006 285.885)  | oklch(0.488 0.243 264.376) |
| sidebar-primary-foreground | oklch(0.985 0 0)           | oklch(0.985 0 0)           |
| sidebar-accent             | oklch(0.967 0.001 286.375) | oklch(0.274 0.006 286.033) |
| sidebar-accent-foreground  | oklch(0.21 0.006 285.885)  | oklch(0.985 0 0)           |
| sidebar-border             | oklch(0.92 0.004 286.32)   | oklch(1 0 0 / 10%)         |
| sidebar-ring               | oklch(0.705 0.015 286.067) | oklch(0.552 0.016 285.938) |

## Ad-hoc Status Colors

Use these Tailwind utilities for status indicators — they are not part of the token system but are used consistently:

- **Success:** `text-green-600 / dark:text-green-400, bg-green-500`
- **Error:** `text-red-600, text-destructive`

## Typography

### Font Families

| Token                          | Font       | Usage                         |
|--------------------------------|------------|-------------------------------|
| <code>--font-geist-sans</code> | Geist      | All UI text (applied to body) |
| <code>--font-geist-mono</code> | Geist Mono | Code, monospace content       |

Body has `font-feature-settings: "rlig" 1, "calt" 1` and antialiased enabled.

### Type Scale

| Class                  | Size | Usage                                              |
|------------------------|------|----------------------------------------------------|
| <code>text-xs</code>   | 12px | Timestamps, shortcuts, helper text, code           |
| <code>text-sm</code>   | 14px | Descriptions, labels, body copy, card descriptions |
| <code>text-base</code> | 16px | Base text, inputs (mobile)                         |
| <code>text-lg</code>   | 18px | Dialog titles, sub-headings                        |
| <code>text-xl</code>   | 20px | Section titles, header logo                        |
| <code>text-2xl</code>  | 24px | Page titles, card titles                           |

|                       |      |                            |
|-----------------------|------|----------------------------|
| <code>text-3xl</code> | 30px | Dashboard/profile headings |
| <code>text-4xl</code> | 36px | Large display text         |
| <code>text-5xl</code> | 48px | Hero title                 |

## Font Weights

| Class                      | Weight | Usage                                                |
|----------------------------|--------|------------------------------------------------------|
| <code>font-medium</code>   | 500    | Buttons, labels, nav items                           |
| <code>font-semibold</code> | 600    | Card titles, section headings, badges, dialog titles |
| <code>font-bold</code>     | 700    | Page titles, hero heading                            |

## Line Heights & Tracking

| Class                        | Usage                 |
|------------------------------|-----------------------|
| <code>leading-none</code>    | Labels, card titles   |
| <code>leading-5</code>       | Code blocks           |
| <code>leading-6</code>       | List items            |
| <code>leading-7</code>       | Paragraphs (markdown) |
| <code>tracking-tight</code>  | Hero/display text     |
| <code>tracking-widest</code> | Keyboard shortcuts    |

## Spacing

### Container Pattern

|                                     |
|-------------------------------------|
| <code>container mx-auto px-4</code> |
|-------------------------------------|

Responsive overrides where needed:

- Header: `px-3 sm:px-4`
- Footer: `px-4 sm:px-6 lg:px-8`

### Max Widths

| Class                 | Value | Usage                             |
|-----------------------|-------|-----------------------------------|
| <code>max-w-sm</code> | 24rem | Auth forms                        |
| <code>max-w-md</code> | 28rem | Login/register cards, error pages |

|           |       |                         |
|-----------|-------|-------------------------|
| max-w-1g  | 32rem | Dialog content (sm+)    |
| max-w-2x1 | 42rem | Large dialogs           |
| max-w-3x1 | 48rem | Embeds, protected state |
| max-w-4x1 | 56rem | Main content pages      |

## Vertical Spacing (space-y)

| Class       | Usage                           |
|-------------|---------------------------------|
| space-y-1   | Tight lists, inline stacks      |
| space-y-1.5 | Card header                     |
| space-y-2   | Form field groups, small stacks |
| space-y-3   | Footer stacks                   |
| space-y-4   | Form sections, dialog content   |
| space-y-6   | Card content sections           |
| space-y-8   | Page-level sections             |

## Padding

| Class | Usage                                  |
|-------|----------------------------------------|
| p-1   | Dropdown content, icon buttons         |
| p-2   | Code blocks, muted backgrounds         |
| p-3   | Chat bubbles, inputs                   |
| p-4   | Grid items, action buttons, list items |
| p-6   | Cards, dialog content                  |

## Page Vertical Padding

| Class        | Usage                  |
|--------------|------------------------|
| py-3 sm:py-4 | Header                 |
| py-4 sm:py-6 | Footer                 |
| py-8         | Standard content pages |
| py-12        | Home page, dashboard   |
| py-16        | Error/not-found pages  |

# Border Radius

| Token       | Value                       | Class        |
|-------------|-----------------------------|--------------|
| --radius    | 0.625rem (10px)             | Base         |
| --radius-sm | calc(--radius - 4px) = 6px  | rounded-sm   |
| --radius-md | calc(--radius - 2px) = 8px  | rounded-md   |
| --radius-lg | var(--radius) = 10px        | rounded-lg   |
| --radius-xl | calc(--radius + 4px) = 14px | rounded-xl   |
| —           | 9999px                      | rounded-full |

## Usage:

- rounded-md — Buttons, inputs, textarea, code blocks, dropdowns
- rounded-lg — Cards, dialogs, feature cards, chat bubbles
- rounded-xl — Hero logo container
- rounded-full — Badges, avatars

# Shadows

| Class     | Usage                                       |
|-----------|---------------------------------------------|
| shadow-xs | Inputs, textarea, secondary/outline buttons |
| shadow-sm | Card base                                   |
| shadow-md | Card hover, dropdown content                |
| shadow-lg | Dialogs, dropdown sub-content               |

No custom shadow definitions — all Tailwind defaults.

# Animations

## Custom Keyframes

| Name     | Effect                              | Duration | Easing   |
|----------|-------------------------------------|----------|----------|
| fade-in  | Opacity 0 → 1                       | 0.3s     | ease-out |
| fade-up  | Opacity 0 → 1 + translateY(8px → 0) | 0.4s     | ease-out |
| scale-in | Opacity 0 → 1 + scale(0.97 → 1)     | 0.2s     | ease-out |

Use via: animate-fade-in, animate-fade-up, animate-scale-in

## tw-animate-css Animations

Used on dialogs and dropdowns:

- `animate-in / animate-out`
- `fade-in-0 / fade-out-0`
- `zoom-in-95 / zoom-out-95`
- `slide-in-from-{top|bottom|left|right}-2`

## Transition Classes

| Class                                      | Usage                           |
|--------------------------------------------|---------------------------------|
| <code>transition-colors</code>             | Links, hover color changes      |
| <code>transition-opacity</code>            | Avatar hover, reveal-on-hover   |
| <code>transition-all duration-200</code>   | Card interactive hover, buttons |
| <code>transition-[color,box-shadow]</code> | Input/textarea focus            |

## Utility Classes

```
.card-interactive {
  @apply transition-all duration-200 ease-out;
}
.card-interactive:hover {
  @apply shadow-md -translate-y-0.5;
}
```

```
.auth-bg {
  background-image: radial-gradient(
    circle at 50% 0%,
    var(--accent) 0%,
    transparent 50%
  );
}
```

## Layout

### Root Structure

```
<body class="antialiased min-h-screen flex flex-col">
  <SiteHeader />
  <main id="main-content" class="flex-1">{children}</main>
  <SiteFooter />
  <Toaster />
</body>
```

## Page Layout Patterns

### Auth pages:

```
flex min-h-[calc(100vh-4rem)] items-center justify-center p-4
  → Card w-full max-w-md
```

### Standard content pages:

```
container mx-auto px-4 py-8
  → max-w-4xl mx-auto
```

### Error/not-found pages:

```
container mx-auto px-4 py-16
  → max-w-md mx-auto text-center
```

## Grid Patterns

| Pattern                                              | Usage                 |
|------------------------------------------------------|-----------------------|
| grid grid-cols-1 md:grid-cols-2 lg:grid-cols-4 gap-6 | Feature cards (4-col) |
| grid grid-cols-1 md:grid-cols-2 gap-6                | Dashboard cards       |
| grid grid-cols-1 md:grid-cols-2 gap-4                | Profile info          |
| grid grid-cols-1 md:grid-cols-3 gap-4                | Quick actions         |

## Responsive Breakpoints

Standard Tailwind breakpoints:

- `sm`: (640px) — Padding adjustments, text alignment, button sizing
- `md`: (768px) — Grid column changes (→ 2 col), input font size
- `lg`: (1024px) — Grid column changes (→ 4 col), wide padding

## Icons

Library: Lucide React

## Sizing Convention

| Size | Classes                  | Usage              |
|------|--------------------------|--------------------|
| XS   | <code>h-3 w-3</code>     | Inline badge icons |
| SM   | <code>h-3.5 w-3.5</code> | Copy buttons       |

|         |                                             |                                 |
|---------|---------------------------------------------|---------------------------------|
| Default | <code>h-4 w-4</code> or <code>size-4</code> | Standard UI icons               |
| MD      | <code>h-5 w-5</code>                        | Header logo icon                |
| LG      | <code>h-7 w-7</code>                        | Hero logo icon                  |
| XL      | <code>h-16 w-16</code>                      | Error/empty state illustrations |

## Commonly Used Icons

Bot, User, Lock, Shield, Mail, Calendar, Copy, Check, Loader2, LogOut, Sun, Moon, Github, ArrowLeft, RefreshCw, AlertCircle, FileQuestion, Database, Palette, Video

## Components (shadcn/ui)

All components live in `src/components/ui/`. They use `data-slot` attributes, accept `className` for overrides via `cn()`, and follow either `React.forwardRef` or functional component patterns.

### Button

6 variants, 4 sizes (CVA-based):

| Variant                  | Usage                 |
|--------------------------|-----------------------|
| <code>default</code>     | Primary actions       |
| <code>secondary</code>   | Secondary actions     |
| <code>outline</code>     | Tertiary actions      |
| <code>ghost</code>       | Subtle/icon actions   |
| <code>destructive</code> | Delete/danger actions |
| <code>link</code>        | Inline text links     |

| Size                 | Height              | Padding           |
|----------------------|---------------------|-------------------|
| <code>sm</code>      | <code>h-8</code>    | <code>px-3</code> |
| <code>default</code> | <code>h-9</code>    | <code>px-4</code> |
| <code>lg</code>      | <code>h-10</code>   | <code>px-6</code> |
| <code>icon</code>    | <code>size-9</code> | —                 |

### Card

6 sub-components: `Card`, `CardHeader`, `CardTitle`, `CardDescription`, `CardContent`, `CardFooter`

Base: `rounded-lg border bg-card text-card-foreground shadow-sm`



## Input / Textarea

- Height: `h-9` (input), `min-h-16` (textarea)
- Border: `border bg-transparent rounded-md shadow-xs`
- Focus: `focus-visible:border-ring focus-visible:ring-ring/50 focus-visible:ring-[3px]`
- Validation: `aria-invalid:border-destructive aria-invalid:ring-destructive/20`
- Responsive font: `text-base md:text-sm`

## Badge

4 variants: default, secondary, destructive, outline

Base: `rounded-full border px-2.5 py-0.5 text-xs font-semibold`

## Dialog

Radix-based with overlay (`bg-black/50`), fade + zoom animations, optional close button.

## DropdownMenu

Radix-based. Content: `rounded-md border p-1 shadow-md min-w-[8rem]`. Items support a destructive variant.

## Spinner

Sizes: `sm` (h-4 w-4), `md` (h-6 w-6), `lg` (h-8 w-8). Uses `Loader2` with `animate-spin`.

## Toast (Sonner)

Custom icons per state (success, info, warning, error, loading). Themed via CSS variable overrides.

## Focus & Interaction States

### Focus Ring (Global)

```
outline-2 outline-offset-2 outline-ring/70
```

Component-level override:

```
focus-visible:ring-ring/50 focus-visible:ring-[3px]
```

## Disabled

```
disabled:pointer-events-none disabled:opacity-50
```

## Interactive Card Hover

```
transition-all duration-200 ease-out
hover:shadow-md hover:-translate-y-0.5
```

**Dark Mode**

- **Method:** Class-based via `next-themes` with `attribute="class"` and `disableTransitionOnChange`
- **Default:** System preference
- **Toggle:** 3-way dropdown — Light / Dark / System
- All semantic color tokens swap automatically via `.dark` CSS selector
- Use `dark:` prefix for component-specific overrides (e.g., `dark:bg-input/30`)

## Branding

**Logo Text**

```
bg-gradient-to-r from-primary to-primary/70 bg-clip-text text-transparent
```

### Logo Icon Container

```
w-8 h-8 rounded-lg bg-primary/10 flex items-center justify-center
```

Hero variant: w-12 h-12 rounded-xl



# Atlas — Production Readiness Audit Report

Date: 2026-06-28

**Remediation status (applied 2026-06-28):** The safe high-priority tranche (both Criticals + all 13 Highs + several low-risk Mediums/Lows) has been implemented via coordinated sub-agents and verified with `tsc --noEmit (clean)`, `npm run lint (0 errors)`, and `next build (all routes compile)`. See § **Remediation Applied** at the end of this report for the full list of fixes, files changed, and remaining items. **One blocker is environmental:** the DB index migration `drizzle/0018_regular_karnak.sql` is generated but **not yet applied** — the dev Neon credentials in `.env` return `28P01 password authentication failed (password rotated)`. Refresh `POSTGRES_URL` and run `npx drizzle-kit migrate` to apply it.

## Executive Summary

Atlas has a fundamentally sound architecture: tenant isolation via `requireAgencyUser()` and double-condition `and(eq(agencyId, ...), ...)` scoping is consistent across all 20 server-action files, all inputs are Zod-validated, and there is no SQL injection or `dangerouslySetInnerHTML` exposure. However, the application is **not yet production-ready** without remediation. The two genuine blockers are silent fake-PNR fallback on failed airline bookings (agents see a confirmation for a flight that was never booked) and missing try/catch on Stripe webhook DB writes (crashes cause endless Stripe retries with duplicate side-effects). Counting across all six audits: **2 critical blockers, 11 high-severity, 21 medium-severity, and ~20 low-severity** issues. The high-severity cluster is dominated by missing error handling (no try/catch in 10 action files, no `error.tsx` in client-facing segments), absent `fetch` timeouts on all external supplier calls, RTL breakage in the mobile drawer, and a prompt-injection vector in the AI chat endpoint.

## Critical Issues (must fix before production)

| #  | Category       | File                                                                                        | Issue                                                                                                                                                                 | Fix                                                                                                                                          |
|----|----------------|---------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------|
| C1 | Error Handling | <code>src/lib/suppliers/duffel.ts:278</code> , <code>src/lib/actions/bookings.ts:382</code> | Failed Duffel order silently falls back to a fake <code>DF-xxxxxxx</code> reference and marks booking <code>confirmed</code> — agent sees success, no real PNR exists | Return discriminated provisional/confirmed result; never set <code>status: "confirmed"</code> on a provisional; surface a visible UI warning |

|    |                |                                                                                                      |                                                                                                                                                                                                                         |                                                                                                                                                 |
|----|----------------|------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|
| C2 | Error Handling | src/app/api/stripe/webhook/<br>route.ts:75-86, src/app/api/stripe/connect-webhook/<br>route.ts:40-46 | No try/catch around webhook DB updates; a DB throw returns 500 and Stripe retries indefinitely → duplicate side-effects. connect-webhook also returns 400 (stops retries) when temporarily misconfigured instead of 503 | Wrap DB writes in try/catch returning 500 only on genuine transient errors; return 503 (not 400) for misconfiguration so Stripe retries cleanly |
|----|----------------|------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|

## High Severity Issues

| #  | Category       | File                                                                                                                                                                                 | Issue                                                                                                                                | Fix                                                |
|----|----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------|
| H1 | Security       | src/app/api/chat/<br>route.ts:28-36                                                                                                                                                  | Zod schema accepts role: "system" from client → prompt injection / jailbreak of the AI agent, including spurious createBooking calls | Drop "system" from the client                      |
| H2 | Security       | src/app/api/portal/auth/signout/<br>route.ts:2,15;<br>src/app/portal/layout.tsx:39                                                                                                   | Portal signout is GET via <a href> → CSRF forced-logout (CWE-352)                                                                    | Change handler to POST; use                        |
| H3 | Error Handling | 10 action files<br>(clients.ts:45, team.ts, suppliers.ts, invites.ts, opportunities.ts, portal-proposals.ts, notifications.ts, proposals-public.ts, portal-invite.ts, onboarding.ts) | Zero try/catch around DB calls → unhandled exceptions crash the action with a generic 500, no logging                                | Wrap DB writes; return { ok console.error("[action |
| H4 | Error Handling | src/app/api/chat/<br>route.ts:77-277                                                                                                                                                 | streamText and all tool execute bodies have no try/catch; createBooking item loop silently skips failed items                        | Wrap streamText and each                           |

|     |                |                                                                                                                                                                                                                             |                                                                                                                                            |                                                         |
|-----|----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------|
| H5  | Error Handling | src/lib/suppliers/duffel.ts:39, hotelbeds.ts:159, payments/stripe.ts                                                                                                                                                        | No <code>fetch</code> timeout (AbortController) on any external call → function hangs to Vercel's 30s timeout                              | Add <code>AbortController</code> + s base fetch wrapper |
| H6  | Error Handling | src/app/portal/, src/app/platform/, src/app/(app)/bookings/, src/app/(app)/clients/                                                                                                                                         | No segment-level <code>error.tsx</code> ; errors bubble to root boundary whose "Go home" link points to / (inaccessible to portal clients) | Add <code>error.tsx</code> per segment (portal/login)   |
| H7  | Performance    | src/app/(app)/dashboard/page.tsx:83-85                                                                                                                                                                                      | Counts done by fetching every row's <code>id</code> then <code>.length - 3</code> full-table scans for 3 integers                          | Use <code>db.\$count(table, w</code>                    |
| H8  | Performance    | src/lib/actions/bookings.ts:53, src/lib/actions/products.ts:36                                                                                                                                                              | <code>nextReference()</code> fetches all references in the agency and finds max in JS on every create                                      | Single <code>max(...)</code> aggregation                |
| H9  | Performance    | src/lib/actions/products.ts:57-67                                                                                                                                                                                           | <code>recalcTotals()</code> issues one sequential UPDATE per item (N+1 writes)                                                             | Batch CASE-WHEN update c                                |
| H10 | Performance    | src/app/(app)/dashboard/page.tsx, src/app/(app)/finance/page.tsx                                                                                                                                                            | No Suspense boundaries; user sees blank screen until all DB queries finish (finance loads 1,000 bookings first)                            | Split into shell + Suspense as                          |
| H11 | UX / RTL       | src/components/app/app-shell.tsx:223                                                                                                                                                                                        | Mobile nav drawer hardcodes <code>slide-in-from-left-2 / left-0 / border-r</code> → appears from wrong side in Arabic (RTL)                | Conditional <code>rtl: variants</code> or               |
| H12 | UX / A11y      | All auth forms ( <code>sign-in-button.tsx:83</code> , <code>sign-up-form.tsx:107</code> , <code>forgot-password-form.tsx:70</code> , <code>reset-password-form.tsx:25,100</code> , <code>accept-invite-form.tsx:99</code> ) | Error <code>&lt;p&gt;</code> has no <code>role="alert"/aria-live</code> → screen readers never announce login failures                     | Add <code>role="alert"</code> to error                  |

|     |          |                                                          |                                                                                   |                                              |
|-----|----------|----------------------------------------------------------|-----------------------------------------------------------------------------------|----------------------------------------------|
| H13 | Database | src/lib/<br>schema.ts:188-198<br>(verification<br>table) | No index on<br>verification.identifier<br>→ full scan on every login/OTP<br>check | Add<br>index("verification_id<br>+ migration |
|-----|----------|----------------------------------------------------------|-----------------------------------------------------------------------------------|----------------------------------------------|

## Medium Severity Issues

| #  | Category          | File                                              | Issue                                                                                                                                           | Fix                                                   |
|----|-------------------|---------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------|
| M1 | Security          | src/lib/actions/<br>commissions.ts:28-31, 74-76   | recordCommission does not<br>cross-tenant validate<br>supplierId,<br>bookingItemId,<br>agentUserId                                              | Add ownership c<br>scoped to user.<br>before INSERT   |
| M2 | Security          | src/lib/notifications/<br>templates.ts:14, 52-72  | User strings (clientName,<br>agencyName, roleLabel)<br>interpolated into HTML email<br>without escaping                                         | Add h() HTML-<br>helper, apply to s<br>sourced values |
| M3 | Security          | src/lib/storage.ts:117                            | File upload validates extension<br>only, not MIME magic bytes;<br>SVG payloads accepted; risky<br>on local-fs fallback path                     | Add file-type<br>byte check; reject<br>user images    |
| M4 | Error<br>Handling | src/lib/actions/<br>commissions.ts:505-508        | autoGenerateCommissions<br>failure swallowed with<br>console.error; booking<br>advances with no commissions<br>and no warning                   | Return structured<br>surface non-bloc<br>warning      |
| M5 | Error<br>Handling | src/lib/actions/<br>platform.ts:134-136           | Stripe customer-create failure<br>only logged; agency saved<br>without stripeCustomerId<br>→ later "Agency not found" or<br>duplicate customers | Structured log +<br>for reconciliation                |
| M6 | Error<br>Handling | src/lib/suppliers/<br>index.ts:78-97 (safeSearch) | degraded: true from mock<br>fallback not forwarded to AI in<br>chat route (api/chat/<br>route.ts:96-101) → mock<br>prices presented as real     | Include source/<br>in tool result                     |
| M7 | Error<br>Handling | entire src/                                       | No structured logging library;<br>only<br>console.error(string)<br>→ cannot query/alert per<br>tenant or correlate digests                      | Adopt Pino/Senti<br>line logging conv                 |
| M8 | Performance       | src/lib/actions/<br>bookings.ts:459-468           | advanceStatus ticketing<br>loop: sequential external API +<br>DB write per item                                                                 | Promise.all tl<br>calls; batch DB v                   |

|     |                 |                                                                                                                                                                                    |                                                                                                                                   |                                                               |
|-----|-----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------|
| M9  | Performance     | src/lib/actions/<br>commissions.ts:258-277                                                                                                                                         | getCommissions() has no<br>.limit() → unbounded<br>result set                                                                     | Add .limit(50<br>pagination                                   |
| M10 | Performance     | src/app/(app)/dashboard/<br>page.tsx:109                                                                                                                                           | Loads up to 500 bookings with<br>payments+travellers to<br>compute KPIs in JS                                                     | Move aggregatio<br>scope traveller q<br>passport alerts o     |
| M11 | Performance     | src/app/(app)/finance/<br>page.tsx:74                                                                                                                                              | Loads up to 1,000<br>bookings+payments; KPIs<br>computed via JS reductions                                                        | Replace with SQ<br>aggregates; load<br>only when balan        |
| M12 | Performance     | src/lib/schema.ts                                                                                                                                                                  | Missing composite indexes for<br>(agencyId, status) /<br>(agencyId, createdAt)<br>on booking, opportunity,<br>product, commission | Add composite in<br>migrate                                   |
| M13 | Performance     | src/lib/export/<br>datasets.ts:74+                                                                                                                                                 | Export loaders are unbounded<br>findMany → OOM/timeout on<br>large agencies                                                       | Per-dataset limit:<br>streaming export                        |
| M14 | Performance     | src/components/app/app-<br>shell.tsx:1                                                                                                                                             | Entire nav shell is "use<br>client" only for<br>usePathname()                                                                     | Extract active-lin<br>tiny <NavLink><br>as server compo       |
| M15 | Performance     | src/app/(app)/dashboard/<br>page.tsx:33                                                                                                                                            | Recharts (~150KB) eagerly<br>imported for all users                                                                               | next/dynamic<br>ssr:false } C<br>imports                      |
| M16 | Performance     | src/components/hotels/<br>hotel-details-<br>view.tsx:221, src/<br>components/search/hotel-<br>details-dialog.tsx:123                                                               | Raw <img> bypassing next/<br>image optimization                                                                                   | Use next/imag<br>Hotelbeds CDN i<br>images.remot              |
| M17 | Code<br>Quality | 9+ files (billing.ts,<br>invites.ts, payments.ts,<br>portal-payments.ts, portal-<br>invite.ts, api/portal/auth/<br>request/route.ts,<br>robots.ts, sitemap.ts, auth-<br>client.ts) | NEXT_PUBLIC_APP_URL ??<br>"localhost:3000"<br>duplicated, bypassing env.ts                                                        | Export single AP<br>config.ts                                 |
| M18 | Code<br>Quality | src/lib/suppliers/ vs src/<br>lib/suppliers/providers/                                                                                                                             | Two parallel supplier<br>abstractions coexist;<br>providers/ registry is empty<br>and unused                                      | Mark provider<br>progress or com<br>migration and re<br>layer |
| M19 | Code<br>Quality | src/lib/actions/**/*ts                                                                                                                                                             | 62 of 94 exported server<br>actions lack explicit return-type<br>annotations                                                      | Annotate with ;<br>Promise<Acti                               |



|     |           |                                                                                                                          |                                                                                                                                                                                     |                                                                                   |
|-----|-----------|--------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|
| M20 | UX / A11y | client-form.tsx:62,<br>booking-form.tsx:87,<br>opportunity-form.tsx                                                      | Server errors only via auto-dismissing toast, no persistent inline error                                                                                                            | Add <code>serverError</code> <code>role="alert"</code> message                    |
| M21 | UX / A11y | All forms                                                                                                                | <code>aria-invalid</code> never set on fields after failed validation                                                                                                               | Set <code>aria-invalid</code> on empty/invalid rec                                |
| M22 | UX / A11y | clients/page.tsx:92,<br>suppliers/page.tsx:85                                                                            | Search <code>&lt;Input&gt;</code> has no <code>&lt;Label&gt;</code> (placeholder is not an accessible name)                                                                         | Add <code>&lt;Label&gt;</code> <code>htmlFor="q"</code> <code>className="s</code> |
| M23 | UX / A11y | mode-toggle.tsx:19, create-agency-form.tsx:66,<br>travellers-<br>manager.tsx:303, payments-manager.tsx:227               | Icon-only buttons missing <code>aria-label/aria-haspopup</code> ; save buttons lack <code>aria-busy</code>                                                                          | Add <code>aria-label</code> <code>haspopup="me</code> <code>busy</code>           |
| M24 | UX / RTL  | ~62 <code>mr-/ml-</code> + ~94 <code>pl-/pr-/border-l/border-r</code> across components                                  | Physical CSS directions don't flip in RTL                                                                                                                                           | Refactor to logic: <code>ps-/pe-/border-e</code>                                  |
| M25 | UX / RTL  | List pages with tables ( <code>clients</code> , <code>bookings</code> , <code>products</code> , <code>suppliers</code> ) | <code>text-right</code> numeric columns don't flip in RTL                                                                                                                           | Replace with <code>te</code>                                                      |
| M26 | Database  | src/lib/schema.ts                                                                                                        | No <code>createdAt</code> index on booking/client/opportunity/product/payment/notification                                                                                          | Add <code>createdAt</code> (priority: booking opportunity > prc payment)          |
| M27 | Database  | src/lib/schema.ts:423,510-536                                                                                            | <code>bookingItem.supplierId</code> / <code>productItem.supplierId</code> FKs have no index → full scans in commission/supplier joins                                               | Add <code>supplierId</code> on both tables                                        |
| M28 | Database  | src/lib/schema.ts                                                                                                        | No soft-delete ( <code>deletedAt</code> ) anywhere; hard deletes orphan commissions, no undelete/audit                                                                              | Add <code>deletedAt</code> booking/opportunity/product; filter list               |
| M29 | Database  | src/lib/schema.ts:985-1001                                                                                               | <code>supplierRate</code> has no <code>denormalized</code> <code>agencyId</code> → direct agency-scoped queries need double join                                                    | Add <code>agencyId</code> <code>supplierCont</code> (matches pattern)             |
| M30 | Database  | src/lib/actions/bookings.ts:65-78                                                                                        | <code>recalcTotal</code> ( <code>bookingId</code> ) queries <code>bookingItem</code> by <code>bookingId</code> with no agency scope (footgun if called with attacker-controlled id) | Pass <code>agencyId</code> filter <code>booking.e</code> or accept pre-loc        |

## Low Severity Issues / Recommendations

| #   | Category       | File                                                                                                | Issue                                                                                                               |
|-----|----------------|-----------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------|
| L1  | Security       | src/proxy.ts matcher                                                                                | /suppliers, /finance, /reports, /settings, /billing, /commissions not in matcher (still safe via requireAgencyUser) |
| L2  | Security       | src/app/api/portal/auth/request/<br>route.ts                                                        | No rate limiting on magic-link request → resource exhaustion / enumeration                                          |
| L3  | Security       | src/lib/actions/settings.ts:21-27                                                                   | Locale cookie httpOnly:false (intentional; validated via isLocale())                                                |
| L4  | Error Handling | src/lib/actions/settings.ts:34, 65                                                                  | Empty catch {} with no logging                                                                                      |
| L5  | Error Handling | src/lib/suppliers/hotelbeds.ts:467                                                                  | Silent catch {} on availability fallback                                                                            |
| L6  | Error Handling | src/lib/suppliers/content-cache.ts:110, 121                                                         | .catch(() => {}) silent cache-write swallow → cache never heals                                                     |
| L7  | Error Handling | clipboard handlers ai-quote-builder.tsx:79, itinerary-builder.tsx:306, 340, payments-manager.tsx:95 | Silent clipboard catch after success toast → "copied!" shown even on failure                                        |
| L8  | Error Handling | src/app/error.tsx:16-17                                                                             | error.digest shown in UI but not logged server-side → can't correlate                                               |
| L9  | Performance    | src/lib/actions/commissions.ts:284-309                                                              | getCommissionsByBooking() does two sequential queries                                                               |
| L10 | Performance    | src/app/layout.tsx:54                                                                               | All i18n messages shipped to every client                                                                           |
| L11 | Performance    | src/app/proposal/[id]/page.tsx, public token pages                                                  | No ISR/revalidate on public proposal pages                                                                          |
| L12 | Code Quality   | src/lib/session.ts                                                                                  | Entire file is dead code (requireAuth, getOptionalSession, stale protectedRoutes)                                   |
| L13 | Code Quality   | src/proxy.ts                                                                                        | Middleware never executes (not middleware.ts, no middleware/default export) → manifest empty                        |
| L14 | Code Quality   | src/components/ui/github-stars.tsx                                                                  | Exported component with zero consumers                                                                              |

|     |              |                                                                                                                                   |                                                                                                                           |
|-----|--------------|-----------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------|
| L15 | Code Quality | portal-session.ts:20, settings.ts:24, 60, platform.ts:200, 228                                                                    | secure:<br>NODE_ENV==="production"<br>duplicated 5x                                                                       |
| L16 | Code Quality | analytics.ts:62, finance/page.tsx:176                                                                                             | Intl.DateTimeFormat("en-GB", {month: "short"})<br>duplicated                                                              |
| L17 | Code Quality | permissions.ts:53, 193                                                                                                            | session.user as unknown<br>as {...} cast duplicated                                                                       |
| L18 | Code Quality | hotelbeds.ts:81, add-to-booking-dialog.tsx:67, add-to-proposal-dialog.tsx:57, payments-manager.tsx:97, hotel-details-view.tsx:297 | Non-null assertions without guard                                                                                         |
| L19 | Code Quality | api/portal/auth/request/route.ts:67, notifications/email.ts:30                                                                    | console.log violates no-console<br>ESLint rule (warn/error only)                                                          |
| L20 | UX / A11y    | hotel-search-experience.tsx:222-228                                                                                               | Search button loses label text +<br>no aria-label while loading                                                           |
| L21 | UX / Mobile  | Table wrappers in list pages                                                                                                      | No visual scroll hint on<br>overflowing tables                                                                            |
| L22 | UX / Mobile  | pipeline-board.tsx:101-108                                                                                                        | Stage-move trigger ~20px touch<br>target (< WCAG 44px)                                                                    |
| L23 | UX / A11y    | hotel-details-view.tsx:221, hotel-details-dialog.tsx:123                                                                          | Gallery images use empty<br>alt="" though not purely<br>decorative                                                        |
| L24 | UX / A11y    | portal/layout.tsx:25                                                                                                              | <nav> missing aria-label                                                                                                  |
| L25 | Code Quality | billing/stripe.ts:17, payments/stripe.ts:9                                                                                        | isBillingConfigured()/<br>isStripeConfigured()<br>identical                                                               |
| L26 | Database     | src/lib/schema.ts:1023-1025                                                                                                       | commission.bookingItemId<br>uses ON DELETE SET NULL →<br>orphaned, unreconcilable<br>commissions                          |
| L27 | Database     | src/lib/schema.ts:652-868                                                                                                         | Missing Drizzle relations for<br>supplier,<br>supplierContract,<br>supplierRate, commission<br>→ breaks with: eager loads |

|     |          |                                                 |                                                                                                                                                             |
|-----|----------|-------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|
| L28 | Database | drizzle/0006_add_agency_tenancy.sql:343         | Unconditional backfill could set platform-admin <code>agencyId</code> to seed agency (non-issue in practice, all <code>is_platform_admin=false</code> then) |
| L29 | Database | src/lib/schema.ts:882-906                       | hotelContent uses text("code") PK (justified natural key)                                                                                                   |
| L30 | Database | payment.bookingTraveller and other child tables | No denormalized <code>agencyId</code> (defense-in-depth gap)                                                                                                |

## Category Findings

### 1. Security

Overall posture is solid: `agencyId` is always server-derived from `session.user.agencyId`, every action calls `requireAgencyUser()/requireUser()/requirePlatformAdmin()/requireManager()` first, all mutations use `and(eq(id), eq(agencyId))` double-conditions, all inputs are Zod-validated, Drizzle parameterizes all queries, no secrets are in `NEXT_PUBLIC_`, no `dangerouslySetInnerHTML`, Stripe webhooks verify signatures, impersonation gates on `requirePlatformAdmin()` with `httpOnly` cookies, portal actions verify `clientId + agencyId`, and invite registration ties `agencyId/role` to the invite row (`input:false`).

- **H1 — Prompt injection via `role:"system"`** (`api/chat/route.ts:28-36`): client can POST system messages that flow into `convertToModelMessages` and override the server system prompt, including triggering `createBooking`. Restrict the enum to `["user", "assistant"]`.
- **H2 — CSRF forced-logout** (`api/portal/auth/signout/route.ts:2,15, portal/layout.tsx:39`): GET signout via `<a href>` is exploitable with `<img src=...>`. Switch to POST + form. Impact limited to DoS but CWE-352.
- **M1 — Commission FK cross-tenant** (`commissions.ts:28-31, 74-76`): `bookingId` is validated but `supplierId/bookingItemId/agentUserId` are not, allowing foreign-tenant FK references and an existence oracle.
- **M2 — Unescaped HTML email** (`notifications/templates.ts`): `clientName/agencyName/roleLabel` interpolated raw; low risk in mail clients but the same names render on admin pages.
- **M3 — Extension-only upload validation** (`storage.ts:117`): no magic-byte check; SVG/script payloads accepted; risky on local-fs fallback.
- **L1** missing middleware matcher routes, **L2** no rate limit on magic-link request, **L3** intentional non-`httpOnly` locale cookie (validated).

### 2. Error Handling

This is the weakest category and the source of both blockers.

- **C1 — Fake PNR fallback** (`duffel.ts:278, bookings.ts:382`): the single most dangerous

behavior in the app. A failed airline booking is recorded as `confirmed` with a provisional `DF-` reference and no agent-visible warning — agents could issue tickets against a non-existent booking.

- **C2 — Webhook DB writes unguarded** (`stripe/webhook/route.ts:75-86`, `connect-webhook/route.ts:40-46`): DB throw → 500 → Stripe retry storm with duplicate side-effects; `connect-webhook` returns 400 on misconfig (stops legitimate retries).
- **H3 — No try/catch in 10 action files; H4 — chat route + tools unguarded; H5 — no fetch timeouts** on Duffel/Hotelbeds/Stripe (hang to Vercel 30s); **H6 — missing `error.tsx`** in `portal/platform/bookings/clients` (root boundary sends portal clients to inaccessible `/`).
- Medium: **M4** silent commission auto-gen failure, **M5** Stripe customer-create failure unsurfaced, **M6** degraded flag not forwarded to AI, **M7** no structured logging.
- Low: **L4-L8** silent/empty catches and missing digest logging.

### 3. Performance

No catastrophic issues but several scale time-bombs as agencies grow.

- **H7** dashboard counts via full fetch + `.length`; **H8** `nextReference()` fetches all references per create; **H9** `recalcTotals()` N+1 UPDATES; **H10** no Suspense → blank screen on dashboard/finance (finance loads 1,000 bookings first, has no `loading.tsx`).
- Medium: **M8** serial ticketing API/DB loop, **M9** unbounded `getCommissions()`, **M10/M11** large in-memory KPI computation on dashboard (500 bookings) and finance (1,000 bookings) that should be SQL aggregates, **M12** missing composite indexes, **M13** unbounded export → OOM, **M14** unnecessarily-client AppShell, **M15** eager Recharts ~150KB, **M16** raw `<img>`.
- Low: **L9** sequential commission-by-booking queries, **L10** full i18n payload per client, **L11** no ISR on public proposals.

### 4. Code Quality

Clean of TODO/FIXME/HACK markers; main issues are dead code and duplication.

- Dead code: **L12** `session.ts` (entire file), **L13** `proxy.ts` (middleware never executes — wrong filename/export), **L14** `github-stars.tsx`.
- Duplication: **M17** `APP_URL` fallback in 9+ files, **L15** secure cookie option ×5, **L16** `SHORT_MONTH` formatter, **L17** `session.user` cast ×2, **L25** identical Stripe-configured sentinels.
- Architecture: **M18** two parallel supplier systems (`suppliers/` live vs empty `suppliers/providers/`).
- Types: **M19** 62/94 actions lack return-type annotations; **L18** 5 unguarded non-null assertions; **L19** 2 `console.log` ESLint violations.

### 5. UX & Accessibility

Strongest gaps are RTL (Arabic) breakage and screen-reader announcements.

- **H11/H12** — mobile drawer slides from the wrong side in RTL (`app-shell.tsx:223`); auth error `<p>s` lack `role="alert"`.
- RTL: **M24** ~156 physical-direction utilities should be logical (`me-/ms-/ps-/pe-/border-s/border-e`); **M25** `text-right` → `text-end` on numeric table columns.

- Forms: **M20** errors only via dismissible toast, **M21** `aria-invalid` never set, **M22** search inputs missing `<Label>`.
- A11y: **M23** icon-only buttons missing `aria-label/aria-haspopup/aria-busy` (`mode-toggle`, `create-agency-form`, `travellers-manager`, `payments-manager`).
- Loading: **L1.1** 6 list pages have no `loading.tsx` skeleton; low-severity **L20-L24** scroll hint, touch target, gallery alt, nav label.

## 6. Database

Schema is well-structured with consistent UUID PKs and tenant scoping; gaps are indexing and soft-delete.

- **H13** — no index on `verification.identifier` (full scan every login).
- Indexing: **M26** missing `createdAt` indexes on high-volume tables, **M27** missing `supplierId` indexes on `bookingItem/productItem`, **M12** missing (`agencyId`, `status`) composites.
- Tenant isolation: **M29** `supplierRate` missing denormalized `agencyId`; **M30** `recalcTotal` queries `bookingItem` without agency scope (footgun); **L30** child tables (`payment`, `bookingTraveller`) lack defense-in-depth `agencyId`.
- Data lifecycle: **M28** no soft-delete anywhere; **L26** `commission.bookingItemId` SET NULL orphans commissions.
- Hygiene: **L27** missing relations for supplier/commission tables, **L28** migration 0006 backfill edge case, **L29** `hotelContent` text PK (justified).

## Files That Need Immediate Attention

1. `src/lib/suppliers/duffel.ts` — C1 (fake PNR), H5 (no fetch timeout), L5/L18 — the highest-risk file; fake-booking behavior is a business blocker.
2. `src/app/api/stripe/webhook/route.ts` + `connect-webhook/route.ts` — C2 (unguarded DB writes → retry storm).
3. `src/lib/actions/bookings.ts` — C1 (confirmed status on provisional), H8 (`nextReference`), M8 (serial ticketing), M30 (`recalcTotal` scope).
4. `src/app/api/chat/route.ts` — H1 (prompt injection), H4 (no try/catch), M6 (degraded flag).
5. `src/components/app/app-shell.tsx` — H11 (RTL drawer), M14 (client component), M24 (logical props).
6. `src/app/(app)/dashboard/page.tsx` — H7 (count via fetch), H10 (no Suspense), M10/M15.
7. `src/app/(app)/finance/page.tsx` — H10 (no Suspense/loading), M11 (1,000-row JS KPIs).
8. `src/lib/schema.ts` — H13 (verification index), M12/M26/M27/M28/M29, L26/L27.
9. `src/lib/actions/commissions.ts` — M1 (cross-tenant FK), M4 (silent failure), M9 (no limit), L9.
10. `src/lib/actions/products.ts` — H8 (`nextReference`), H9 (`N+1 recalcTotals`).
11. **Auth form components** (sign-in/sign-up/forgot/reset/accept-invite) — H12 (`role="alert"`), M21.

## Remaining Recommendations

- Adopt a structured logging / observability layer (Pino + Vercel log drain, or Sentry) so per-tenant

error rates can be alerted on and user-visible `error.digest` values correlate with server logs (M7, L8).

- Add rate limiting (Upstash or token bucket) to the magic-link request endpoint and any other unauthenticated POST endpoints (L2).
- Introduce a soft-delete strategy (`deletedAt`) on core entities for audit/undelete before data volume makes retrofitting expensive (M28).
- Complete or formally retire the `suppliers/providers/` registry to remove the dual-abstraction ambiguity (M18).
- Centralize repeated config (`APP_URL`, `isProduction`, `formatShortMonth`) in `config.ts/format.ts` and delete confirmed dead code (`session.ts`, `proxy.ts` or `rename`, `github-stars.tsx`) (M17, L12-L17).
- Run a single RTL refactor pass converting physical to logical Tailwind utilities across components, and add `loading.tsx` skeletons to the six list pages lacking them (M24, L1.1).
- Annotate all server actions with explicit `Promise<ActionResult>` return types to catch accidental return-shape drift in CI (M19).
- After schema index/FK changes, run `drizzle-kit generate && migrate` (never push, per AGENTS.md).

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## Remediation Applied (2026-06-28)

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Implemented via parallel coder sub-agents, each owning a disjoint file set. Verification: `npm tsc --noEmit clean · npm run lint 0 errors (48 pre-existing import-order warnings only) · npm next build` all routes compile.

### Critical

- **C1 — Fake-PNR fallback** — `duffel.ts` no longer fabricates a DF-... reference; `bookFlight` throws on provider failure. `bookings.ts` `confirmItemBooking` returns a discriminated result; failed items are set to `pending` (existing enum value), never `confirmed`. `advanceStatus` aborts the ticketing advance and returns `{ ok:false, reasons }` if any item failed. Files: `src/lib/suppliers/duffel.ts`, `src/lib/actions/bookings.ts`.
- **C2 — Stripe webhook retry storm** — DB writes wrapped in `try/catch` in both `webhook/route.ts` and `connect-webhook/route.ts`; misconfiguration in `connect-webhook` now returns 503 (was 400) so Stripe retries; signature failures stay 400; transient DB errors return 500 with logging.

### High

- **H1** chat role enum restricted to `["user", "assistant"]` (no client system).
- **H2** portal signout changed GET→POST + `<form method="POST">` (303 redirect).
- **H3** `try/catch` + `console.error` + friendly error returns added across 10 action files (clients, team, suppliers, invites, opportunities, portal-proposals, notifications, proposals-public, portal-invite, onboarding) and commissions.
- **H4** chat `streamText` + every tool `execute` wrapped; `createBooking` collects failed items/ travellers instead of silently skipping.

- **H5** 15s `AbortController` fetch timeouts on Duffel, Hotelbeds, Stripe wrappers.
- **H6** segment `error.tsx` added for `portal/` ( $\rightarrow$  `/portal/login`), `(app)/` ( $\rightarrow$  `/dashboard`), `platform/` ( $\rightarrow$  `/platform`).
- **H7** dashboard counts now use `db.$count(...)` instead of `fetch + .length`.
- **H8** `nextReference()` uses a single `max(...)` aggregate.
- **H9** `recalcTotals()` N+1 `UPDATE`s replaced with `Promise.all`.
- **H10** Suspense around dashboard insights + new `finance/loading.tsx` skeleton.
- **H11** mobile drawer made RTL-aware via Tailwind `rtl: variants`.
- **H12** `role="alert"` added to all auth-form error messages.
- **H13** `verification_identifier_idx` added (in migration 0018, pending apply).

## Medium / Low (included in tranche)

- **M1** `commissions recordCommission` now cross-tenant-validates `supplierId/bookingItemId/agentUserId` before insert.
- **M2** `escapeHtml()` applied to user-sourced values in email templates.
- **M6** `degraded/source` flag now part of `safeSearch` result type and forwarded to the AI chat tools (mock prices disclosable).
- **M12/M26/M27/L26/L27** composite + `createdAt` + `supplierId` indexes, commission cascade, and supplier/commission Drizzle relations — all in `drizzle/0018_regular_karnak.sql` (**pending apply** — see blocker above).
- **M30** `recalcTotal` now agency-scoped via join on `booking.agencyId`.
- **L12/L13/L14** deleted dead code: `src/lib/session.ts`, `src/proxy.ts` (dormant non-executing middleware; protections already enforced by `requireAgencyUser()`), `src/components/ui/github-stars.tsx`.

## Deferred (explicitly out of this tranche — recommend follow-up)

- **M28** soft-delete (`deletedAt`) strategy across core entities.
- **M18** retire/complete the dual `suppliers/providers/` abstraction.
- **M24/M25** full RTL refactor (~156 physical $\rightarrow$ logical Tailwind utilities).
- **M7** structured logging/observability (Pino/Sentry).
- **L2** rate limiting on the magic-link request endpoint.
- Remaining Medium/Low UX, a11y, and perf items per the tables above.

## Action required to fully close

1. Refresh `POSTGRES_URL` in `.env` (dev) with a valid Neon password, then run `npx drizzle-kit migrate` to apply `0018_regular_karnak.sql`.
2. Apply the same migration to **prod**: `POSTGRES_URL=<prod-url> npx drizzle-kit migrate`.



# Production-Readiness Audit

**Date:** 2026-06-28 · **Auditor:** Lead Software Architect · **Scope:** security, error handling, performance, code quality, UX, database. No new features.

**Note:** the brief mentioned "optimize Prisma" — Atlas uses **Drizzle ORM**, not Prisma. Drizzle findings are reported instead.

## Summary

Atlas has **strong security foundations** — tenant isolation, authorization, and input safety are largely correct. The gaps are in **operational hardening** (rate limiting, error handling, loading states) and **reversibility** (no soft delete). One genuine horizontal-privilege-escalation issue was found **and fixed** during this audit.

| Area           | Verdict                                           |
|----------------|---------------------------------------------------|
| Security       | ● Solid; 1 escalation bug fixed, 2 hardening gaps |
| Error handling | ● Inconsistent try/catch; few error boundaries    |
| Performance    | ● Well-indexed; pagination missing                |
| Code quality   | ● Clean; minor lint/dup                           |
| UX             | ● Loading/error/mobile gaps                       |
| Database       | ● Good FKs/indexes; no soft delete                |

## 1. Security

| #  | Issue                                                                                                                                                                                                                                                    | Severity | Proposed fix                                                                                                                          | Status                                                                   |
|----|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|---------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------|
| S1 | <b>Horizontal privilege escalation</b> — detail pages ( <code>bookings/clients/opportunities/products/[id]</code> ) were tenant-scoped ( <code>agencyId</code> ) but <b>not</b> agent-scoped, so an agent could open a colleague's record by direct URL. | High     | Add agent own-scope ( <code>createdById/ownerId/assignedToId</code> ) to each detail query, gated by <code>seesAllData(role)</code> . | ✓ <b>Fixed</b> (4 files this audit; list pages fixed earlier in 134ceb3) |
| S2 | <b>No rate limiting</b> on login, the public proposal token endpoints ( <code>/p/[token], acceptProposalByToken</code> ), or API routes — brute-force / abuse vector.                                                                                    | Medium   | Add an IP+route limiter (e.g. Upstash ratelimit) in middleware for auth + public token routes.                                        | ☐ Recommended                                                            |

|    |                                                                                                          |               |                                                                                 |                                      |
|----|----------------------------------------------------------------------------------------------------------|---------------|---------------------------------------------------------------------------------|--------------------------------------|
| S3 | <b>Demo credentials committed</b> in docs/deployment.md (incl. platform-admin).                          | <b>Medium</b> | Rotate the demo passwords; move them out of the repo or behind a private note.  | <input type="checkbox"/> Recommended |
| S4 | Supplier-detail booking-item count query not explicitly agency-scoped (logically safe via FK ownership). | <b>Low</b>    | Add <code>agencyId</code> to the count <code>where</code> for defense-in-depth. | <input type="checkbox"/> Optional    |

#### Verified GOOD (no action):

- **Tenant isolation** — `requireAgencyUser()` on every (app) page (enforced in the layout) and every server action; `getSupplierById`, all `[id]` detail queries, and exports filter by `agencyId`. `scripts/test-tenant-isolation.ts` exists.
- **Authorization** — all 20 action files guard their exports (billing via a `requireBillingAdmin` wrapper; platform via `requirePlatformAdmin`; `platform.exitAgencyView` only clears cookies; `proposals-public` is token-authorized by design).
- **No SQL injection** — Drizzle parameterizes; no `sql.raw/string`-built queries.
- **No XSS sinks** — zero `dangerouslySetInnerHTML`; React auto-escapes.
- **Secrets** — `.env*` gitignored; no `.env` tracked; Stripe webhooks verify signatures.

## 2. Error Handling

| #  | Issue                                                                                                     | Severity      | Proposed fix                                                                                                                                                            |
|----|-----------------------------------------------------------------------------------------------------------|---------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| E1 | Only <b>10/20</b> action files use <code>try/catch</code> ; uncaught throws surface a generic Next error. | <b>Medium</b> | Standardize on the <code>ActionResult</code> return shape with user-friendly messages; wrap external calls (Stripe/Duffel/Hotelbeds/Resend) in <code>try/catch</code> . |
| E2 | Only <code>assistant/</code> has an <code>error.tsx</code> ; most routes have no error boundary.          | <b>Medium</b> | Add <code>error.tsx</code> per route group (or one in (app) /) with a retry action.                                                                                     |
| E3 | No structured logging; <code>logActivity</code> covers ~11/21 action files.                               | <b>Low</b>    | Add a logger; extend <code>logActivity</code> to every mutation.                                                                                                        |

## 3. Performance

| #  | Issue                                                                                                               | Severity      | Proposed fix                                                                      |
|----|---------------------------------------------------------------------------------------------------------------------|---------------|-----------------------------------------------------------------------------------|
| P1 | Hardcoded <code>limit</code> (200/500) on list queries — <b>silent truncation</b> and unbounded scan as data grows. | <b>Medium</b> | Real pagination (cursor/offset) — pairs with the planned <code>DataTable</code> . |
| P2 | No obvious N+1 — queries use <code>with</code> joins and grouped counts.                                            | —             | None.                                                                             |

|    |                                                                                                                                     |     |                                                                                    |
|----|-------------------------------------------------------------------------------------------------------------------------------------|-----|------------------------------------------------------------------------------------|
| P3 | Mostly Server Components (good); bundle not analyzed; no <code>next/dynamic</code> for heavy client islands (charts, search sheet). | Low | Run <code>@next/bundle-analyzer</code> ; lazy-load chart/search client components. |
|----|-------------------------------------------------------------------------------------------------------------------------------------|-----|------------------------------------------------------------------------------------|

DB is well-indexed (**53 indexes**) and Drizzle is parameterized — no slow-query red flags in static review (runtime profiling recommended).

## 4. Code Quality

| #  | Issue                                                                                                                                | Severity | Proposed fix                                      |
|----|--------------------------------------------------------------------------------------------------------------------------------------|----------|---------------------------------------------------|
| Q1 | Redundant auth: <code>suppliers/[id]</code> calls <code>requireAgencyUser()</code> then <code>getSupplierById</code> calls it again. | Low      | Drop the page-level call (helper already guards). |
| Q2 | 47 ESLint <code>import/order</code> warnings.                                                                                        | Low      | <code>pnpm lint --fix</code> (auto-fixable).      |
| Q3 | No dead code or duplicated logic found; <code>src/lib</code> (domain/queries/analytics) and domain-folded components are clean.      | —        | None.                                             |

## 5. UX

| #  | Issue                                                                           | Severity | Proposed fix                                                                          |
|----|---------------------------------------------------------------------------------|----------|---------------------------------------------------------------------------------------|
| U1 | Loading states missing on most routes (only <code>assistant/dashboard</code> ). | Medium   | Add <code>loading.tsx</code> per route using the existing <code>skeleton.tsx</code> . |
| U2 | Error states missing (see E2).                                                  | Medium   | Add <code>error.tsx</code> .                                                          |
| U3 | Tables overflow on mobile (no <code>overflow-x-auto</code> ).                   | Medium   | Wrap tables; ship the responsive <code>DataTable</code> .                             |
| U4 | Accessibility not deeply audited.                                               | Low      | Run the web-interface-guidelines pass (focus order, labels, contrast).                |
| U5 | Empty states are present and filter-aware.                                      | —        | None (good).                                                                          |

## 6. Database

| #  | Issue                                                                                                                                                         | Severity | Proposed fix                                                                         |
|----|---------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|--------------------------------------------------------------------------------------|
| D1 | <b>No soft delete</b> ( <code>deletedAt</code> ) anywhere; FK cascades perform <b>hard</b> deletes — conflicts with the "everything is reversible" principle. | Medium   | Add nullable <code>deletedAt</code> to core tables + filter reads (needs migration). |
| D2 | <code>reference</code> only on <code>booking/product</code> ; <code>updatedAt/status/notes</code> inconsistent on child tables.                               | Low      | Backfill per the entity standard (needs migration).                                  |

|    |                                                                                                                                                       |   |                |
|----|-------------------------------------------------------------------------------------------------------------------------------------------------------|---|----------------|
| D3 | FKs (45 <code>onDelete</code> : 25 cascade / 20 set null), indexes (53), sequential migrations (latest 0017), <code>generate→migrate</code> workflow. | — | Verified good. |
|----|-------------------------------------------------------------------------------------------------------------------------------------------------------|---|----------------|

## 7. Files modified (this audit)

- `src/app/(app)/bookings/[id]/page.tsx` — agent own-scope on detail query
- `src/app/(app)/clients/[id]/page.tsx` — agent own-scope on detail query
- `src/app/(app)/opportunities/[id]/page.tsx` — agent own-scope on detail query
- `src/app/(app)/products/[id]/page.tsx` — agent own-scope on detail query

(Earlier this session: list-page agent scoping 134ceb3; quality-gate repair 7d5f4e1.) Verified: `npm run check (lint + tsc)` passes clean.

## Remaining recommendations (priority order)

1. **Rate limiting** (S2) — highest-value security add; small, middleware-level.
2. **Soft delete migration** (D1) — unblocks reversibility *and* GDPR erasure.
3. **Error boundaries + loading states** (E2/U1/U2) — mechanical, big UX win; `skeleton.tsx` already exists.
4. **Real pagination** (P1) — fold into the shared `DataTable` build.
5. **Rotate committed demo credentials** (S3).
6. **Standardize action error handling** (E1) on the `ActionResult` shape.
7. **Lint --fix + bundle analysis** (Q2/P3).

All items are tracked in the spec-vs-reality gap tracker.

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# Owner's-Eye UX Audit

**Date:** 2026-06-28 · **Lens:** "I own this travel agency and use Atlas all day." Walking every workflow looking for friction. **Recommendations only — nothing implemented.**

How to read severity: **P0** = trips me up daily / costs real time · **P1** = recurring annoyance · **P2** = polish.

## Top findings (prioritized)

| # | Issue                                                                       | Sev | Why it hurts                                                                                                                                                           |
|---|-----------------------------------------------------------------------------|-----|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 | <b>Opportunities page is orphaned from the nav</b>                          | P0  | My core sales list isn't in the sidebar; I can only reach deals via a client or the "Operations" board. I lose my pipeline.                                            |
| 2 | <b>"Operations" is really the sales pipeline</b>                            | P0  | The label says fulfillment/ops; it's actually the deal board. I waste clicks hunting for where my pipeline lives — and it overlaps Opportunities + the Bookings board. |
| 3 | <b>Can't create a booking without first creating a client</b>               | P0  | A walk-in/phone booking forces me to stop, go create a client, come back. Two screens for one sale.                                                                    |
| 4 | <b>Nav order contradicts daily work</b>                                     | P1  | Finance/Commissions/Reports sit <i>above</i> Bookings/Clients. My most-used screens are buried under money screens I open weekly.                                      |
| 5 | <b>Three money destinations</b> (Finance, Commissions, Reports)             | P1  | I never know which to open for "how much did we make?" Overlapping, unlabeled difference.                                                                              |
| 6 | <b>Fragmented sourcing</b> (Search vs Hotels vs Assistant vs inline)        | P1  | Four ways to look up inventory. I forget which finds what.                                                                                                             |
| 7 | <b>Sales→Proposal is manual</b>                                             | P1  | Won an opportunity? I re-enter trip details into a new proposal. Duplicated work.                                                                                      |
| 8 | <b>Inconsistent terminology</b> (Proposal/Product, Opportunity, Operations) | P2  | Same thing has different names across URL, page, and conversation.                                                                                                     |
| 9 | <b>No nav grouping</b> (15 flat items)                                      | P2  | A wall of links; no Sales / Ops / Finance / Admin sections to scan.                                                                                                    |

## Workflow walkthroughs

### Finding new business → my pipeline (P0)

**What I see:** The sidebar has **Operations** (which opens a Pipeline kanban) but **no Opportunities link** — even though there's a full `/opportunities` page. So my deals live in two places with two names,

and the "real" one isn't in the menu.

**Why it hurts:** My pipeline is the heartbeat of the agency. Making me guess between "Operations" and a hidden Opportunities page costs clicks every single day and makes the tool feel unfinished.

**Better workflow:** One nav item — **"Sales"** or **"Pipeline"** — that opens the board with a list/board toggle (the toggle already exists on Bookings). Retire the "Operations" name. One pipeline, one entry point.

## Taking a booking (P0)

**What I see:** On *New booking*, if the client doesn't exist I get "No clients yet. Add a client first" — I must leave, create the client, and return.

**Why it hurts:** Half my bookings start as a phone call from someone not yet in the system. The client-first rule is right for *data*, but it shouldn't be a *detour*. That's two screens and a lost context switch for one sale.

**Better workflow:** Inline " + New client" inside the booking form (a quick create that returns the new client selected), or a combined "New booking" that captures a minimal client on the same screen. Create the client record *as a side effect* of the sale, not a prerequisite step.

## Building the quote (P1)

**What I see:** I win an opportunity, then open *New proposal* and re-type the destination, dates, and value I already captured on the opportunity. Convert proposal→booking is one click (great) — but opportunity→proposal isn't.

**Why it hurts:** Re-entering the same trip details is duplicated work and a place for transcription errors. It breaks the "never ask twice" principle right in the middle of my money-making flow.

**Better workflow:** A **"Create proposal" button on the opportunity** that pre-fills client, destination, dates, and value (the AI quote builder already exists — wire it to the opportunity). Won → proposal in one click, mirroring proposal → booking.

## Seeing the money (P1)

**What I see:** Finance, Commissions, and Reports are three separate nav items, consecutive, with no stated difference.

**Why it hurts:** As the owner the #1 question is "how are we doing?" Three doors for that answer means I learn-and-forget which is which.

**Better workflow:** Group them under one **"Finance"** section with sub-tabs (Overview · Commissions · Export). One mental model, drill in for detail.

## Sourcing flights/hotels (P1)

**What I see:** A **Search** page, a separate **Hotels** module, the **Assistant**, and an inline search sheet on the booking — four entry points.

**Why it hurts:** I can't form a habit. "Where do I price a hotel?" has four answers depending on context.

**Better workflow:** Make the **in-booking search the primary path** (I'm almost always sourcing *for a trip*), and frame standalone Search/Hotels as "explore without a booking." Fold the Assistant's search into the same sheet.

## Terminology inconsistencies (P2)

Same concept, different words across URL / page title / everyday speech:

| Concept           | URL                              | UI label                 | Agency speak                             | Fix                                                                     |
|-------------------|----------------------------------|--------------------------|------------------------------------------|-------------------------------------------------------------------------|
| Quote/<br>offer   | /products                        | "Proposals"              | "quote" /<br>"proposal"                  | Rename route+code to <code>proposals</code> ; pick one word everywhere. |
| Sales<br>pipeline | /operations + /<br>opportunities | "Operations" +<br>(none) | "pipeline" /<br>"deals" /<br>"enquiries" | One name ("Pipeline"), one page.                                        |
| Deal              | /opportunities                   | "Opportunity"            | often "enquiry" /<br>"lead"              | Confirm the agency's word; use it consistently.                         |

Mismatched names make training new staff slower and make the product feel stitched together.

## Missing automation (P1 — productivity multipliers)

As an owner I want the system to do the obvious next step:

- New client → **welcome email** (not sent today)
- Opportunity won → **draft proposal** (manual today)
- Booking confirmed → **invoice** (on-demand PDF today)
- Travel completed → **review request** (not built)

Each is a manual chore I'll forget under load. These are the automation triggers — building them is the difference between a record-keeping tool and one that runs the agency.

## What already feels good (keep it)

- **Getting-started checklist** — I knew what to do on day one.
- **Booking lifecycle stepper** with "Advance to next" — I always know the state and the next action. This is exactly the "what do I do next?" principle done right.
- **Convert proposal** → **booking** in one click — more of this, please.
- **Client timeline** (activity + payments + notifications in one view) — this is how I think about a client.

- **Empty states with a CTA** — never a dead end.
- 

## Recommended sequence

---

1. **Fix the pipeline confusion** (#1, #2) — rename/merge Operations + Opportunities into one "Pipeline" nav item. Cheap, high daily relief.
2. **Inline client create in booking** (#3) — removes the worst detour.
3. **Reorder + group the nav** (#4, #9) — daily work on top, grouped sections.
4. **Opportunity** → **proposal generation** (#7) — wire the existing AI quote builder.
5. **Consolidate Finance** (#5) and **terminology pass** (#8).
6. **Automation triggers** (welcome email, invoice, review) as capacity allows.

These are UX/workflow recommendations; the engineering gaps behind some of them (automation, pagination, DataTable) are tracked in the gap tracker.



# CTO Review — Atlas

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**Date:** 2026-06-28 · **Reviewer:** CTO (pre-first enterprise customer) **Verdict:** Atlas is a well-structured, ambitious product with real engineering quality — but it is not enterprise-ready today. Several issues would cause visible failures under real-world load or scrutiny. This report is honest, not diplomatic.

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## Executive Summary

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| Dimension            | Grade | One-line verdict                                                                                   |
|----------------------|-------|----------------------------------------------------------------------------------------------------|
| Architecture         | B+    | Solid Next.js 16/Drizzle/Neon foundation; DB connection model wrong for serverless                 |
| Security             | B     | Tenant isolation real; no middleware, no CSP, no rate limiting                                     |
| Database             | B+    | Good schema; soft delete missing; 11 tables missing <code>updatedAt</code>                         |
| Folder structure     | A-    | Clean and predictable; minor naming confusion (products = proposals)                               |
| UI                   | B     | Design system consistent; no loading/error states; mobile overflow                                 |
| UX                   | C+    | Core flows work; 3 P0 navigation problems; no automation                                           |
| Business logic       | B     | Core RBAC solid; reference generation has race condition                                           |
| Performance          | C+    | Finance page does 4 sequential awaits; DB connection wrong driver                                  |
| Scalability          | C     | No connection pooler; no caching layer; no queue; single DB point of failure                       |
| API design           | B-    | REST-style routes adequate; no versioning; chat endpoint has no body size limit                    |
| Developer experience | C+    | No CI/CD; one test file; no seed script for fresh dev; two DB drivers in <code>package.json</code> |
| Maintainability      | B     | <code>bookings.ts</code> at 976 lines needs splitting; products/proposals naming confusing         |
| Testing              | D     | One test file in the entire codebase. Not acceptable for enterprise.                               |
| Deployment           | B-    | Vercel is fine; no health endpoint; no rollback plan documented                                    |
| CI/CD                | F     | Zero — no <code>.github/workflows/</code> . No automated checks before merge.                      |
| Documentation        | A-    | Unusually thorough for this stage; gap tracker is a standout                                       |
| Code quality         | B+    | Consistent patterns; 45 raw <code>console</code> calls instead of a real logger                    |

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## CRITICAL ISSUES

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*Must be fixed before the first enterprise customer goes live.*

---

## C1 — No database connection pooler

**File:** `src/lib/db.ts`

```
const client = postgres(connectionString); // wrong driver for serverless
```

Atlas uses `postgres` (the `postgres.js` driver) on Vercel serverless functions. Every function invocation opens a **new TCP connection** to Neon. At 50 concurrent users that's 50 open connections. At 200 it exhausts Neon's connection limit and every query starts failing with "too many clients."

Neon provides a **serverless pooler** (`-pooler` suffix in the connection string) that multiplexes thousands of connections through ~10 persistent ones — but you need to switch to the `@neondatabase/serverless` driver to use it correctly over HTTP (not TCP), which is mandatory for Vercel edge/serverless.

You also have **two DB drivers** (`postgres` + `pg`) in `package.json` with no clear owner — `pg` is for Drizzle migrations, `postgres` for runtime. This needs to be explicit and documented.

**Impact:** 200 concurrent users = database down. Enterprise customer demos in front of their IT team = connection pool exhaustion at the worst moment.

**Fix:**

```
npm install @neondatabase/serverless
```

```
// src/lib/db.ts
import { neon } from "@neondatabase/serverless";
import { drizzle } from "drizzle-orm/neon-http";
const sql = neon(process.env.POSTGRES_URL!);
export const db = drizzle(sql, { schema });
```

Keep `pg` only for `drizzle-kit` migrations (it needs a real TCP connection).

---

## C2 — Zero CI/CD pipeline

**Finding:** No `.github/workflows/` directory. Nothing runs automatically on `git push`.

This means:

- A broken build can be pushed to `main` and deployed to production instantly
- No automated type checking, linting, or tests before deploy
- The quality gate (`npm run check`) only runs if someone remembers to run it
- No way to enforce code review before merge

**Impact:** The first enterprise customer will eventually hit a regression caused by an unreviewed push.

At that point you have no safety net and no audit trail.

**Fix:** Create `.github/workflows/ci.yml`:

```
name: CI
on: [push, pull_request]
jobs:
  check:
    runs-on: ubuntu-latest
    steps:
      - uses: actions/checkout@v4
      - uses: actions/setup-node@v4
        with: { node-version: 20 }
      - run: npm install --legacy-peer-deps
      - run: npm run check          # lint + typecheck
      - run: npm run build:ci       # next build
```

Vercel already deploys on push — CI just adds the gate before it reaches Vercel.

---

## C3 — No test coverage

**Finding:** One test file exists: `.claude/worktrees/*/src/lib/references.test.ts` — not even in the main source tree. Zero tests for:

- Server actions (RBAC, tenant scoping, business logic)
- API routes (auth, input validation)
- Domain logic (`domain.ts` — roles, capabilities, booking lifecycle rules)
- Utility functions

**Impact:** Any refactor silently breaks business rules. Enterprise customers expect vendors to have a test suite. Without it, you cannot safely hire additional engineers (they will introduce bugs you won't catch).

**Fix (prioritized, not everything at once):**

1. Integration tests for the 5 most critical server actions: `createBooking`, `advanceBookingStatus`, `acceptProposalByToken`, `createCommissions`, `createAgency`
  2. Unit tests for `domain.ts` (RBAC logic is pure functions — trivial to test)
  3. Tenant isolation test already exists as a script (`test-tenant-isolation.ts`) — promote it to the CI pipeline
- 

## C4 — No middleware (rate limiting, security headers enforcement)

**Finding:** No `src/middleware.ts`. This means:

- Login endpoint (`/api/auth/sign-in`) has no rate limiting — brute-forceable
- Public proposal endpoint (`/p/[token]`) has no rate limiting
- Portal magic-link request has no rate limiting — email flooding

- Security headers (set in `next.config.ts`) work but a middleware could short-circuit abuse before it hits server functions

CSP (Content-Security-Policy) is also absent from `next.config.ts`. Every major browser security audit flags missing CSP — an enterprise customer's security team will catch this immediately.

**Impact:** Any motivated attacker can brute-force login, flood the magic-link email, or scrape tokenized public proposals. An enterprise security review fails.

**Fix:**

```
// src/middleware.ts
import { NextResponse } from "next/server";
import type { NextRequest } from "next/server";
// Use Upstash ratelimit (Redis-backed, works on Vercel Edge)
export function middleware(req: NextRequest) {
  // 1. Rate limit /api/auth and /api/portal/auth/request
  // 2. Add Content-Security-Policy header
  return NextResponse.next();
}
export const config = { matcher: ["/api/auth/:path*", "/api/portal/:path*"] };
```

## C5 — Race condition in booking/proposal reference generation

**File:** `src/lib/actions/bookings.ts:55-68`

```
// Fetches ALL booking references for the agency, iterates in JS to find max
const rows = await db.select({ reference: booking.reference })
  .from(booking).where(eq(booking.agencyId, agencyId));
let max = 1000;
for (const r of rows) {
  const n = Number.parseInt(r.reference.replace(/\D/g, ""), 10);
  if (Number.isFinite(n) && n > max) max = n;
}
return `BKG-${max + 1}`;
```

At 10,000 bookings per agency this loads the full reference list into memory on every creation. Worse: two simultaneous booking creations can read the same max and produce duplicate references (unique constraint catches it, but the UX is a silent retry loop, not an atomic increment).

**Impact:** Agencies with high booking volume (enterprise customers) hit slow booking creation and occasional silent retries. Could silently fail under load.

**Fix:** Atomic SQL max:

```
import { max as sqlMax } from "drizzle-orm";
const [row] = await db
  .select({ max: sqlMax(booking.numericReference) })
  .from(booking)
  .where(eq(booking.agencyId, agencyId));
```

```
const next = (row?.max ?? 1000) + 1;
return `BKG-${next}`;
```

Requires adding a `numericReference` integer column (same migration as soft delete).

---

## HIGH PRIORITY

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*Fix within the first sprint after launch.*

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### H1 — No soft delete

**File:** `src/lib/schema.ts`

No `deletedAt` column exists anywhere. The "everything is reversible" founding principle is aspirational only. An agency that accidentally deletes a booking has no recovery path. This also blocks GDPR erasure (you can't "erase" what you can't soft-delete and then hard-purge on schedule).

**Fix:** Migration adding `deletedAt: timestamp("deleted_at")` to core tables (booking, client, opportunity, product), filter all reads with `.where(isNull(entity.deletedAt))`, and a restore action per entity.

---

### H2 — `requireAgencyUser` makes a DB query on EVERY page render

**File:** `src/lib/permissions.ts:117-132`

```
const ag = await db.query.agency.findFirst({
  where: eq(agency.id, user.agencyId),
  columns: { status: true, subscriptionStatus: true },
});
```

This query fires on every authenticated server render. At 1,000 concurrent users that's 1,000 extra agency-status queries per second. The agency status doesn't change often — it's a perfect cache candidate.

**Fix:** Cache the agency status in the session token (updated on suspension/ reactivation) or use `unstable_cache` with a short TTL and `revalidateTag` on suspension events.

---

### H3 — Finance page does 4 sequential DB queries (633 lines)

**File:** `src/app/(app)/finance/page.tsx`

The page is enormous (633 lines) and makes 4+ DB queries, some sequential. It will be slow as the dataset grows and impossible to maintain.

**Fix:** Split into smaller async components using React Suspense, parallelizing queries with `Promise.all` or per-section server components. Cap the page at 150 lines of layout; extract `<PaymentsSummary>`, `<CommissionsSummary>`, etc.

---

## H4 — No structured logging (45 raw `console.*` calls in actions)

**Finding:** 45 `console.log/error/warn` calls scattered across action files. On Vercel these show up as unstructured text in Function Logs with no correlation IDs, no severity levels, no timestamps, no request tracing.

**Impact:** When an enterprise customer files a support ticket saying "booking creation failed at 14:32 UTC," you cannot find the relevant log entry.

**Fix:** A one-file logger:

```
// src/lib/logger.ts
export const logger = {
  info: (msg: string, meta?: object) =>
    console.log(JSON.stringify({ level: "info", msg, ...meta, ts: new Date().toISOString() })),
  error: (msg: string, meta?: object) =>
    console.error(JSON.stringify({ level: "error", msg, ...meta, ts: new Date().toISOString() })),
};
```

Replace all `console.*` calls. Ship a correlation ID from the request context.

---

## H5 — `bookings.ts` is 976 lines — unmaintainable

**File:** `src/lib/actions/bookings.ts`

Nearly 1,000 lines covering booking CRUD, lifecycle advancement, traveller management, item management, itinerary, and commission generation. A new engineer cannot reason about it. Every bug fix risks breaking something else in the file.

**Fix:** Split into:

- `bookings/crud.ts` — create, update, delete
  - `bookings/lifecycle.ts` — advance status, prerequisites
  - `bookings/travellers.ts` — passenger CRUD
  - `bookings/items.ts` — trip service items
  - `bookings/commissions.ts` — commission generation
- 

## H6 — No health check endpoint

**Finding:** No `/api/health` route. Vercel's uptime monitors, load balancers, and the enterprise customer's infrastructure team expect one.

**Fix:**

```
// src/app/api/health/route.ts
export async function GET() {
  return Response.json({ status: "ok", ts: new Date().toISOString() });
}
```

---

## H7 — Products are called "proposals" everywhere but the URL and the DB

**Finding:** The concept is **Proposal** in all UI, docs, and conversation. But the database table is `product`, the URL is `/products`, the actions file is `products.ts`, the type is `ProductInput`. An enterprise customer's developer integrating via the API will be confused, and new engineers will be too.

**Fix:** Rename — DB table to `proposal`, route to `/proposals`, action file to `proposals.ts`. One migration, one find-replace. Do it now before the codebase gets larger.

---

## H8 — No input validation on chat endpoint (body size / message length)

**File:** `src/app/api/chat/route.ts`

No body size limit, no message length cap, no token budget per request. A user (or attacker) can send a 100KB message that gets passed to the LLM, costing significant API spend.

**Fix:** Add `export const maxDuration = 30` (Vercel), validate message array length and individual message size before calling the LLM, and add the rate limiter from C4.

---

## MEDIUM PRIORITY

*Address in the first 60 days.*

---

### M1 — 11 tables missing `updatedAt`

11 of 24 tables have no `updatedAt` — impossible to answer "when was this record last changed?" for audit, support, or sync scenarios.

### M2 — No caching layer whatsoever

Dashboard, finance, analytics — every render hits the DB fresh. `unstable_cache` with `revalidateTag` on mutations would cut DB load by ~60% for read-heavy pages.

### M3 — Opportunities page not in navigation

The core sales pipeline (`/opportunities`) is not in the sidebar — agents can only reach it via the operations board or a client page. Discovered in the owner UX audit.

#### **M4 — "Operations" label wrong**

`/operations` is the pipeline kanban — "Operations" implies fulfillment/logistics. Enterprise buyers will be confused during demos.

#### **M5 — No background jobs / queue**

Sending emails, generating PDFs, syncing hotel content, processing webhooks — all done synchronously in request handlers. Slow operations block the user. Vercel serverless has a 10s default timeout.

#### **M6 — Amadeus marked as "legacy" but still in the active provider selection**

`getFlightSupplier()` still falls back to Amadeus. The codebase comment says "decommissioned 2026-07-17". Remove it before it becomes a dead code maintenance burden.

#### **M7 — No pagination on any list page**

All list queries use hardcoded `LIMIT 200` or `LIMIT 500`. An agency with 5,000 clients or 10,000 bookings silently gets a truncated list with no indication that data is missing. This is a data integrity perception problem for enterprise.

#### **M8 — Portal session expiry not enforced on the server**

The portal session expiry check (`gt(portalSession.expiresAt, new Date())`) exists but the TTL is not clearly documented and there's no cleanup job to remove expired sessions. Old sessions accumulate forever.

#### **M9 — No GDPR compliance path**

No data export per client, no erasure flow, no data retention policy, no consent tracking. EU travel agencies (your primary market) are legally required to provide these. Blocked by missing soft delete (H1).

#### **M10 — Two PostgreSQL drivers in package.json**

`postgres` (runtime) and `pg` (drizzle-kit migrations) coexist with no explanation. Confusing for new engineers and a potential version conflict risk.

---

## **NICE-TO-HAVE**

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*Post-stabilization improvements.*



---

## N1 — No PR template or contribution guidelines

No `PULL_REQUEST_TEMPLATE.md`, no `CONTRIBUTING.md`. Enterprise customers who want to contribute integrations or customizations have no guidance.

## N2 — `picsum.photos` in production image whitelist

`next.config.ts` whitelists `picsum.photos` (lorem ipsum placeholder images) in production. This is a dev-only domain — remove it from production config.

## N3 — `OPENROUTER_MODEL` defaults to `openai/gpt-5-mini`

The env default in `env.ts` references `gpt-5-mini` — a model that may not exist or may be deprecated. This should be an explicit, versioned model ID.

## N4 — No API versioning

All API routes are unversioned (`/api/chat`, `/api/export`). When you change the chat protocol or export format, existing integrations break with no migration path. Add `/api/v1/` prefixing now while there are no external consumers.

## N5 — Finance page at 633 lines is both a performance and maintainability problem

Covered in H3 — also a nice-to-have style cleanup even after the Suspense split.

## N6 — No Storybook or component documentation

UI components have no visual documentation. When a designer or new frontend engineer joins, they have no way to browse the component library.

## N7 — `PROTECTED_DB_HOSTS` is a soft guard

The protection against running destructive scripts against prod is a hostname string comparison. It can be bypassed. Not critical today, but consider a proper DB IAM role with no DROP/TRUNCATE in production.

---

## What is genuinely good

I want to be clear: most codebases I review at this stage are far messier. These are the things that show real engineering judgment:

- **Tenant isolation** is correctly implemented and tested — `agencyId` on every table, enforced in every query, with a script that verifies no leaks.
- `ActionResult<T>` is a clean, consistent error-handling contract.

- **Zod validation** exists and is used, even if not universal.
- **The domain model** (`domain.ts`) is well-organized — roles, capabilities, enums, status metadata in one place.
- **The provider abstraction** (`src/lib/suppliers/`) is well-designed — the new `providers/` layer with capability-segmented interfaces is production-quality.
- **Security headers** in `next.config.ts` (X-Frame-Options, X-Content-Type-Options, Referrer-Policy) — most startups forget these.
- **The documentation** is the best I have seen at this stage. The gap tracker, the owner UX audit, the architecture decisions — this saves weeks of onboarding.
- `env.ts` validates all environment variables at startup with Zod — catches misconfigured deploys immediately.

## Priority matrix

| Priority | Issue                            | Effort  | Risk if ignored                   |
|----------|----------------------------------|---------|-----------------------------------|
| Critical | C1: No connection pooler         | 1 day   | DB down at 200 users              |
| Critical | C2: No CI/CD                     | 1 day   | Broken builds reach prod          |
| Critical | C3: No tests                     | 2 weeks | Regressions on every change       |
| Critical | C4: No middleware/rate limiting  | 2 days  | Brute force, abuse, CSP fail      |
| Critical | C5: Reference race condition     | 2 hours | Duplicate refs under load         |
| High     | H1: No soft delete               | 3 days  | Unrecoverable deletes, GDPR block |
| High     | H2: Agency query on every render | 4 hours | Slow at scale                     |
| High     | H3: Finance page 633 lines       | 2 days  | Slow, unmaintainable              |
| High     | H4: No structured logging        | 4 hours | Blind in production               |
| High     | H5: bookings.ts 976 lines        | 2 days  | Bugs, slow PRs                    |
| High     | H6: No health endpoint           | 30 min  | Monitoring blind                  |
| High     | H7: products = proposals naming  | 1 day   | Developer confusion forever       |
| High     | H8: Chat no input limit          | 2 hours | LLM cost abuse                    |
| Medium   | M1–M10                           | Various | Quality and compliance debt       |

## Recommended 30-day plan before enterprise onboarding

### Week 1 — Non-negotiable hardening

- C1: Switch to `@neondatabase/serverless` driver
- C2: Add GitHub Actions CI (lint + typecheck + build)
- C4: Create `middleware.ts` with rate limiting + CSP header

- C5: Fix reference generation with atomic SQL max
- H6: Add `/api/health` endpoint

## Week 2 — Reliability

- H1: Soft delete migration for core tables
- H4: Replace `console.*` with structured logger
- H8: Chat endpoint body size + rate limit
- H2: Cache agency status

## Week 3 — Code health

- H5: Split `bookings.ts` into 5 files
- H7: Rename `products`→`proposals` (DB + routes + components)
- M3/M4: Fix navigation (add Opportunities, rename Operations)
- C3: Start: integration tests for 5 critical actions

## Week 4 — Enterprise readiness

- M9: Skeleton GDPR data export per client
- M5: Evaluate Vercel background tasks or QStash for async work
- H3: Split finance page with Suspense
- C3: Continue test coverage

This plan gets Atlas from "impressive prototype" to "defensible enterprise product" in 30 days. The bones are good. The hardening is what separates a product people trust with their revenue from one they pilot and abandon.

# Decision 0001 — Navigation IA restructure

---

**Date:** 2026-06-28 **Status:** accepted **Deciders:** Product owner + CTO review

## Context

---

Atlas had 15 flat navigation items in an order that contradicted the golden workflow (Finance appeared above Clients; Opportunities was entirely missing from the nav; Search and Hotels were two separate items for the same user task; Operations was a duplicate of the Bookings board view). An IA audit (docs/owner-ia-audit.md) identified these as P0/P1 usability issues.

## Decision

---

Restructure navigation into 5 labelled sections following the golden workflow: **WORK** (Clients → Pipeline → Proposals → Bookings) · **SOURCING** (Flights · Hotels) · **FINANCE** · **TOOLS** · **ADMIN**. Rename `/products` → `/proposals` canonically (URL rewrite). Retire Operations as a nav item. Create `/sourcing/flights` as a dedicated page.

## Reasoning

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- Section-grouped nav reduces cognitive load from 15 flat items to 4-5 scannable groups
- Workflow order (Clients first, Finance last) matches how agents think
- Sourcing section scales cleanly to Cars, Cruises, Packages without nav restructure
- `/proposals` label matches what agents, clients and every competitor call the concept
- Operations was a duplicate of the Bookings board toggle — removing it eliminates confusion

## Alternatives considered

---

| Option                              | Why rejected                                                                    |
|-------------------------------------|---------------------------------------------------------------------------------|
| Keep flat nav, just reorder         | Doesn't solve the 15-item cognitive load or the Search/Hotels duplication       |
| Merge Search + Hotels into one page | Reasonable but defers the Sourcing section scaling problem                      |
| Rename Operations → Pipeline        | The real pipeline is Opportunities; Operations was a board view, not a pipeline |

## Consequences

---

- Easier: adding new sourcing verticals (Cars, Cruises) — one line in NAV\_SECTIONS
- Easier: onboarding new agents — nav reads like the job

- Harder: old `/products` and `/search` bookmarks need redirects (added to `next.config.ts`)
- Old URLs preserved via 301 redirects (`products`) and 307 redirects (`search`, `hotels`, `operations`)

## Related

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- `docs/owner-ia-audit.md` (full IA analysis)
- `src/components/app/app-shell.tsx` (implementation)
- `next.config.ts` (rewrites + redirects)
- `src/app/(app)/sourcing/flights/page.tsx` (new page)

# Decision 0002 — Agency Network is a Stage 3 Direction

---

**Date:** 2026-06-28 **Status:** accepted (direction), deferred (implementation) **Deciders:** Founding team + board

## Context

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Board feedback on the strategic audit introduced the concept of Atlas as infrastructure connecting agencies to each other, not just software serving agencies individually. Agency A cannot fulfil a complex request; Atlas surfaces Agency B — a specialist — who can. Revenue sharing, inventory exchange, trusted subcontractor relationships, shared local suppliers. Network effects from participant relationships, not code.

This was not in the original strategic audit and represents a meaningfully stronger moat than data accumulation alone. Network effects in B2B do not decay; an agency earning revenue through Atlas referrals will not leave the platform.

## Decision

---

Name this as the Stage 3 strategic direction. Begin designing for it in the data model now at minimal cost. Delay the product surface until Atlas has 300+ agencies (Stage 3 threshold).

## Reasoning

---

Network effects require critical mass before generating value. At 50 agencies, the network is too thin for meaningful referral flow. Building the product surface before critical mass exists creates engineering overhead without corresponding user value and fragments attention from the supplier booking and automation work that must happen first to reach the scale where the network becomes possible.

Designing for it in the data model costs almost nothing. Agency records should carry specialisation attributes, destination expertise tags, and capacity signals from the beginning — these enable future matching without requiring a current product surface.

## What this means now (Stage 1–2)

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- Agency schema should include `specialisations`, `destinations_served`, and `agency_type` as structured fields (not free text) to enable future matching
- Activity and booking data should be structured to allow anonymised capability inference over time
- Do not build any product surface for inter-agency connection yet

- Do not market this capability yet — announcing a network that does not exist creates expectation debt

## What this means at Stage 3 (300+ agencies)

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- Surface: "Agency B specialises in this destination and has capacity"
- Revenue sharing mechanism for fulfilled referrals
- Trusted subcontractor relationships formalised in the platform
- Inventory and resource exchange between agencies

## Alternatives considered

---

| Option                                | Why deferred/rejected                                                                                           |
|---------------------------------------|-----------------------------------------------------------------------------------------------------------------|
| Build the network now                 | Critical mass is required; building before scale creates overhead without value                                 |
| Never build the network               | This is the strongest long-term moat available; intentionally deferring is different from discarding            |
| Build it as a marketplace immediately | Marketplace dynamics require both sides; agencies are not yet numerous enough to form a functioning marketplace |

## Consequences

---

- Stronger: when Atlas reaches 300 agencies, the network surface can be activated on data infrastructure already in place
- Harder: resisting the temptation to build this early when it is clearly a good idea
- Unchanged: no current engineering work required beyond schema attributes

## Related

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- docs/strategic-audit.md — the original moat analysis this extends
- docs/strategy-board-response.md — the board feedback that introduced this direction
- docs/domain.md — agency entity where specialisation attributes should be added

# Decision 0003 — Sprint 1 Booking Architecture

---

**Date:** 2026-06-28 **Status:** accepted **Deciders:** engineering (Atlas Sprint 1)

## Context

---

Atlas had functional search (Duffel flights, Hotelbeds hotels) but no complete booking workflow. Supplier references were stored as untyped JSONB. There was no price revalidation, no idempotency, no analytics event stream, and no clear environment separation between development, sandbox, and production. The `BookingProvider` interface existed in `providers/types.ts` but nothing used it.

The goal: build the full booking architecture in sandbox so that production credentials can be plugged in with configuration changes only — no code changes.

## Decision

---

Build a layered booking architecture: **config** → **registry** → **provider adapters** → **booking services** → **server actions** → **UI**. Each layer is independent and testable. The UI never sees supplier types. Environment is resolved once in a config module.

## Reasoning

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### Why consolidate to the new `BookingProvider` interface now

The legacy `SupplierProvider` (search-only, both verticals in one class) was a stepping stone. The new interface is capability-segmented and multi-step (search → quote → book → cancel). Building Sprint 1 on top of the new interface means provider adapters written now are the same ones production uses — no "sandwich migration" later.

### Why four new database tables

`booking_supplier_ref` — supplier references must be queryable (status polling, cancellation, reporting). Storing them in JSONB is schema-less and unindexable.

`booking_event` — `activity_log` is too coarse for a booking funnel. A dedicated append-only event table serves as both the compliance audit trail and the analytics source without a third-party SDK dependency. One write, two consumers.

`booking_document` — vouchers and e-tickets are not line items. Separating them into their own table allows typed retrieval and future Vercel Blob integration.

`booking_idempotency` — without key tracking, a browser retry or a server-action re-execution after a timeout can create duplicate supplier orders. The key is derived deterministically so it is stable across retries for the same booking attempt.



## Why a single config module (`suppliers/config.ts`)

Scattered `isDuffelConfigured()` / `process.env.*` checks across multiple files make it impossible to reason about which provider is active. A single config module resolves `EnvironmentTier` once and returns typed `FlightProviderConfig` / `HotelProviderConfig` objects. Providers receive config objects — they never read `process.env` directly.

## Why in-process retry (no job queue)

Sprint 1 is sandbox-only. Async job queues (pg-boss, BullMQ) add operational complexity that is not justified until production traffic. The `BookingService` retry handler is synchronous and typed by error code (retryable errors get exponential backoff; non-retryable errors surface immediately to the UI).

## Why post-booking stubs (not omitted)

Defining `cancel`, `refreshStatus`, `getVoucher`, `modifyPassengers` as typed interfaces with `NotImplementedError` now means Sprint 2 only fills in implementations — the UI is already wired against stable types. Omitting the interfaces forces Sprint 2 to design and implement simultaneously.

## Alternatives considered

| Option                                                         | Why rejected                                                                                                            |
|----------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------|
| Continue using <code>SupplierProvider</code> (legacy)          | Would require a second migration when production arrives; two abstraction layers for the same concern                   |
| Store supplier refs in <code>booking_item.details</code> JSONB | Unqueryable, no index, no schema guarantee, impossible to cancel/poll                                                   |
| Third-party analytics SDK (PostHog, Segment)                   | External dependency with no offline fallback; <code>booking_event</code> table is queryable and sufficient for Sprint 1 |
| Redis for idempotency keys                                     | Redis is not in the stack; DB table with TTL expiry achieves the same guarantee without a new service                   |
| Async job queue for retry                                      | Premature for sandbox; adds ops burden before production traffic justifies it                                           |

## Consequences

### Easier:

- Adding a new provider (TravelgateX, Expedia) requires one new adapter file and one `register()` call — no changes to services, actions, or UI.
- Switching sandbox → production is `ATLAS_ENV=production` + production tokens.
- Booking funnel analytics are queryable from day one via `booking_event`.
- Duplicate booking prevention is guaranteed at the DB level (idempotency PK).

### Harder:

- The `booking_idempotency` table needs a periodic cleanup job (expired rows). Acceptable trade-off; a simple cron (`DELETE WHERE expires_at < NOW()`) suffices.
- Post-booking stubs surface `NotImplementedError` in the UI — must be surfaced gracefully ("Coming soon") rather than silently swallowed.

## Related

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- `src/lib/suppliers/config.ts` — environment config module (new, Sprint 1)
- `src/lib/suppliers/providers/types.ts` — `BookingProvider` interface (existing)
- `src/lib/suppliers/providers/registry.ts` — provider registry (existing, now wired)
- `src/lib/schema.ts` — schema additions (migration 0019)
- `docs/database.md` — updated table and migration entries
- `docs/api-integrations.md` — updated provider status

# Phase 1 — "Sellable" Implementation Plan

---

Goal: take Atlas from demo to sellable. Seven features, dependency-ordered.

Decisions locked:

- **Billing:** vendor bills agencies (SaaS subscriptions) **now**; agencies bill travelers (Stripe Connect) **later**.
- **Email:** Resend · **Flights:** Amadeus · **Hotels:** Hotelbeds.

## Key findings (much is already partially built)

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- **Email** — a SendGrid adapter already exists at `src/lib/notifications/email.ts` with the right `sendEmail()/EmailResult` contract. Resend = one-file swap. Gaps: invites never actually send (toast is cosmetic, delivery is copy-paste); both Better Auth hooks (`auth.ts:85-97`) only `console.log`.
- **Stripe** — a fetch-based Checkout integration already exists for *traveler* booking payments (`src/lib/payments/stripe.ts`); `STRIPE_SECRET_KEY` is in `.env`. SaaS subscriptions are a **separate billing plane** on the agency table.
- **Travel APIs** — `SupplierProvider` abstraction exists with a working **Amadeus implementation for flights AND hotels** (`src/lib/suppliers/`). Flights ~done.
- **Bug** — `products/[id]/page.tsx:61` links to `/products/${id}/proposal`, a route that doesn't exist (real one is `/proposal/[id]`). Fix during PDF/e-sign work.

## Sequence

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1. DB split — foundation; before billing & real customers
2. Email (Resend) — unblocks invites, billing receipts, e-sign, proposals
3. Stripe SaaS billing — needs email + agency gating
4. PDF proposals — pair
5. E-signature — e-sign needs public token + email
6. Hotelbeds + flights — parallel track, independent of 1-5

## 1. Separate databases — S (operational)

---

- Route `src/lib/db.ts:5-12` through `getServerEnv()` (currently bypasses zod).
- Add prod guard to destructive scripts (`seed-demo-data.ts`, `db:reset`, `test-tenant-isolation.ts`) — refuse unless `ALLOW_PROD=1` / `host ≠ prod`.
- User action: create a dev Neon branch; prod `POSTGRES_URL` only in Vercel env.
- Vercel `build:ci` skips migrations — confirm out-of-band migrate process.

## 2. Email delivery (Resend) — S

---

- Rewrite `src/lib/notifications/email.ts` to call Resend, keep the signature (zero caller changes); add optional `html?` for React Email later.
- Add `RESEND_API_KEY` + `EMAIL_FROM` to `src/lib/env.ts` (schema + `checkEnv`).
- Wire invites: capture `createInvite()` at `actions/invites.ts:67`, build link from `NEXT_PUBLIC_APP_URL + /invite/${token}`, send, persist to notification.
- Fill the two auth hooks (`auth.ts:85-97`): password reset + verification.
- Extend `notification.kind` (`invite`, `password_reset`, `proposal`) as an outbox.
- Watch: auth emails may have no `agencyId` (NOT NULL on notification).

### 3. Stripe SaaS billing (vendor → agencies) — M

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- Migration 0009: add to agency — `stripeCustomerId`, `stripeSubscriptionId`, `subscriptionStatus`, `priceId`, `currentPeriodEnd`, `trialEndsAt`.
- Gate in `requireAgencyUser` (`permissions.ts:136-144`) — already checks `agency.status`; add subscription-state block (`past_due/canceled`).
- `createAgency` (`actions/platform.ts:74`) creates Stripe customer + trial sub.
- New route `src/app/api/stripe/webhook/route.ts` (POST, `raw req.text()` body).
- Add the official stripe npm SDK (subscriptions + signature verification).
- Env: `STRIPE_WEBHOOK_SECRET` + price IDs. Surface status on platform detail page.
- Stripe Connect (traveler payments) = deferred to Phase 1.5, separate module on the `payment/booking` plane.

### 4. PDF proposals — M

---

- Today = HTML + `window.print()`. Add real renderer: Puppeteer + `@sparticuz/chromium` on Vercel serverless, store via `@vercel/blob` (`src/lib/storage.ts`). Alt: `react-pdf`. Fix the broken proposal link.

### 5. E-signature acceptance — M (greenfield)

---

- Add to product: `shareToken` (unique), `acceptedAt`, `declinedAt`, `signerName`, `signerEmail`, `signatureData`, `signerIp`, `signerUserAgent`.
- Public token route mirroring `/i/[token]:generateProposalLink` (copy `generateShareLink` `bookings.ts:383`), new unauth `/p/[token]`.
- New unauth `acceptProposalByToken` (first token-authenticated write) → set signature, `product.status=accepted`, `opportunity.stage=won`, send confirmation email.

### 6. Hotelbeds (hotels) + Amadeus (flights) — M

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- New `src/lib/suppliers/hotelbeds.ts` mirroring `amadeus.ts` (Api-key + SHA-256).
- Split `getSupplier()` (`suppliers/index.ts:20`) into `getFlightSupplier()` / `getHotelSupplier()`; update `safeSearch` + call sites (`actions/search.ts`, `api/chat/route.ts`).
- Add `rateKey` to `HotelOffer`; persist supplier offers into `bookingItem.details` (AI

`createBooking` currently drops them; `confirmItemBooking` expects them).

- Add `HOTELBEDS_API_KEY/HOTELBEDS_SECRET` to `env.ts`.

## Effort

| # | Feature               | Effort | Status                                    |
|---|-----------------------|--------|-------------------------------------------|
| 1 | DB split (code parts) | S      | ✓ done — Neon branch is a manual step     |
| 2 | Email (Resend)        | S      | ✓ done                                    |
| 3 | Stripe SaaS billing   | M      | ✓ done (subscriptions); Connect deferred  |
| 4 | PDF proposals         | M      | ✓ done (@react-pdf, server-rendered)      |
| 5 | E-signature           | M      | ✓ done (public token + sign + audit)      |
| 6 | Hotelbeds + flights   | M      | ✓ done (per-vertical providers + rateKey) |

## Manual setup required (you)

- **Neon dev branch:** create a separate Neon branch/database for local dev; put its URL in local `.env`, keep prod `POSTGRES_URL` only in Vercel env. Set `PROTECTED_DB_HOSTS=<prod-neon-host>` in prod env so destructive scripts refuse to touch it (override with `ALLOW_PROD=1`).
- **Resend:** add `RESEND_API_KEY` + `EMAIL_FROM` (verified domain) to `.env` and Vercel. Without them, emails log to console (status logged).
- **Stripe billing:** add `STRIPE_SECRET_KEY`, `STRIPE_PRICE_ID` (the SaaS plan price), and `STRIPE_WEBHOOK_SECRET` to `.env` and Vercel. Register the webhook endpoint `POST /api/stripe/webhook` in the Stripe dashboard, subscribing to `customer.subscription.{created, updated, deleted}`. Enable the Billing Portal.
- **Suppliers:** add `AMADEUS_CLIENT_ID/AMADEUS_CLIENT_SECRET` (flights) and `HOTELBEDS_API_KEY/HOTELBEDS_SECRET` (hotels); optional `*_HOSTNAME=production`. Without them, search falls back to sample data. Hotelbeds keys availability by its own destination codes — pass a resolved `cityCode` for live hotel search (city name works against the mock).

## Known follow-ups (not blocking Phase 1)

- AI `createBooking` tool (`api/chat/route.ts`) still doesn't persist the supplier offer into `bookingItem.details`, so AI-built items can't be auto-booked by `confirmItemBooking`. The human search→"Add to booking" flow does persist it.
- Hotelbeds destination-code resolution (content API) for city-name → code.

# UI Polish & Responsive Improvements

## Overview

Improve the visual quality, component consistency, and responsive behavior of the boilerplate UI across all pages. This includes refreshing the color palette with a subtle blue accent, replacing bare HTML elements with shadcn components, adding hover/transition effects via a reusable utility class, and introducing `sm:` breakpoints throughout for better tablet/mobile scaling. All custom styling goes in `globals.css` — no inline or custom CSS on components.

## Quick Links

- Requirements — full requirements and acceptance criteria
- Action Required — manual steps needing human action

## Dependency Graph

```
graph TD
  task-01-globals-css["01: globals.css Foundation"]
  task-02-layout["02: Layout Sticky Footer"]
  task-03-site-header["03: Site Header Responsive"]
  task-04-site-footer["04: Site Footer Responsive"]
  task-05-home-page["05: Home Page Overhaul"]
  task-06-dashboard["06: Dashboard Page"]
  task-07-chat-page["07: Chat Page"]
  task-08-profile-page["08: Profile Page"]
  task-09-auth-pages["09: Auth Pages"]
  task-10-setup-checklist["10: Setup Checklist"]
  task-11-starter-prompt-modal["11: Starter Prompt Modal"]
  task-01-globals-css --> task-05-home-page
  task-01-globals-css --> task-06-dashboard
  task-01-globals-css --> task-07-chat-page
  task-01-globals-css --> task-08-profile-page
  task-01-globals-css --> task-09-auth-pages
  task-01-globals-css --> task-10-setup-checklist
  task-01-globals-css --> task-11-starter-prompt-modal
  task-02-layout --> task-09-auth-pages
```

## Waves

| Wave | Tasks                              | Description                                                                           |
|------|------------------------------------|---------------------------------------------------------------------------------------|
| 1    | task-01, task-02, task-03, task-04 | Foundation: globals.css enhancements, layout flex structure, header/footer responsive |

|   |                                                               |                                                                   |
|---|---------------------------------------------------------------|-------------------------------------------------------------------|
| 2 | task-05, task-06, task-07, task-08, task-09, task-10, task-11 | All pages: Card adoption, responsive breakpoints, component swaps |
|---|---------------------------------------------------------------|-------------------------------------------------------------------|

## Task Status

---

### Wave 1

- ☒ task-01-globals-css — globals.css color palette, animations, and utility classes
- ☒ task-02-layout — Root layout sticky footer with flex column
- ☒ task-03-site-header — Header responsive padding and sizing
- ☒ task-04-site-footer — Footer responsive padding

### Wave 2

- ☐ task-05-home-page — Home page Card adoption + responsive typography
- ☐ task-06-dashboard — Dashboard Card adoption + responsive grid
- ☐ task-07-chat-page — Chat Input swap + sticky form + responsive
- ☐ task-08-profile-page — Profile responsive stacking + grid fixes
- ☐ task-09-auth-pages — Auth pages animation, shadow, and background
- ☐ task-10-setup-checklist — Setup checklist Card adoption
- ☐ task-11-starter-prompt-modal — Textarea swap + responsive buttons

# Action Required: UI Polish & Responsive Improvements

---

No manual steps required for this feature. All tasks can be implemented automatically.



# Requirements: UI Polish & Responsive Improvements

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## Summary

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The boilerplate UI is functional but visually flat — bare `<div>` elements instead of shadcn Cards, a monotone neutral-gray palette, no hover/transition effects, and minimal responsive breakpoints (most pages only use `md:`). The goal is to meaningfully improve visual quality, component consistency, and responsive behavior across all pages and components.

All changes must use Tailwind utilities and shadcn standards. Custom styling (animations, utilities, color tokens) goes in `globals.css`. No custom inline styles or CSS-in-JS on individual components. The result should look like a polished, high-quality default that can be customized by users of the boilerplate.

## Goals

---

- Refresh the color palette with a subtle cool blue accent (shift primary/ring/accent from flat gray to hue 270)
- Replace all bare `<div>` cards with shadcn Card components for consistency
- Replace native `<input>`, `<button>`, and `<textarea>` elements with their shadcn equivalents
- Add hover/transition effects to interactive cards via a reusable `card-interactive` utility
- Introduce `sm:` breakpoints throughout for proper tablet/mobile scaling
- Add entrance animations to auth pages
- Fix the sticky footer layout so it works on short-content pages
- Ensure all pages scale well from 320px to 1536px+ viewports

## Non-Goals

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- No hamburger/mobile menu for the header — the current content (avatar + theme toggle) is minimal enough
- No new shadcn component installations — all needed components are already installed
- No changes to business logic, API routes, or authentication flows
- No new pages or routes
- No changes to the Geist font choice
- No Storybook or component documentation

## Acceptance Criteria

---

- [ ] All pages render correctly in both light and dark modes
- [ ] No horizontal scroll on any page at 320px viewport width

- [ ] Feature cards, dashboard cards, and next steps cards use shadcn Card components
- [ ] Chat input uses shadcn Input, not a native `<input>`
- [ ] Footer sticks to the bottom on short-content pages (auth pages)
- [ ] Primary color has a visible cool-blue tint (not flat gray)
- [ ] Cards have hover lift effect with shadow transition
- [ ] All grids have `sm:` breakpoints for tablet-size viewports
- [ ] `pnpm lint` and `pnpm typecheck` pass with no errors

## Assumptions

---

- Tailwind CSS v4 is in use (confirmed: `@import "tailwindcss"` syntax, `@theme inline block`)
- `tw-animate-css` is installed in dependencies but currently unused (confirmed in `package.json`)
- The `ThemeProvider` from `next-themes` renders children directly without a wrapping `<div>` (confirmed by reading the component)
- All shadcn components needed (Card, Input, Button, Textarea, Badge, Avatar, Dialog) are already installed

## Technical Constraints

---

- All custom CSS (animations, utilities, color tokens) must go in `src/app/globals.css`
- Component files should only use Tailwind utility classes and shadcn components — no `style={{}} props`
- Must maintain both light and dark mode support via the existing OKLch CSS variable system
- Must not break existing functionality (auth flows, chat, diagnostics)

# Task 01: globals.css Foundation

---

## Status

---

complete

## Wave

---

1

## Description

---

Update `globals.css` to establish the visual foundation for the entire UI improvement. This includes importing the unused `tw-animate-css` package, refreshing the color palette with a subtle blue accent, adding custom keyframe animations, expanding the base layer with better defaults, and defining reusable utility classes. All subsequent tasks depend on the utilities and color tokens defined here.

## Dependencies

---

**Depends on:** None (Wave 1) **Blocks:** task-05-home-page.md, task-06-dashboard.md, task-07-chat-page.md, task-08-profile-page.md, task-09-auth-pages.md, task-10-setup-checklist.md, task-11-starter-prompt-modal.md

**Context from dependencies:** N/A — this is the foundation task.

## Files to Modify

---

- `src/app/globals.css` — Add `tw-animate-css` import, update color tokens, add animations, expand base layer, add utility classes

## Technical Details

---

### Implementation Steps

1. **Add `tw-animate-css` import** — The package `tw-animate-css@1.4.0` is already installed (in `package.json`) but never imported. Add the import right after the `tailwindcss` import:

```
@import "tailwindcss";
@import "tw-animate-css";
```

1. **Add animation tokens to `@theme inline`** — Add these at the end of the existing `@theme`

inline block (after the `--radius-xl` line):

```
--animate-fade-in: fade-in 0.3s ease-out;
--animate-fade-up: fade-up 0.4s ease-out;
--animate-scale-in: scale-in 0.2s ease-out;
```

**1. Update light mode color tokens in `:root`** — Change these 3 values (leave all other tokens unchanged):

```
/* CHANGE these 3 lines: */
--primary: oklch(0.21 0.034 270);           /* was oklch(0.21 0.006 285.885) */
--ring: oklch(0.705 0.06 270);             /* was oklch(0.705 0.015 286.067) */
--accent: oklch(0.96 0.012 270);           /* was oklch(0.967 0.001 286.375) */
```

**1. Update dark mode color tokens in `.dark`** — Change these 3 values (leave all other tokens unchanged):

```
/* CHANGE these 3 lines: */
--primary: oklch(0.92 0.02 270);           /* was oklch(0.92 0.004 286.32) */
--ring: oklch(0.552 0.05 270);            /* was oklch(0.552 0.016 285.938) */
--accent: oklch(0.28 0.018 270);          /* was oklch(0.274 0.006 286.033) */
```

**1. Add keyframe definitions** — Add these right before the `@layer base` block:

```
@keyframes fade-in {
  from { opacity: 0; }
  to { opacity: 1; }
}

@keyframes fade-up {
  from { opacity: 0; transform: translateY(8px); }
  to { opacity: 1; transform: translateY(0); }
}

@keyframes scale-in {
  from { opacity: 0; transform: scale(0.97); }
  to { opacity: 1; transform: scale(1); }
}
```

**1. Expand the `@layer base` block** — Replace the existing `@layer base` block with:

```
@layer base {
  * {
    @apply border-border;
  }
  html {
    scroll-behavior: smooth;
  }
  body {
    @apply bg-background text-foreground;
    font-feature-settings: "rlig" 1, "calt" 1;
  }
}
```

```

}
:focus-visible {
  @apply outline-2 outline-offset-2 outline-ring;
}
}

```

### 1. Add utility classes — Add a new `@layer utilities` block after the `@layer base` block:

```

@layer utilities {
  .card-interactive {
    @apply transition-all duration-200 ease-out;
  }
  .card-interactive:hover {
    @apply shadow-md -translate-y-0.5;
  }
  .auth-bg {
    background-image: radial-gradient(
      circle at 50% 0%,
      var(--accent) 0%,
      transparent 50%
    );
  }
}

```

## Final file structure (order of sections)

1. `@import "tailwindcss";`
2. `@import "tw-animate-css";`
3. `@theme inline { ... }` (with animation tokens added at end)
4. `:root { ... }` (with 3 updated color values)
5. `.dark { ... }` (with 3 updated color values)
6. `@keyframes` definitions (3 keyframes)
7. `@layer base { ... }` (expanded)
8. `@layer utilities { ... }` (new)

## Acceptance Criteria

- `[] tw-animate-css` is imported and animation utilities like `animate-fade-up` are available
- `[] :root` has `--primary: oklch(0.21 0.034 270), --ring: oklch(0.705 0.06 270), --accent: oklch(0.96 0.012 270)`
- `[] .dark` has `--primary: oklch(0.92 0.02 270), --ring: oklch(0.552 0.05 270), --accent: oklch(0.28 0.018 270)`
- `[]` Three keyframe animations (`fade-in`, `fade-up`, `scale-in`) are defined
- `[] @theme inline` includes `--animate-fade-in`, `--animate-fade-up`, `--animate-scale-in`
- `[] .card-interactive` utility class is defined with hover shadow and translate
- `[] .auth-bg` utility class is defined with radial gradient
- `[] html` has `scroll-behavior: smooth`
- `[] :focus-visible` has global outline styling
- `[]` All other existing color tokens remain unchanged

# Task 02: Layout Sticky Footer

---

## Status

---

complete

## Wave

---

1

## Description

---

Update the root layout to use a flex column structure that pushes the footer to the bottom of the viewport on short-content pages. Currently, pages like login/register have minimal content and the footer floats up, leaving blank space below. Adding `min-h-screen flex flex-col` to the body and `flex-1` to the main element creates a proper sticky footer layout.

## Dependencies

---

**Depends on:** None (Wave 1) **Blocks:** task-09-auth-pages.md (auth pages rely on the flex-1 main for proper vertical centering)

**Context from dependencies:** N/A — this is a foundation task.

## Files to Modify

---

- `src/app/layout.tsx` — Add flex classes to body and main elements

## Technical Details

---

### Implementation Steps

1. Add flex layout classes to the `<body>` element (line 92). The current className is:

```
className={` ${geistSans.variable} ${geistMono.variable} antialiased`}
```

Change it to:

```
className={` ${geistSans.variable} ${geistMono.variable} antialiased min-h-screen fle
```

1. Add `flex-1` to the `<main>` element (line 101). The current element is:

```
<main id="main-content">{children}</main>
```

Change it to:

```
<main id="main-content" className="flex-1">{children}</main>
```

## Why this works

The `ThemeProvider` component (`src/components/theme-provider.tsx`) is a pass-through wrapper around `next-themes ThemeProvider` — it renders children directly without adding a wrapping `<div>`. So the flex column chain from `<body>` flows directly through to `<SiteHeader>`, `<main>`, and `<SiteFooter>`.

The `min-h-screen` ensures the body fills at least the full viewport. `flex flex-col` creates a vertical flex container. `flex-1` on `<main>` makes it expand to fill all available space, pushing `<SiteFooter>` to the bottom.

## Acceptance Criteria

---

- [ ] `<body>` has classes `min-h-screen flex flex-col` in addition to existing font variable classes
- [ ] `<main>` has `className="flex-1"`
- [ ] On short-content pages (e.g. `/login`), the footer sits at the bottom of the viewport
- [ ] On long-content pages (e.g. `/profile`), the footer sits below the content as usual
- [ ] No visual regressions on any page

# Task 03: Site Header Responsive

---

## Status

---

complete

## Wave

---

1

## Description

---

Update the site header component with responsive padding and sizing so it scales better on mobile devices. Currently the header uses fixed `px-4 py-4` padding and `text-2xl` logo text that could be tighter on small phones. The action buttons also use a fixed `gap-4` that is generous on narrow screens.

## Dependencies

---

**Depends on:** None (Wave 1) **Blocks:** None

**Context from dependencies:** N/A

## Files to Modify

---

- `src/components/site-header.tsx` — Update Tailwind classes for responsive sizing

## Technical Details

---

### Implementation Steps

1. **Update nav padding** (line 19). Change:

```
container mx-auto px-4 py-4 flex justify-between items-center
```

To:

```
container mx-auto px-3 sm:px-4 py-3 sm:py-4 flex justify-between items-center
```

1. **Update logo text size** (line 21). Change the `<h1>` `className` from:

```
text-2xl font-bold
```



To:

```
text-xl sm:text-2xl font-bold
```

1. **Update actions group gap** (line 38). Change:

```
flex items-center gap-4
```

To:

```
flex items-center gap-2 sm:gap-4
```

## Current file reference

The full current file is at `src/components/site-header.tsx` (46 lines). The structure is:

- Skip-to-main-content link (accessibility)
- `<header className="border-b">` wrapper
- `<nav>` with logo on left, UserProfile + ModeToggle on right

## Acceptance Criteria

---

- [ ] Nav padding is `px-3 sm:px-4 py-3 sm:py-4`
- [ ] Logo text is `text-xl sm:text-2xl`
- [ ] Actions gap is `gap-2 sm:gap-4`
- [ ] Header looks good at 320px viewport width (no overflow, no cramping)
- [ ] Header looks unchanged at desktop widths (1024px+)

# Task 04: Site Footer Responsive

---

## Status

---

complete

## Wave

---

1

## Description

---

Update the site footer with responsive padding to match the header's responsive scaling and maintain consistent container padding across the app. Currently the footer uses fixed `py-6` and `px-4` that could be tighter on mobile and more generous on large screens.

## Dependencies

---

**Depends on:** None (Wave 1) **Blocks:** None

**Context from dependencies:** N/A

## Files to Modify

---

- `src/components/site-footer.tsx` — Update Tailwind classes for responsive padding

## Technical Details

---

### Implementation Steps

1. **Update footer padding** (line 5). Change:

```
border-t py-6 text-center text-sm text-muted-foreground
```

To:

```
border-t py-4 sm:py-6 text-center text-sm text-muted-foreground
```

1. **Update container padding** (line 6). Change:

```
container mx-auto px-4
```

To:

```
container mx-auto px-4 sm:px-6 lg:px-8
```

## Current file reference

The file is at `src/components/site-footer.tsx` (24 lines). It renders a `<footer>` with a `GitHubStars` component and an attribution link.

## Acceptance Criteria

---

- [ ] Footer padding is `py-4 sm:py-6`
- [ ] Container padding is `px-4 sm:px-6 lg:px-8`
- [ ] Footer is slightly more compact on mobile (`py-4` vs `py-6`)
- [ ] Footer has more generous side padding on large screens (`lg:px-8`)

# Task 05: Home Page Overhaul

---

## Status

---

pending

## Wave

---

2

## Description

---

Overhaul the home page with responsive typography, shadow Card components replacing bare divs, hover effects on interactive cards, and better grid breakpoints. This is the highest-visibility page and sets the visual tone for the entire app. The feature cards and next steps cards are currently plain `<div>` elements with `border rounded-lg` — they need to be replaced with proper shadow Card components with the `card-interactive` hover utility.

## Dependencies

---

**Depends on:** task-01-globals-css.md **Blocks:** None

**Context from dependencies:** Task 01 defines the `card-interactive` utility class in `globals.css` which provides `transition-all duration-200 ease-out` with hover effects (`shadow-md -translate-y-0.5`). This task uses that class on all interactive cards. Task 01 also updates the color palette so `text-primary` on icons will show the new blue-tinted primary color.

## Files to Modify

---

- `src/app/page.tsx` — Replace bare divs with shadow Cards, update typography responsive classes, fix grid breakpoints

## Technical Details

---

### Implementation Steps

1. **Add Card imports** at the top of the file. Add this import:

```
import {  
  Card,  
  CardContent,  
  CardHeader,
```

```
CardTitle,  
} from "@/components/ui/card";
```

**1. Update container spacing** (line 13). Change:

```
className="flex-1 container mx-auto px-4 py-12"
```

To:

```
className="flex-1 container mx-auto px-4 sm:px-6 lg:px-8 py-8 sm:py-12"
```

**1. Update hero h1 typography** (line 20). Change:

```
className="text-5xl font-bold tracking-tight bg-gradient-to-r from-primary via-prima
```

To:

```
className="text-3xl sm:text-4xl md:text-5xl font-bold tracking-tight bg-gradient-to-
```

**1. Update hero h2 subtitle** (line 24). Change:

```
className="text-2xl font-semibold text-muted-foreground"
```

To:

```
className="text-xl sm:text-2xl font-semibold text-muted-foreground"
```

**1. Update hero description paragraph** (line 27). Change:

```
className="text-xl text-muted-foreground"
```

To:

```
className="text-base sm:text-lg md:text-xl text-muted-foreground"
```

**1. Update "Video Tutorial" heading** (line 35). Change:

```
className="text-2xl font-semibold flex items-center justify-center gap-2"
```

To:

```
className="text-xl sm:text-2xl font-semibold flex items-center justify-center gap-2"
```

**1. Add shadow to video embed container** (line 43). Change:

```
className="relative pb-[56.25%] h-0 overflow-hidden rounded-lg border"
```

To:

```
className="relative pb-[56.25%] h-0 overflow-hidden rounded-lg border shadow-md"
```

1. **Update feature grid** (line 55). Change:

```
className="grid grid-cols-1 md:grid-cols-2 lg:grid-cols-4 gap-6 mt-12"
```

To:

```
className="grid grid-cols-1 sm:grid-cols-2 lg:grid-cols-4 gap-4 sm:gap-6 mt-12"
```

1. **Replace all 4 feature card divs** (lines 56-92) with shadcn Card components. Each card currently looks like:

```
<div className="p-6 border rounded-lg">
  <h3 className="font-semibold mb-2 flex items-center gap-2">
    <Shield className="h-4 w-4" />
    Authentication
  </h3>
  <p className="text-sm text-muted-foreground">
    Better Auth with Google OAuth integration
  </p>
</div>
```

Replace each with:

```
<Card className="card-interactive">
  <CardHeader className="pb-3">
    <CardTitle className="text-base flex items-center gap-2">
      <Shield className="h-4 w-4 text-primary" />
      Authentication
    </CardTitle>
  </CardHeader>
  <CardContent>
    <p className="text-sm text-muted-foreground">
      Better Auth with Google OAuth integration
    </p>
  </CardContent>
</Card>
```

Do this for all 4 cards (Authentication/Shield, Database/Database, AI Ready/Bot, UI Components/Palette). Add `text-primary` to each icon.

1. **Update "Next Steps" heading** (line 97). Change:

```
className="text-2xl font-semibold"
```

To:

```
className="text-xl sm:text-2xl font-semibold"
```

1. **Update next steps grid** (line 98). Change:

```
className="grid grid-cols-1 md:grid-cols-2 gap-4 text-left"
```

To:

```
className="grid grid-cols-1 sm:grid-cols-2 gap-4 text-left"
```

1. **Replace all 4 next steps card divs** (lines 99-169) with shadcn Card components. Each currently looks like:

```
<div className="p-4 border rounded-lg">
  <h4 className="font-medium mb-2">1. Set up environment variables</h4>
  ...content...
</div>
```

Replace each wrapper with:

```
<Card className="card-interactive">
  <CardContent className="pt-6">
    <h4 className="font-medium mb-2">1. Set up environment variables</h4>
    ...content stays the same...
  </CardContent>
</Card>
```

Use `pt-6` on `CardContent` since there's no `CardHeader` for these simpler cards.

## Acceptance Criteria

- [ ] Hero h1 scales: `text-3xl` on mobile, `text-4xl` on sm, `text-5xl` on md+
- [ ] Hero h2 and description also scale responsively
- [ ] Feature grid uses `sm:grid-cols-2` breakpoint (2 columns at 640px)
- [ ] All 4 feature cards use shadcn `Card/CardHeader/CardContent` with `card-interactive`
- [ ] All feature card icons have `text-primary` class
- [ ] All 4 next steps cards use shadcn `Card/CardContent` with `card-interactive`
- [ ] Video embed has `shadow-md`
- [ ] Container has responsive padding: `px-4` `sm:px-6` `lg:px-8`
- [ ] No TypeScript errors (Card imports are correct)

# Task 06: Dashboard Page

---

## Status

---

pending

## Wave

---

2

## Description

---

Update the dashboard page to use shadcn Card components instead of bare divs, add responsive breakpoints, improve the loading state, and add visual polish. The dashboard currently has two plain `<div>` cards with `border border-border rounded-lg` and a simple "Loading..." text.

## Dependencies

---

**Depends on:** task-01-globals-css.md **Blocks:** None

**Context from dependencies:** Task 01 defines the `card-interactive` utility class in `globals.css` and updates the primary color to a blue-tinted hue. This task uses `card-interactive` on cards and `text-primary` on new icon accents.

## Files to Modify

---

- `src/app/dashboard/page.tsx` — Replace divs with Cards, add responsive classes, improve loading

## Technical Details

---

### Implementation Steps

1. **Add new imports.** The file currently imports: `Link`, `Lock`, `UserProfile`, `Button`, `useDiagnostics`, `useSession`. Add these:

```
import { Lock, Bot, User } from "lucide-react";
import {
  Card,
  CardContent,
  CardDescription,
  CardHeader,
```



```
    CardTitle,  
  } from "@components/ui/card";  
  import { Spinner } from "@components/ui/spinner";
```

Remove the existing `Lock` from the `lucide-react` import (it's being re-imported in the combined import).

**1. Update loading state** (lines 15-19). Change:

```
return (  
  <div className="flex justify-center items-center h-screen">  
    Loading...  
  </div>  
);
```

To:

```
return (  
  <div className="flex justify-center items-center h-screen">  
    <Spinner size="lg" />  
  </div>  
);
```

**1. Update unauthenticated state container** (line 24). Change:

```
className="container mx-auto px-4 py-12"
```

To:

```
className="container mx-auto px-4 sm:px-6 lg:px-8 py-8 sm:py-12"
```

**1. Wrap unauthenticated content in Card** (lines 26-34). Change:

```
<div className="mb-8">  
  <Lock className="w-16 h-16 mx-auto mb-4 text-muted-foreground" />  
  <h1 className="text-2xl font-bold mb-2">Protected Page</h1>  
  <p className="text-muted-foreground mb-6">  
    You need to sign in to access the dashboard  
  </p>  
</div>
```

To:

```
<Card className="mb-8 text-center">  
  <CardContent className="pt-6">  
    <Lock className="w-16 h-16 mx-auto mb-4 text-muted-foreground" />  
    <h1 className="text-2xl font-bold mb-2">Protected Page</h1>  
    <p className="text-muted-foreground mb-6">  
      You need to sign in to access the dashboard  
    </p>  
  </CardContent>
```

```
</Card>
```

### 1. Update authenticated container (line 40). Change:

```
className="container mx-auto p-6"
```

To:

```
className="container mx-auto p-4 sm:p-6 max-w-5xl"
```

### 1. Update title section (lines 41-43). Change:

```
<div className="flex justify-between items-center mb-8">
  <h1 className="text-3xl font-bold">Dashboard</h1>
</div>
```

To:

```
<div className="flex flex-wrap justify-between items-center gap-4 mb-6 sm:mb-8">
  <h1 className="text-2xl sm:text-3xl font-bold">Dashboard</h1>
</div>
```

### 1. Update card grid (line 45). Change:

```
className="grid grid-cols-1 md:grid-cols-2 gap-6"
```

To:

```
className="grid grid-cols-1 sm:grid-cols-2 gap-6"
```

### 1. Replace AI Chat card div (lines 46-59). Change:

```
<div className="p-6 border border-border rounded-lg">
  <h2 className="text-xl font-semibold mb-2">AI Chat</h2>
  <p className="text-muted-foreground mb-4">
    Start a conversation with AI using the Vercel AI SDK
  </p>
  ...button...
</div>
```

To:

```
<Card className="card-interactive">
  <CardHeader>
    <CardTitle className="flex items-center gap-2">
      <Bot className="h-5 w-5 text-primary" />
      AI Chat
    </CardTitle>
```

```

    <CardDescription>
      Start a conversation with AI using the Vercel AI SDK
    </CardDescription>
  </CardHeader>
  <CardContent>
    ...button stays the same...
  </CardContent>
</Card>

```

### 1. Replace Profile card div (lines 62-76). Change:

```

<div className="p-6 border border-border rounded-lg">
  <h2 className="text-xl font-semibold mb-2">Profile</h2>
  <p className="text-muted-foreground mb-4">
    Manage your account settings and preferences
  </p>
  <div className="space-y-2">
    <p><strong>Name:</strong> {session.user.name}</p>
    <p><strong>Email:</strong> {session.user.email}</p>
  </div>
</div>

```

To:

```

<Card className="card-interactive">
  <CardHeader>
    <CardTitle className="flex items-center gap-2">
      <User className="h-5 w-5 text-primary" />
      Profile
    </CardTitle>
    <CardDescription>
      Manage your account settings and preferences
    </CardDescription>
  </CardHeader>
  <CardContent>
    <div className="space-y-2">
      <p><strong>Name:</strong> {session.user.name}</p>
      <p><strong>Email:</strong> {session.user.email}</p>
    </div>
  </CardContent>
</Card>

```

## Acceptance Criteria

- [ ] Both dashboard cards use shadow `Card/CardHeader/CardTitle/CardDescription/CardContent`
- [ ] Both cards have `card-interactive` class
- [ ] AI Chat card has Bot icon with `text-primary`, Profile card has User icon with `text-primary`
- [ ] Loading state shows `Spinner` component instead of "Loading..." text
- [ ] Container has `max-w-5xl` to prevent overly wide content
- [ ] Grid uses `sm:grid-cols-2` breakpoint

- ☐ Title scales: `text-2xl` `sm:text-3xl`
- ☐ Unauthenticated state wraps in Card
- ☐ No TypeScript errors

# Task 07: Chat Page

---

## Status

---

pending

## Wave

---

2

## Description

---

Update the chat page to use shadcn Input/Button components instead of native HTML elements, add responsive breakpoints to the header and message bubbles, make the input form sticky at the bottom, and improve the empty state. This page has the most interactive elements and benefits significantly from consistent component usage.

## Dependencies

---

**Depends on:** task-01-globals-css.md **Blocks:** None

**Context from dependencies:** Task 01 updates the color palette (so `text-primary`, `bg-primary` references use the new blue-tinted primary). No utility classes from Task 01 are used on this page (chat doesn't use `card-interactive`).

## Files to Modify

---

- `src/app/chat/page.tsx` — Replace native elements with shadcn components, add responsive classes, sticky input

## Technical Details

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### Implementation Steps

1. **Add Input import.** Add to the existing imports:

```
import { Input } from "@/components/ui/input";
```

The file already imports `Button` from `@/components/ui/button`.

1. **Also import Bot icon** for the improved empty state. Add `Bot` to the existing `lucide-react` import:

```
import { Copy, Check, Loader2, Bot } from "lucide-react";
```

1. **Replace native `<button>` in `CopyButton`** (lines 165-177). Change the function body from:

```
return (
  <button
    onClick={handleCopy}
    className="p-1 hover:bg-muted rounded transition-colors"
    title="Copy to clipboard"
  >
    {copied ? (
      <Check className="h-3.5 w-3.5 text-green-500" />
    ) : (
      <Copy className="h-3.5 w-3.5 text-muted-foreground" />
    )}
  </button>
);
```

To:

```
return (
  <Button
    variant="ghost"
    size="icon"
    className="h-7 w-7"
    onClick={handleCopy}
    title="Copy to clipboard"
  >
    {copied ? (
      <Check className="h-3.5 w-3.5 text-green-500" />
    ) : (
      <Copy className="h-3.5 w-3.5 text-muted-foreground" />
    )}
  </Button>
);
```

1. **Update container** (line 247). Change:

```
className="container mx-auto px-4 py-8"
```

To:

```
className="container mx-auto px-4 sm:px-6 lg:px-8 py-6 sm:py-8"
```

1. **Update chat header** (lines 249-261). Change:

```
<div className="flex justify-between items-center mb-6 pb-4 border-b">
  <h1 className="text-2xl font-bold">AI Chat</h1>
  <div className="flex items-center gap-4">
    <span className="text-sm text-muted-foreground">
      Welcome, {session.user.name}!
    </span>
  </div>
</div>
```

```
</span>
```

To:

```
<div className="flex flex-col sm:flex-row sm:justify-between sm:items-center gap-3 s
  <h1 className="text-xl sm:text-2xl font-bold">AI Chat</h1>
  <div className="flex items-center gap-2 sm:gap-4">
    <span className="text-sm text-muted-foreground hidden sm:inline">
      Welcome, {session.user.name}!
    </span>
```

Key changes: flex stacks on mobile, welcome text hidden on mobile (`hidden sm:inline`), smaller gaps.

1. **Update message bubble max-widths** (line ~289-290). For user messages, change:

```
bg-primary text-primary-foreground ml-auto max-w-[80%]
```

To:

```
bg-primary text-primary-foreground ml-auto max-w-[85%] sm:max-w-[80%] md:max-w-2xl s
```

For AI messages, change:

```
bg-muted max-w-[80%]
```

To:

```
bg-muted max-w-[85%] sm:max-w-[80%] md:max-w-2xl shadow-sm
```

1. **Update message area** (line 271). Change:

```
className="min-h-[50vh] overflow-y-auto space-y-4 mb-4"
```

To:

```
className="min-h-[50vh] overflow-y-auto space-y-4 mb-4 pb-4"
```

1. **Improve empty state** (lines 272-275). Change:

```
<div className="text-center text-muted-foreground py-12">
  Start a conversation with AI
</div>
```

To:

```
<div className="text-center text-muted-foreground py-16 space-y-3">
```

```

<Bot className="h-12 w-12 mx-auto text-muted-foreground/50" />
<p className="text-lg font-medium">Start a conversation with AI</p>
<p className="text-sm">Type a message below to begin chatting</p>
</div>

```

### 1. Make input form sticky and replace native input (lines 317-344). Change the form from:

```

<form
  onSubmit={...}
  className="flex gap-2"
>
  <input
    value={input}
    onChange={(e) => setInput(e.target.value)}
    placeholder="Type your message..."
    className="flex-1 p-2 border border-border rounded-md focus:outline-none focus:r
    disabled={isStreaming}
  />
  <Button type="submit" disabled={!input.trim() || isStreaming}>
    ...
  </Button>
</form>

```

To:

```

<form
  onSubmit={...}
  className="sticky bottom-0 bg-background py-4 border-t z-10 flex gap-2 sm:gap-3"
>
  <Input
    value={input}
    onChange={(e) => setInput(e.target.value)}
    placeholder="Type your message..."
    className="flex-1 min-w-0"
    disabled={isStreaming}
  />
  <Button type="submit" disabled={!input.trim() || isStreaming}>
    ...
  </Button>
</form>

```

**Key changes:** `sticky bottom-0 bg-background py-4 border-t z-10` makes the form stick. `Input` replaces native `<input>` for consistent styling. `min-w-0` prevents flex overflow.

## Acceptance Criteria

- [ ] Chat input uses shadcn `Input` component, not a native `<input>`
- [ ] CopyButton uses shadcn `Button variant="ghost" size="icon"`, not a native `<button>`
- [ ] Chat header stacks vertically on mobile (`flex-col`) and is horizontal on `sm+`
- [ ] Welcome text is hidden on mobile (`hidden sm:inline`)
- [ ] Message bubbles have `shadow-sm` for depth



- [ ] Message widths scale: `max-w-[85%]` on mobile, `max-w-[80%]` on sm, `max-w-2xl` on md+
- [ ] Input form is sticky at bottom with `bg-background` and `border-t`
- [ ] Empty state shows Bot icon, title text, and subtitle
- [ ] Container has responsive padding: `px-4 sm:px-6 lg:px-8`
- [ ] No TypeScript errors (Input import is correct)

# Task 08: Profile Page

---

## Status

---

pending

## Wave

---

2

## Description

---

Update the profile page with responsive stacking, improved grid breakpoints, and flex-wrap on tight rows. The avatar section needs to stack vertically on mobile, the quick actions grid needs an intermediate `sm:` breakpoint, and all the `flex items-center justify-between` rows in cards and dialogs need `flex-wrap` to prevent overflow on narrow screens.

## Dependencies

---

**Depends on:** task-01-globals-css.md **Blocks:** None

**Context from dependencies:** Task 01 updates the color palette. No utility classes from Task 01 are specifically used on this page (profile cards are informational, not interactive).

## Files to Modify

---

- `src/app/profile/page.tsx` — Update responsive classes throughout

## Technical Details

---

### Implementation Steps

1. **Update container** (line 67). Change:

```
className="container max-w-4xl mx-auto py-8 px-4"
```

To:

```
className="container max-w-4xl mx-auto py-6 sm:py-8 px-4 sm:px-6 lg:px-8"
```

1. **Update title row** (line 68). Change:

```
className="flex items-center gap-4 mb-8"
```

To:

```
className="flex items-center gap-2 sm:gap-4 mb-6 sm:mb-8"
```

1. **Update title text** (line 78). Change:

```
className="text-3xl font-bold"
```

To:

```
className="text-2xl sm:text-3xl font-bold"
```

1. **Update avatar section** (line 85). Change:

```
className="flex items-center space-x-4"
```

To:

```
className="flex flex-col items-center sm:flex-row sm:items-center gap-4"
```

Important: remove `space-x-4` entirely — it conflicts with `flex-col`. The `gap-4` handles spacing in both directions.

1. **Update avatar size** (line 86). Change:

```
className="h-20 w-20"
```

To:

```
className="h-16 w-16 sm:h-20 sm:w-20"
```

1. **Update text container** next to avatar (line 96). Change:

```
className="space-y-2"
```

To:

```
className="space-y-2 text-center sm:text-left"
```

1. **Update name heading** (line 97). Change:

```
className="text-2xl font-semibold"
```

To:

```
className="text-xl sm:text-2xl font-semibold"
```

1. **Update email row** (line 98). Change:

```
className="flex items-center gap-2 text-muted-foreground"
```

To:

```
className="flex flex-wrap items-center justify-center sm:justify-start gap-2 text-mu
```

1. **Update Account Status grid** (line 160). Change:

```
className="grid grid-cols-1 md:grid-cols-2 gap-4"
```

To:

```
className="grid grid-cols-1 sm:grid-cols-2 gap-4"
```

1. **Update all status/activity flex rows** — add `flex-wrap gap-2` and responsive padding. These appear at multiple locations. For each row that has `flex items-center justify-between p-4 border rounded-lg`, change to:

```
flex flex-wrap items-center justify-between gap-2 p-3 sm:p-4 border rounded-lg
```

Apply this to the following locations:

- Line 161 (Email Verification status)
- Line 172 (Account Type status)
- Line 196 (Current Session activity)

1. **Update Quick Actions grid** (line 224). Change:

```
className="grid grid-cols-1 md:grid-cols-3 gap-4"
```

To:

```
className="grid grid-cols-1 sm:grid-cols-2 lg:grid-cols-3 gap-4"
```

1. **Update dialog inner rows** — For the Security Settings dialog (lines 325, 342, 357), each row with `flex items-center justify-between p-4 border rounded-lg`, change to:

```
flex flex-wrap items-center justify-between gap-2 p-3 sm:p-4 border rounded-lg
```

1. **Update Email Preferences dialog rows** (lines 388, 396) — same pattern, change:

```
flex items-center justify-between p-4 border rounded-lg
```

To:

```
flex flex-wrap items-center justify-between gap-2 p-3 sm:p-4 border rounded-lg
```

## Acceptance Criteria

---

- [ ] Avatar section stacks vertically on mobile (flex-col) and is horizontal on sm+
- [ ] Avatar is smaller on mobile (h-16 w-16) and full size on sm+ (h-20 w-20)
- [ ] Text next to avatar is centered on mobile, left-aligned on sm+
- [ ] Email row wraps on narrow screens (flex-wrap)
- [ ] Quick actions grid has intermediate breakpoint: 1 col → 2 at sm → 3 at lg
- [ ] Account status grid uses sm:grid-cols-2 instead of md:grid-cols-2
- [ ] All flex justify-between rows in cards and dialogs have flex-wrap and gap-2
- [ ] Container and title have responsive padding and sizing
- [ ] No overflow or horizontal scroll at 320px viewport
- [ ] Profile page title scales: text-2xl sm:text-3xl

# Task 09: Auth Pages

---

## Status

---

pending

## Wave

---

2

## Description

---

Polish all 4 auth pages (login, register, forgot-password, reset-password) with entrance animations, card shadows, a subtle background gradient, and improved vertical centering that works with the new flex layout. These pages are the first thing new users see, so visual polish matters.

## Dependencies

---

**Depends on:** task-01-globals-css.md, task-02-layout.md **Blocks:** None

**Context from dependencies:** Task 01 defines the `auth-bg` utility class (radial gradient from accent color) and the `animate-fade-up` animation in `globals.css`. Task 02 adds `flex-1` to the `<main>` element, so auth pages can use `flex-1` to fill available vertical space properly.

## Files to Modify

---

- `src/app/(auth)/login/page.tsx` — Add animation, shadow, background, update min-height
- `src/app/(auth)/register/page.tsx` — Same changes
- `src/app/(auth)/forgot-password/page.tsx` — Same changes
- `src/app/(auth)/reset-password/page.tsx` — Same changes

## Technical Details

---

### Implementation Steps

All 4 files follow the exact same pattern. Each has an outer wrapper div with a Card inside.

**For each file, make these 2 changes:**

**Change 1: Update the wrapper div.** Each file has:

```
<div className="flex min-h-[calc(100vh-4rem)] items-center justify-center p-4">
```

Change to:

```
<div className="auth-bg flex flex-1 min-h-[calc(100vh-8rem)] items-center justify-ce
```

Changes:

- `auth-bg` — adds the subtle radial gradient background from `globals.css`
- `flex-1` — fills available space in the flex column layout (from Task 02)
- `min-h-[calc(100vh-8rem)]` — accounts for both header and footer height (was just 4rem for header)

**Change 2: Add shadow and animation to the Card.** Each file has:

```
<Card className="w-full max-w-md">
```

Change to:

```
<Card className="w-full max-w-md shadow-lg animate-fade-up">
```

Changes:

- `shadow-lg` — adds elevation/depth to the card
- `animate-fade-up` — entrance animation (defined in `globals.css` keyframes, registered in `@theme inline`)

## File-specific locations:

### **login/page.tsx:**

- Wrapper div: line 27
- Card: line 28

### **register/page.tsx:**

- Wrapper div: line 20
- Card: line 21

### **forgot-password/page.tsx:**

- Wrapper div: line 21
- Card: line 22

### **reset-password/page.tsx:**

- Wrapper div: line 21
- Card: line 22

## Acceptance Criteria

---

- [ ] All 4 auth pages have `auth-bg` class on the wrapper div
- [ ] All 4 auth pages have `flex-1 min-h-[calc(100vh-8rem)]` on the wrapper div
- [ ] All 4 auth cards have `shadow-lg` for depth
- [ ] All 4 auth cards have `animate-fade-up` for entrance animation
- [ ] Cards animate upward on page load (smooth fade + translate)
- [ ] Subtle radial gradient visible in the background (especially visible in dark mode)
- [ ] Footer stays at the bottom of the viewport on all auth pages
- [ ] No visual regressions to auth form functionality



# Task 10: Setup Checklist Card Adoption

---

## Status

---

pending

## Wave

---

2

## Description

---

Replace the bare `<div>` wrapper in the `SetupChecklist` component with a shadcn `Card` component for visual consistency with the rest of the app. The checklist is displayed on the home page and currently uses `p-6 border rounded-lg` which is the same pattern as the feature cards that are being upgraded to shadcn `Cards`.

## Dependencies

---

**Depends on:** task-01-globals-css.md **Blocks:** None

**Context from dependencies:** Task 01 updates the color palette. The setup checklist does not use `card-interactive` since it's an informational display, not a clickable card.

## Files to Modify

---

- `src/components/setup-checklist.tsx` — Replace wrapper div with shadcn `Card`

## Technical Details

---

### Implementation Steps

1. **Add Card imports.** Add at the top:

```
import {
  Card,
  CardContent,
  CardHeader,
  CardTitle,
} from "@/components/ui/card";
```

1. **Replace the outer wrapper** (line 124). The current structure is:

```

<div className="p-6 border rounded-lg text-left">
  <div className="flex items-center justify-between mb-4">
    <div>
      <h3 className="font-semibold">Setup checklist</h3>
      <p className="text-sm text-muted-foreground">
        {completed}/{steps.length} completed
      </p>
    </div>
    <Button size="sm" onClick={load} disabled={loading}>
      {loading ? "Checking..." : "Re-check"}
    </Button>
  </div>

  {error ? <div className="text-sm text-destructive">{error}</div> : null}

  <ul className="space-y-2">
    ...checklist items...
  </ul>

  {data ? (
    <div className="mt-4 text-xs text-muted-foreground">
      Last checked: {new Date(data.timestamp).toLocaleString()}
    </div>
  ) : null}
</div>

```

Replace with:

```

<Card className="text-left">
  <CardHeader className="pb-4">
    <div className="flex items-center justify-between">
      <div>
        <CardTitle className="text-base">Setup checklist</CardTitle>
        <p className="text-sm text-muted-foreground">
          {completed}/{steps.length} completed
        </p>
      </div>
      <Button size="sm" onClick={load} disabled={loading}>
        {loading ? "Checking..." : "Re-check"}
      </Button>
    </div>
  </CardHeader>
  <CardContent>
    {error ? <div className="text-sm text-destructive mb-4">{error}</div> : null}

    <ul className="space-y-2">
      ...checklist items stay the same...
    </ul>

    {data ? (
      <div className="mt-4 text-xs text-muted-foreground">
        Last checked: {new Date(data.timestamp).toLocaleString()}
      </div>
    ) : null}
  </CardContent>

```

```
</Card>
```

Key changes:

- Outer `<div>` → `<Card>`
- Title area wrapped in `<CardHeader>` with the flex row inside
- `h3` → `<CardTitle className="text-base">` (text-base to match current visual size)
- Content wrapped in `<CardContent>`
- Error div gets `mb-4` since `CardContent` doesn't have the same padding structure
- All checklist items and timestamp stay exactly the same

## Acceptance Criteria

---

- ☐ Setup checklist uses shadow `Card/CardHeader/CardTitle/CardContent`
- ☐ Visual appearance is consistent — no jarring size or spacing changes
- ☐ Title uses `CardTitle` with `text-base` to maintain current size
- ☐ Re-check button still works and is positioned to the right of the title
- ☐ Error state still displays correctly
- ☐ Checklist items render identically to before
- ☐ No TypeScript errors

# Task 11: Starter Prompt Modal

---

## Status

---

pending

## Wave

---

2

## Description

---

Replace the native `<textarea>` in the `StarterPromptModal` with the `shadcn Textarea` component for visual consistency, and make the button row responsive so buttons stack on very small screens.

## Dependencies

---

**Depends on:** task-01-globals-css.md **Blocks:** None

**Context from dependencies:** Task 01 updates the color palette. No utility classes from Task 01 are used in this component.

## Files to Modify

---

- `src/components/starter-prompt-modal.tsx` — Replace native textarea, update button layout

## Technical Details

---

### Implementation Steps

1. **Add Textarea import.** Add at the top:

```
import { Textarea } from "@/components/ui/textarea";
```

1. **Replace native `<textarea>`** (lines 161-167). Change:

```
<textarea
  id="project-description"
  placeholder="e.g., A task management app for teams with real-time collaboration, p
  value={projectDescription}
  onChange={ (e) => setProjectDescription(e.target.value) }
```

```
className="w-full h-24 px-3 py-2 border rounded-md resize-none text-sm"
/>
```

To:

```
<Textarea
  id="project-description"
  placeholder="e.g., A task management app for teams with real-time collaboration, p
  value={projectDescription}
  onChange={ (e) => setProjectDescription(e.target.value) }
  className="h-24 resize-none"
/>
```

Key changes:

- `<textarea>` → `<Textarea>` (shadcn component)
- Remove explicit `w-full px-3 py-2 border rounded-md text-sm` — the shadcn `Textarea` handles these styles internally
- Keep `h-24 resize-none` as overrides

1. **Update button row** (line 174). Change:

```
className="flex gap-2"
```

To:

```
className="flex flex-col-reverse sm:flex-row gap-2"
```

This makes the buttons stack on mobile (with Cancel on top via `flex-col-reverse`) and sit side-by-side on sm+.

## Acceptance Criteria

---

- [ ] `Textarea` uses shadcn `Textarea` component, not a native `<textarea>`
- [ ] `Textarea` has consistent focus ring styling matching other form elements
- [ ] Button row stacks on mobile (`flex-col-reverse`) and is horizontal on sm+ (`sm:flex-row`)
- [ ] "Copy Starter Prompt" button appears before "Cancel" button in both layouts
- [ ] `Textarea` placeholder text renders correctly
- [ ] Copy functionality still works (the `projectDescription` state binding is unchanged)
- [ ] No TypeScript errors